



Assessment Study for Identifying Gaps in Infrastructure & Processing Facilities for Development of Banana Value Chain Under Operation Greens Scheme

Diagnostic Report

October 2021

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List of Acronyms

APEDA	Agricultural and Processed Food Products Export Development Authority
APFPS	Andhra Pradesh Food Processing Society
APMIP	Andhra Pradesh Micro Irrigation Project
ATMA	Agricultural Technology Management Agency
CIPHET	Central Institute of Post-Harvest Engineering & Technology
DBT	Department of Biotechnology
FDL	Food and Drugs Laboratory
FPC	Farmer Producer Company
FPO	Farmer Producer Organisation
FTA	Free Trade Agreement
GAIC	Gujarat Agro Industries Corporation
GAP	Good Agricultural Practices
GGRC	Gujarat Green Revolution Company
GI	Geographical Indication
GNFC	Gujarat Narmada Valley Fertilisers & Chemicals
GSFC	Gujarat State Fertilizers and Chemicals
Ha	Hectare
HDP	High Density Planting
ICD	Inland Container Depot
IIHR	Indian Institute of Horticultural Research
INM	Integrated Nutrient Management
IPM	Integrated Pest Management
JNPT	Jawaharlal Nehru Port Trust
KVK	Krishi Vigyan Kendra
MIDH	Mission for Integrated Development of Horticulture
NRCB	National Research Centre - Banana
PMKSY	Pradhan Mantri Krishi Sinchai Yojana
TANHODA	Tamil Nadu Horticulture Development Agency
TNAU	Tamil Nadu Agriculture University

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Executive Summary

Introduction

India is one of the largest producers of fruits and vegetables in the world. It leads globally in production of banana, mango, papaya, okra and is amongst the top producers of potato, peas, lemon, tomato, cabbage amongst other varieties as well. With a 5-year compound annual growth rate (CAGR) of around 3% and the overall horticultural production exceeding 320 Million MT as of 2019-20, the trajectory seems going only upward. However, in spite of a huge production base, traditionally, the sector has been hindered by challenges faced at harvest and post-harvest level, coupled by inefficient processing and logistical practices. In case of fruits and vegetables, ICAR-CIPHET has estimated the post-harvest losses ranges between 4.58 to 15.88%. It is vital to minimise this loss, in order to provide the better supply chain, value addition, exports and reduce losses, through efficient means, one of the flagship objective of the Pradhan Mantri Kisan Sampada Yojana (PMKSY) under the aegis of Ministry of Food Processing Industries, Government of India. Series of policy interventions culminated with the Operation Greens Scheme by MoFPI extended its ambit to numerous other perishables post Union Budget 2021, one of which is Banana fruit with an over-arching objective of developing strong and integrated value chain.

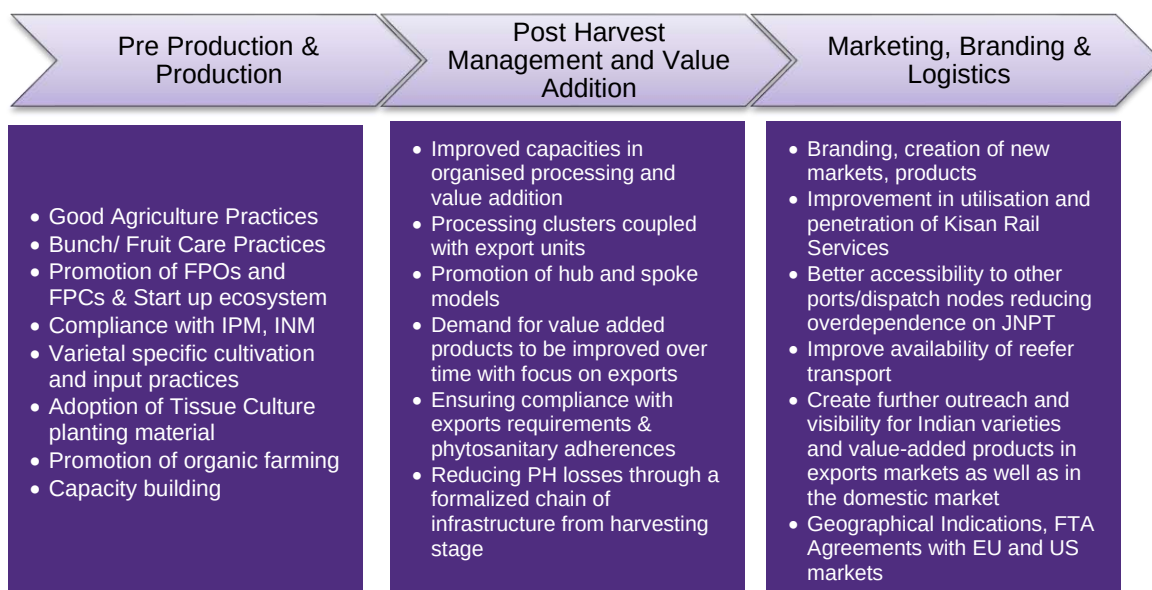
Banana – Global and Indian Perspective

Banana is being cultivated in an area of 878 thousand hectares with total production of 31.5 Million MT across the country (NHB 2018-19). In the last 5 years, the total area and production of banana in the country is steadily growing at a CAGR of around 1.05%. Madhya Pradesh has the highest productivity with 69.5 MT/Ha followed by Gujarat and Andhra Pradesh with 65.5 MT/Ha and 56.2 MT/Ha respectively. In terms of exports, India lags far behind globally with <1% of the total volumes being exported. Over 95% of the consumption of banana in India takes place in fresh form with negligible amount of product is consumed in the processed form.

Though, banana is cultivated in more than 23 states, the total production of Banana is concentrated mainly in Kerala, Andhra Pradesh, Gujarat, Maharashtra, Tamil Nadu and Uttar Pradesh. As a result, a focussed cluster level assessment was mandated in these 5 states, spread across 8 clusters and encompassing 23 districts.

Need for Convergence

Though the focus of the assessment is more on post-harvest infrastructure development and subsequent value chain upgradation, it is imperative to note that the gaps which has a subsequent impact on the post-harvest level starts from the farm and scientific production. It is therefore prudent to devise a holistic vision and strategy for overall sector development in banana from farm to fork, dissecting various sub-stages of which will run concurrently and evolve into a successful model over short, medium and long terms.



The assessment was conducted through a pre-defined sampling strategy, deploying a judicious mix of secondary and primary research, collection, cross validation of information across all clusters in the study region. Through these discussions, the following key recommendations have been derived, which have been elaborated in detail for each cluster subsequently in the report.

1. Pre-Production and Production Stage

- a. **Production of quality planting material:** Banana is a commodity requires intensive cultivation. ; Traditionally banana has been propagated by suckers but realizing the advantages of in vitro propagation, mass multiplication through shoot tip culture has gained popularity for the production of disease-free plants. High multiplication rate per unit time, better establishment with hardening, uniformity at flowering and harvesting, higher returns with increased yield are some of the benefits of micropropagation. Underutilization of tissue culture laboratory and practice of utilizing ratoons from the subsequent plantation cycle reduces productivity and quality of produce. Presence of tissue culture laboratories at cluster level catering to the overall demand of planting material (earlier compensated through sources within and outside state) will bring down transportation costs as well.
- b. **Higher impetus on GAP, Fruit Care and Climate resilient practices:** Training and capacity building being undertaken by farmers was found to be minimal across the clusters. Incessant application of fertilizers and inputs is another challenge which has been highlighted by district level government representatives (DoH, KVK, ATMA etc). Presently, less than 5% of the farmers are aware or practice GAP and fruit care practices. If we take the example of the global leader in banana exports, viz. Ecuador, this figure stands at over 60%. Hence, a holistic strategy to bring area under banana cultivation under GAP and certification compliances like global gap, fair trade, organic gap, HACCP and related certification systems should be developed in coordination with state departments, KVKs, ATMA and SAUs. Influencers and innovative farmers may be approached, success stories to be highlighted and better income generation avenues to be promoted and showcased so as to motivate cultivators to undertake such practices. Crop selection and changes in crop calendars will help farmers adapt to new temperatures and rainfall patterns. Farmers will also need to adopt more systematically measures to respond to increased frequency of floods and more intensive rainfalls. A combination of erosion control actions and better drainage capacities will be needed.
- c. **Promotion of FPOs/ FPCs:** During the primary survey with FPOs in the study region, it has emerged that the formation of FPOs/Cooperative led to increase in farm gate prices to the tune of 15-20% due to better bargaining power, assured offtake and periodicity. FPO membership ensures better input availability, access to market intelligence as well as avenues for capacity building when compared to conventional farmers. It is recommended to have a larger participation of farmers under FPO ecosystem in the case of banana for sustainability and for the efficient use of inputs.
- d. **Promotion of organic farming & conservation agriculture:** Clusters could be identified within the special export zones, to give more impetus on promoting organic banana cultivation. Special varieties other than Grand Naine may also be included in the scheme of things when to promote organic farming in banana. The exclusivity of traditional bananas may be promoted in the European and US market as premium varieties and special vehicles may be mooted for better delivery systems.

2. Post-Harvest handling and Value Addition

- a. **Primary Processing:** Currently, a majority of processing in banana happens at field level itself adopting traditional methods. This reduces quality of banana and may also aid in spread of diseases and insect pests. Hence, improving availability and access to primary processing facilities such as packhouses and collection centres closer to production clusters is vital. Presence of packhouses at taluk/block level will also help in establishing value addition facilities for other horticultural commodities other than banana. Establishment of ripening chambers, packhouses (incl. mobile), collection centres etc. along with suitable sensitisation of farmers and market intermediaries will help in effective utilisation of such facilities and aid in migration from traditional practices to efficient and hygienic handling of banana post farm gate.
- b. **Secondary Value Addition:** Traditionally, banana in India has been majorly consumed as dessert fruit and <5% is consumed in processed form. Secondary and tertiary processing is almost negligible

and thus led to poor adoption and profit realization. Banana puree, green banana flour, ripe banana powder, central core stem powders are gaining popularity in international market due to its versatility, higher resistant starch, low glycaemic properties. There is immense potential domestically as well as in exports markets, hence, integrated pulping lines and manufacturing lines at cluster level should be established. The private sector interest in this regard is evident and the same should be tapped through concentrated efforts and incentives. Subsequently efforts in branding and marketing space, the export can also be enhanced considerably catering to existing and new untapped markets.

- c. **Leverage Waste to Wealth Opportunities:** Banana is one of the topmost 'high volume' fruit crops in India generating mammoth volumes of resultant waste at farm level. The potential to leverage this waste is immense, in terms of farm level use as well as at customer level through value addition. Banana fibre extraction and organic fertilizer generation through banana pseudostem is an untapped opportunity currently waiting to be exploited. District and cluster level interventions elaborated in this regard can help in generating a cumulative increase in farmer income (INR 1100 crore+), reduction in use of NPK fertilizers (2.3 Mn MT) as well as additional employment generation potential (20000+) in the study region.

3. Marketing, Branding and Logistics

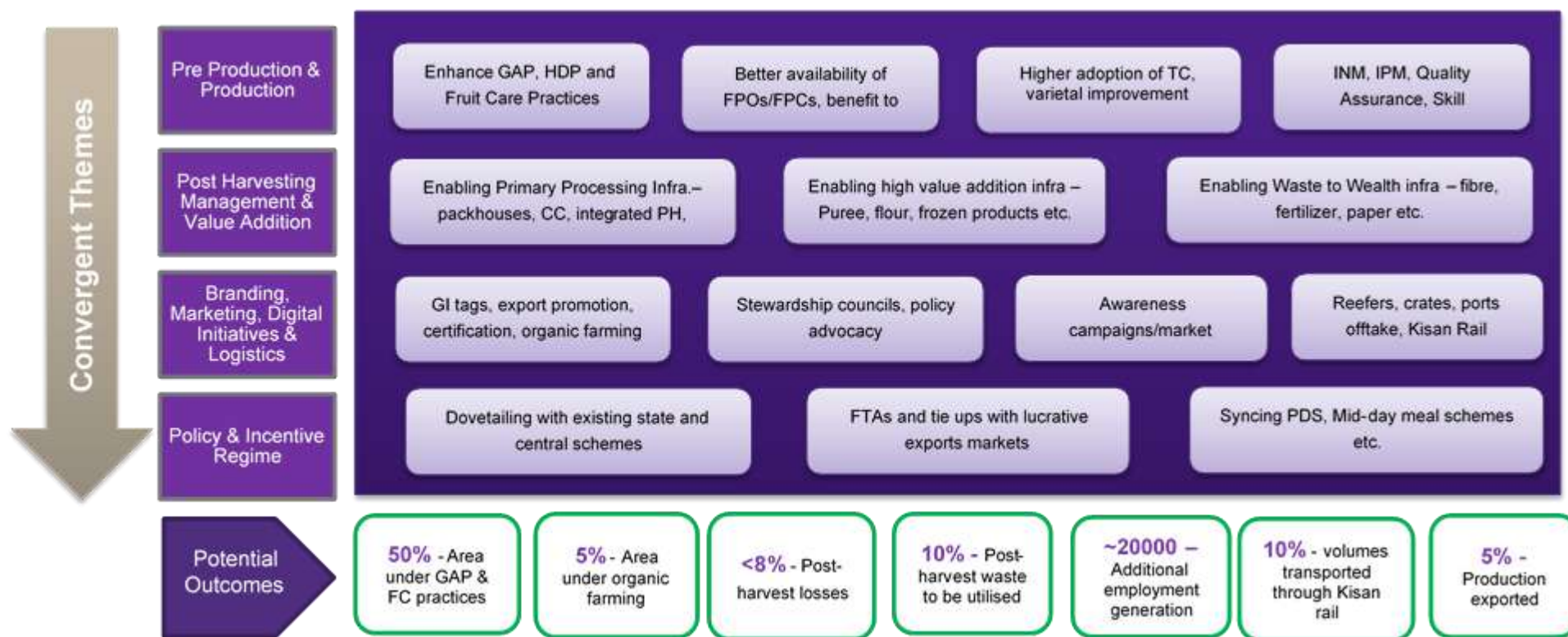
- a. **Improved Availability of Kisan Rail:** Through Kisan Rail, the transportation cost is reduced by INR 1.5-2/kg. However, it's current penetration is limited and pocketed to limited clusters. Also, the awareness with regards to its availability, schedule, costs involved and process to be followed needs to be highlighted among the stakeholders. There is high potential for utilisation of Kisan Rail as the cost of transit and resultant losses will get substantially reduced when compared to road transport through a mix of adoption of crates, reefer-based transit etc. Overall, better promotion of Kisan Rail availability, new stops at key production junctions and clusters, cost structures and process to be followed needs to be undertaken suitably.
- b. **Changes in banana production and market driven high through-put cultivation:** Unlike other countries where the banana is grown in large area and monitored by big firms, Indian banana industry is highly fragmented. Cluster formation or co-operative farming for procurement of all the inputs on collective basis, better implementation of the technologies, credit approval and better price realization need to be strengthened and encouraged. Traceability of the farm should be given importance with the stricter adherence to good agricultural and handling practices. Farmers need to be educated about export requirements and international quality standards. It is essential that the farmers work together to reduce the input cost and enhance uniform maintenance of plantations and thus able to get uniform quality, exportable bunches.
- c. **Branding and marketing Initiatives:** Though banana is one of the mostly widely consumed fruits in India, there is still immense potential to leverage an already strong foothold through improved value addition, product diversification and accessibility to newer markets. Creation of specific demand and branding to Indian banana and promotion of traditional banana are essential to have a foot print in the international market. Value added products such as banana flour can be substituted to at least 30% with conventional wheat flour to cater to the diabetic population due to its low glycaemic index. Integration through Mid-day Meal schemes and PDS system may be promoted to boost the utilization of banana-based products. Registration and acquiring of GI tag and focussed marketing in exports markets, especially EU and USA should also be focused upon other than traditional markets like west Asia. Specific impetus on branding initiatives as a separate component should be laid upon in subsequent policy and subsidy regimes.
- d. **Enabling IoT/ AI based ecosystems along with traceability:** Developing methodologies based on input-output models which provide information about the cluster wise production of banana and could identify the major catchments for supplying the fresh produces. This will allow in calculating the impact of any change in the supply chain. Enabling start-ups, skill development, identifying innovations through value chain approach and developing IoT/ AI based ecosystems, based on the demand of present and future markets. Leveraging Blockchain enabled traceability solution which will help in ascertaining source of production, variety of produce, farming practices followed, organic certification as well as help in mapping entire chain of transfer of produce. This will help in ascertaining price trends, peak and lean seasons of a crop, anticipating glut/scarcity of crop in an area and hence in

inventory management on basis of dashboards and reports developed. Farm level blockchain technology is the need of the hour to compete globally for exports and becoming a leader in exports.

4. Others

- a. **Exports Facilitation and Sea Protocols:** India lags behind in fresh and value-added banana exports due to underdeveloped trade routes, practices and penetration in lucrative markets. Through all of the farm level initiatives summarised above, as well as focussed marketing efforts, over time, availability of exports compliant produce can be achieved. Development of sea protocols and favourable access to distribution networks and sea routes should be undertaken. Orientation of Free Trade Agreements (FTAs) and trade policy in favour of banana exports should be evolved (for eg. Ecuador, Columbia and Mozambique have trade agreements with EU). Banana could also be suggested as a tradable commodity in bilateral agreements to reduce the customs duty upon arrival. India should get the Most favoured Nations (MFN) when trading banana with other countries with zero or reduced tariff to be competitive in the market. The entrepreneurs need to be encouraged with incentives such as tax holidays, reduction in GST, providing special green channel at ports, reduction in handling time, etc. Being a perishable commodity, fast clearances at port level through a single point of inspection should be prioritised.
- b. **Promotion and Branding of traditional bananas and creation of export window:** Specification and grade standards should be made available for traditional banana varieties. Protocol for postharvest handling of Nendran, Ney Poovan and Red banana need to be perfected for shipping by sea. Export window may be created in the nodal place like ICAR- NRC Banana, to cater to need of exporters. Indian bananas should be marketed with specific brand name to get renowned global visibility. Phytosanitary manual, SOPs, GAPs should be prepared, involving all the stakeholders and exports standards for traditional varieties may also be developed.
- c. **Twining Value Chain Actors for Synergy:** Synergizing with the large number of value chain stakeholders such as the MoFPI, MoA, APEDA, NHB, FIEO, EIC, FSSAI as well as food processing industry associations across relevant clusters is key. This will enable in leveraging mutual strengths and mitigating challenges which may be faced across the value chain. These include exports related certification, best practices and compliances, quality testing and assurance as well as visibility in target markets. Policy advocacy through chamber engagement or common commodity specific committee/stewardship council may also be explored. Support from NRCB and IFFPT may be sought across all value chain interventions to access various prototypes, models, access to market intelligence and latest sectoral trends.
- d. **IT infrastructure/ Dashboard:** It is suggested that a real time dashboard highlighting details of post-harvest management infrastructure pertaining to banana is maintained. The portal can capture the beneficiaries of schemes where number of marginal, small and large farmer beneficiaries is maintained separately. The portal will help a farmer in gaining market intelligence. An IT portal will help in identifying the lacunae and address the same simultaneously for SML farmers. The portal can also maintain information on market intelligence, Kisan Rail availability and timetable as well as track movement of produce through such channels. It will help in identifying demand and supply from particular clusters/states. Also, such a portal will provide access to plant and machinery suppliers, maintenance experts as well as available technologies for primary and secondary processing. Stakeholders widely suggested that an e-NAM like portal may be developed for better marketing of banana to benefit in terms of price discovery; payment guarantee and access to nationwide market and traders' benefits from reduced transaction costs and delivery guarantee. This becomes even more imperative since negligible banana trade takes place through APMC ecosystem.

Vision 2030



Indicative Investments proposed in study region across various components (excluding waste to wealth processing)

Proposed Intervention Component	Clusters/States under Assessment					Indicative Costs for proposed Interventions				
	Gujarat	Uttar Pradesh	Maharashtra	Andhra Pradesh	Tamil Nadu	Numbers Proposed	Units	Total Investment (Cr. INR)	Mode of implementation	Available Scheme/Incentives Convergence
Pre-Production and Production										
Tissue Culture Lab with Virus Indexing Facility & Hardening Center	LP	HP	LP	MP	MP	41	Nos.	102.5	Private Sector	MIDH, NHM
Promotion of Good Agricultural Practices	MP	MP	MP	MP	MP	133,813	Ha.	134	State Horticulture Departments, KVK, ATMA etc.	MIDH, NHM
Promotion of Farmer Producer Organisations	MP	MP	MP	MP	MP	138	Nos.	59	NABARD, SFAC and state govt.	SFAC, NABARD, State Governments
MRL, Pesticide Residue testing lab	MP	HP	MP	MP	MP	8	Nos.	32	State Government & Private Sector	MIDH, MoFPI, FSSAI
Basic testing facility at district level	MP	HP	MP	MP	MP	23	Nos.	6	State Government & Pvt. sector	MIDH, MoFPI, FSSAI
Sub Total								333		
Post-harvest and value addition										
Adoption of Crates	MP	MP	MP	MP	MP	2195000	Nos.	55	Private Sector	MIDH, state schemes
Ripening Chamber Facility	LP	MP	MP	LP	MP	137	Nos.	58	Pvt. Sector and FPO	MIDH, state schemes
APEDA Certified Packhouse	MP	HP	LP	HP	HP	7	Nos.	28	Pvt. Sector, Government, FPOs	MIDH, state schemes
Chips/Flour making units	HP	LP	HP	HP	HP	20	Nos.	15	Private Sector, FPOs	MIDH, state schemes
Banana Puree / value addition manufacturing unit	HP	MP	HP	HP	LP	8	Nos.	120	Private Sector	MIDH, state schemes
Collection centres	HP	HP	HP	HP	HP	252	Nos.	10	Pvt. Sector, Government, FPOs	MIDH, state schemes
Sub Total								286		
Logistics marketing and branding										
Refrigerated Transport Vehicles	HP	HP	HP	HP	HP	171	Nos.	51.3	State Governments, Logistics Service Providers	MIDH
Branding, digital and Marketing Initiatives	MP	MP	MP	MP	MP	8	-	49	State Governments	No scheme
Sub Total								101		
TOTAL								720		

Source: GT Analysis

HP – High Priority

MP – Medium Priority

LP – Low Priority

Component wise Indicative Costs

S. No.	Stage of Intervention	Total Indicative Cost (INR Cr.)
1	Pre-Production and Production	333
2	Post-harvest Management and Value Addition	286
3	Logistics, Marketing and Branding	101
	Total Project Cost	720

Note: Proposed 'waste to wealth' value addition investment and costs is provided in the respective segment separately.

Crop Profile: Banana

1.1. Overview

Banana is one of the oldest cultivated crops and considered as the native of southeast Asia, with India as one of its centres of origin. There are over 1000 different varieties of banana growing around the world. In India, banana is the second most important fruit crop after mango and also the most consumed amongst the fruits cultivated in India. It accounts for about 32%¹ of the total fruit production in India, majorly because of its year-round availability, affordability, varietal range, taste, nutritive and medicinal values among all classes of people. Banana is popular in a global scale not only for its nutritional value but also because of its economic importance especially to smallholding farmers in the developing countries. Bananas and plantains are the fourth most important staple crop in the world and are critical for food security in many tropical countries. Though banana is an important tropical fruit because of its valuable application in food industry, its by-products possess an enormous untapped potential. The by-products are an excellent source of highly valuable raw materials for other industries which can prevent an ultimate loss of huge amount of untapped biomass and environmental issues². In terms of world export trade, banana ranks as the 5th most important crop after coffee, cereal, sugar and cocoa.

Industrial/Commercial	Medicinal	Culinary
• Single banana plant waste constitutes 80 % of the total plant mass. This biomass can be used to obtain natural fibre as raw material for textile industry, pulp and paper production • It can also be used for biofuel production such as ethanol, methane, briquettes etc • Banana waste can be used to generate organic fertilizer, banana fibres and paper manufacturing	• Banana/plantain by-products is a huge source of important bioactive compounds which can be used as potential nutraceuticals • Products such as pseudostem juice and other nutraceutical products can also be manufactured	• Banana has various uses than can be turned into many processed products such as: • Banana puree • Banana powder • Banana Jam • Banana juice • Banana sauce • Banana vinegar etc.

1.2. Production and Productivity Analysis

Global Scenario

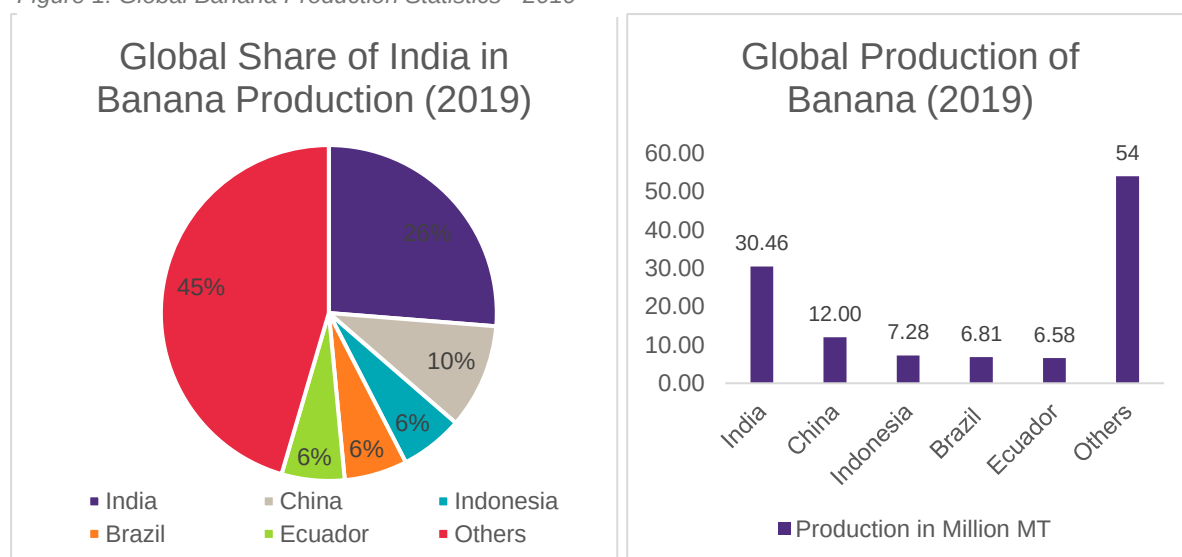
Banana is the 4th most important staple crop in the world after wheat, maize and rice. It is produced in more than 135 countries in tropical and sub-tropical regions globally. World-wide the total production of banana exceeded 116 Million MT for the year 2019 across an area of 5.15 Million ha (FAO). India is the largest producer with 27% share in global production with total production of 31.50 million MT as seen in the graph below³. Other than India, the leading producers are China, Brazil, Philippines, Ecuador, Indonesia, Mexico, Guatemala and Colombia in terms of area and production. India, Indonesia, China and Philippines contribute more than 60% of banana production.

¹ Source: National Horticulture Board, All India Area and Production of Horticulture Crops – 2019-2020

² Padam BS, Tin HS, Chye FY, Abdullah MI. Banana by-products: an under-utilized renewable food biomass with great potential. J Food Sci Technol. 2014;51(12):3527–3545. doi:10.1007/s13197-012-0861-2

³ <http://www.fao.org/faostat/en/#data/QC>

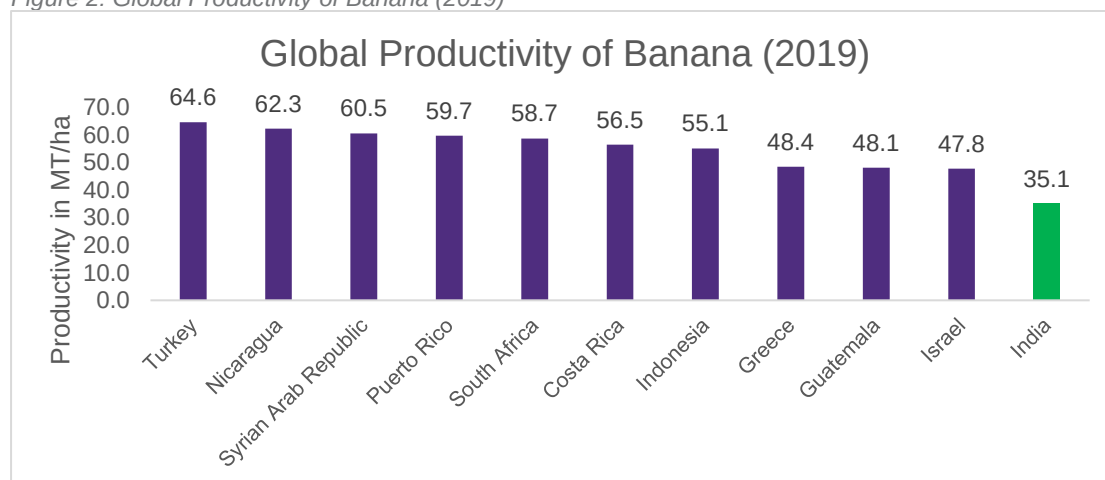
Figure 1: Global Banana Production Statistics - 2019



Source: FAO Stat and GT Analysis

Though India is the largest producer of Banana in the world, it ranks 23rd in terms of productivity (35.17 MT/ha) which is still competitive when compared to other major producing countries such as China, Philippines etc. Globally, Turkey has the highest productivity of 64.59 MT/ha followed by Nicaragua, Syria and Puerto Rico with average productivity of 64.59 MT/ha, 62.22 Mt/ha and 60.50 MT/ha respectively.

Figure 2: Global Productivity of Banana (2019)



Source: FAO Stat and GT Analysis

Indian Scenario

India has made enormous achievements in the enhancement of banana production with both horizontal and vertical expansion. In 1990s, the crop was cultivated in less than 5 lakh ha concentrated mostly on southern and eastern part of India with the lesser productivity. In the last decade, the crop was introduced to new areas like Madhya Pradesh, Uttar Pradesh and thus jump in area along with higher productivity was noticed. Currently, among the fruit crops, banana accounts for 13% and 33% of the total area and production respectively. In Tamil Nadu and Kerala, more than 15 different varieties of banana are grown commercially whereas the states like Gujarat and Maharashtra follows almost monoclonal cultivation of Grand Naine. The other major banana producing states are Karnataka, Kerala, West Bengal and Assam (NHB, 2018). Within India, Madhya Pradesh has the highest productivity of 69 metric tonnes /ha as against the national average of 30.5 tonnes/ha. The comparatively lower productivity (38.8 MT/ha) in Tamil Nadu is due to its adaptation of polyclonal cultivation of banana. Although location specific varieties like Rasthali (Silk), Poovan (Mysore), Ney Poovan, Thella Chakkarakeli, Karpuravalli (Pisang Awak), Nendran (French Plantain), Hill Banana (Pome-AAB) and Monthan (Cooking banana) are grown in large quantity in different regions of the country. The Cavendish clones occupies 52% of the total area under banana cultivation and contributes 64% of the total banana production.

Table 1: Geographical distribution of major Banana growing states in India

State	Area ('000 Ha)	Quantity (Million MT)	Productivity (MT/ha)	Share %
Andhra Pradesh	88.96	5.00	56.2	16%
Gujarat	68.15	4.47	65.6	15%
Maharashtra	80.88	4.21	52.0	14%
Tamil Nadu	82.63	3.20	38.8	10%
Uttar Pradesh	69.38	3.17	45.7	10%
Karnataka	110.55	2.33	21.1	8%
Madhya Pradesh	26.38	1.83	69.5	6%
Bihar	31.07	1.40	44.9	5%
West Bengal	49.3	1.20	24.3	4%
Kerala	109.26	1119.16	10.2	4%

Source: National Horticulture Board (2018)

Seasonality and Varietal Distribution

Banana is a tropical crop and grows well in a temperature range of 15°C – 35°C with relative humidity of 75-85%. It prefers tropical humid lowlands and is grown from the sea level to an elevation of 2000m above sea level. In India, this crop is being cultivated in climates ranging from humid tropical to dry mild subtropics through selection of appropriate varieties.

Table 2: Seasonality of Banana crop in India

Lean period	Peak period					Throughout year						
States	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Andhra Pradesh												
Gujarat												
Maharashtra												
Tamil Nadu												
Uttar Pradesh												
Karnataka												

Characteristics of major varieties in India:

- **Grand Naine:** This is the most accepted and traded international variety . Produces a bunch weighing 25 kg and may go up to 32-35 kg with 8-10 hands and 200-220 fruits/bunch.
- **Robusta:** It has black, brown blotches on the stem. Bunches weigh around 20 kg having 8-10 hands/bunch. It's length is 15-20 cm and girth is 12 cm with thick fruit skin.

- **Dwarf Cavendish:** The plant is a dwarf with large bunches weighing about 20 kg. It's compactly arranged 8-10 hands and length is 13-14 cm and girth is 8-10 cm.
- **Red Banana:** The plant is tall and its colour is purplish red. The bunch weight is 20-25 kg with 6-7 hands and 80 fruits/bunch. The length of this banana variety is 16-18 cm.
- **Nendran:** The plant of this banana variety has bunch of 5-6 hands weighing about 6-12 kg. Rich in carotenoids.
- **Rasthali:** This variety is mostly found in South India. The size of banana of this variety is medium. Its flesh is white and firm with a characteristic flavour. Its tree is short in length and each bunch weighs not more than 20 kg. Cropping cycle is usually longer than 12 months.
- **Poovan:** Banana of this variety is slightly acidic, firm and has sour-sweet aroma. Bunch of this variety are medium in size, its fruits are closely packed and keeping quality is good.
- **Panchanadan:** Variety is popular for its cooling effects in hot tracts in summer. Bunch weighs around 12-15 kg.

1.3. Market Analysis

Global consumption for bananas is estimated to grow at a CAGR of 1.21% for the next 5 years with the Asia-Pacific region expected to lead the market with a 61% share of global consumption. The 2nd largest producer, China has a per capita consumption of 10 kg/year while the largest exporter of banana, Ecuador has a per capita consumption 18 kg/year. On the other hand, in India, the table topper, consumption is considerably high with over 95%, followed by processed products at 1.8% and <0.10% exports. The per capita consumption of fresh banana in India is high at 24 kg/year. India has a high consumption of Banana owing to its availability in abundance across the nation.

In India, banana as an enterprise is generating around \$7 billion and providing livelihood to millions of farmers. India contributes 21% world banana production with the production of 31.74 million tons in an area of 8.98 lakh hectares. Banana alone contributes 2.3% to the Agricultural GDP.

Bananas are also used in a wide variety of food products such as breakfast cereals, ice cream and other desserts apart from raw consumption. The demand for bananas owing to its various health benefits is expected to rise in the coming years. Post-harvest loss in bananas is a more serious problem in developing countries than those in developed countries. Improper handling and storage causing physical damages and mechanical losses including bruising, cracking cuts, microbial spoilage by fungi and bacteria and physiological losses play a critical role in reducing the market value. Hence, processing and value addition is the best way of utilizing surplus production.

The processed Banana market in India is at a very nascent stage. There is a good market demand for different banana products which is currently being catered by SME segment primarily to the domestic market in India. The processed banana market consists primarily of banana chips, banana powder, banana puree and to some extent banana flour. Since >95% of banana produced in India is consumed domestically as fresh fruit, it leaves a lesser scope for processing, which is primarily met through low (C Grade) banana in most geographies. These variants are priced low compared to other variants (A&B) and hence, aids SME units and mid-sized processors in meeting bottom lines.

1.4. Global Trade Scenario

According to FAO estimates, the major banana exporting countries are Ecuador, Colombia, Costa Rica and Philippines and major importing countries are the US, Belgium, Germany, UK and Russia. World exports of bananas show a high level of concentration, with developing countries in Latin America and Caribbean region accounting for the bulk of exports. The four leading banana exporting countries in 2019 accounted for 62 per cent of world exports with Ecuador alone providing more than 30 per cent. The international trade operates with the export price nearing to US \$1000 per ton of produce. It is equivalent to \$1/kg of produce, making it INR 12- 14/ fruit.

With the presence of bigger domestic population, the banana produced are predominantly consumed within the country itself by the major banana producers like India, China and Brazil. Besides, the banana exports are governed by multinational fruit companies with the organized business environment whereas

in India, the business is totally unorganized and is handled by multiple authorities in bits and pieces. Five multinational fruit companies alone controlled 44% of the international banana trade.

When it comes to imports, the major importing countries are countries of EU collectively (Germany, the United Kingdom and Italy) followed by United States (US) with an estimated global volume share of 22 percent in 2019 which is further followed by China which expanded its volume share to an estimated 12 percent of global net imports, overtaking the Russian Federation as the third largest importer of bananas globally. Similarly, to Japan, China procures the majority of its bananas from the Philippines, and benefited from higher Pilipino supplies in 2019 following the recovery of production.

- **Global Exports of fresh banana fruit**

Total global exports of fresh Banana amounted to USD 12.31 billion in 2018 of which India contributed to less than 1% in the international Banana trade. The global export market grew by 11% between 2016 and 2020. The leading exporting countries include Ecuador, Philippines, Costa Rica, Guatemala and Colombia (2020) whereas the leading import destinations include USA, Russia, Japan, Germany and China (2020).

Table 3: World export of fresh banana vis a vis India

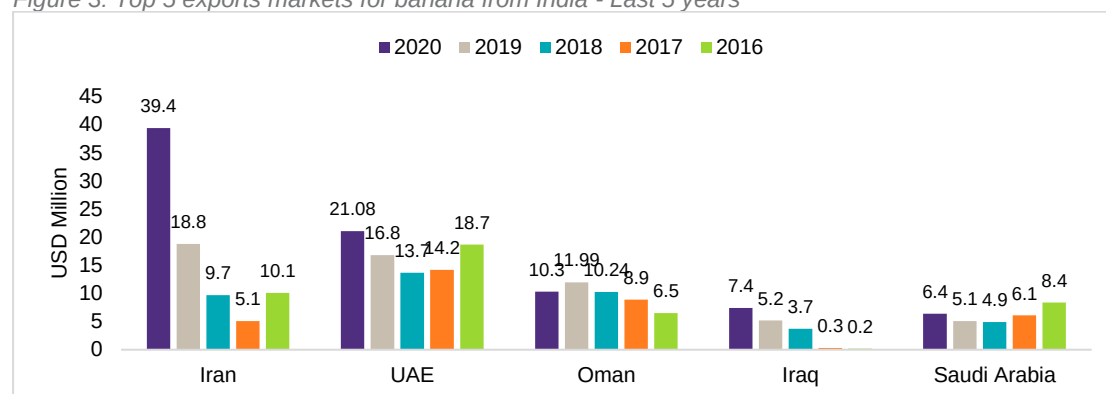
Leading Exporting Countries	Global Ranking (Value)	Exports Volume and Value (2020)	Exports Volume and Value (2019)	Exports Volume and Value (2018)
Ecuador	1	\$3577 Mn 7.03 Mn MT	\$3185 Mn 6.67 Mn MT	\$3115 Mn 6.55 Mn MT
Philippines	2	\$1607 Mn 3.72 Mn MT	\$1930 Mn 4.35 Mn MT	\$1504 Mn 3.38 Mn MT
Costa Rica	3	\$1080 Mn 2.62 Mn MT	\$998 Mn 2.38 Mn MT	\$1028 Mn 2.48 Mn MT
Colombia	4	\$913 Mn 2.03 Mn MT	\$870 Mn 1.89 Mn MT	\$809 Mn 1.74 Mn MT
Guatemala	5	\$845 Mn 2.5 Mn MT	\$841 Mn 2.43 Mn MT	\$811 Mn 2.36 Mn MT
India	21	\$100 Mn 0.20 Mn MT	\$75 Mn 0.17	\$55 Mn 0.11 Mn MT

Source – ITC Trademap, Banana = HS code – 080390



Though India's exports between 2016 and 2020 grew at 14% in value terms and it ranking first globally in production of banana amounting to 27% of the global production, India's share in global trade is miniscule – 0.14% of volume and 0.05 % of value. The majority of banana exports (90%) from India are to West Asian and North African countries like UAE, Bahrain, Egypt, Saudi Arabia, Qatar and Iran.

Figure 3: Top 5 exports markets for banana from India - Last 5 years



Source: ITC Trademap HS Code – 080390 (fresh or dried bananas, excl plantains)

The top 5 exporting states from India are Maharashtra (73%), Tamil Nadu (3%), Kerala (10%), Uttar Pradesh (11%) and West Bengal (2%).

Even though, nearly 20 per cent of total world output is produced in India, the export is negligible. Nonetheless, India, in recent times, increased its export volume by 47 percent due to further expansion in the harvested area for traded varieties. Supply shortages in the Philippines, the main competing exporter, meant that shipments from India could benefit from the high demand in Gulf countries, European market and Southeast Asia. Another supporting factor was the low price of Indian bananas, which is being sold at a 50 % discount compared to bananas from Ecuador and the Philippines at the Dubai auction centre. India exports maximum bananas to middle east countries with the lion share to United Arab Emirates (41%), followed by Saudi Arabia, Iran, Kuwait and Oman.

The increased uptake of voluntary sustainability standards (VSSs) in the banana sector is another important development to watch, especially given the pace of this uptake in recent years. In 2016, 7% of banana production was VSS-compliant—a notable development given that these standards were almost nonexistent in the sector about a decade ago. That same year, 1% of production was potentially VSS compliant, while conventional banana production accounted for 92% of the market. VSS-compliant banana production, which at first primarily came from smallholders, expanded significantly at the turn of the century, as large plantations owned by Dole, Fyffe, and Chiquita became certified. Worldwide banana that comply with voluntary sustainability standards (VSSs) accounted only seven percent of total production. Among the standards Rainforest alliance handles the maximum area followed by organic and fairtrade international. VSSs continue to play an important role in helping to address the sector's sustainability challenges, since the criteria embedded in each standard, which certified plantations and producers need to comply with, encourage the improvement of agricultural practices. These criteria include soil conservation measures that can enhance crop yields and the resilience of banana plantations to climate change, the definition of minimum and living wages for workers, and the application of integrated pest management measures.

Some of the major obstacles faced by the banana exporters are;

- Insufficient coordination among border agencies, multiple border controls
- Improper pre and post-harvest practices like, non-compliance with GAP

- Controlled atmosphere, controlled humidity, remote access monitoring, enhanced air flow, better insulation materials and techniques and internal monitoring controls are essential for successfully enhancing the shelf life of banana to more than 30 days.
- Long waiting times loading/unloading port reducing the quality, thus final price of perishables
- Non-availability of online export documentation processing
- Lack of cross-border data harmonization and use of international standards/norms country wise
- Non availability of traceable and certified material for exports
- Basic infrastructure and cold chain management
- Lack of aggressive working by the banking and insurance companies in the fresh agro-export sector and farming sector, largely affecting funding given by government banking/pvt sector. This leads to pushing the farmers for borrowing finance at higher interest, lowering risk taking ability for export
- Export requires numerous documentation during registration process as well traceability charts are required by importers which are time consuming and tedious making Indian farmers hesitant as majority have low literacy. eg: Sedex/ISO/Globalgap. Understanding the minimum residue level standards required in international market by implementing international certifications is required.
- As there is no presence of single window, there are many documentation which are replicated losing time value which is most important in perishable exports. This requires advanced managerial competencies in very different domains—such as trade policies, business process analysis, change management, electronic business and information technology management and standards, legal issues and single window architectures.

Though exports from India are improving, there is far more to be done in terms of realising its true potential with the market potential available currently. The major challenges in this regard are related to:

- High cost of production (Eg. Farm gate price in India is INR 14/kg, Ecuador INR 6/kg)
- High sea freight and terminal handling charges from India
- Lack of focus on export-oriented varieties, standardisation and branding
- Lack of focus on a closed quality assurance system (Ecuador has AGROCALIDAD, which is a dedicated quality control system under central government; it also has regulations to mandate banana production in line with exports standards). Other Latin American markets also have similar structures in place.
- Lack of FTAs between India and EU/GCC markets. (The world's leading importer USA levies tariff of less than 5% in many countries. Notably, leading exporters like Ecuador enjoy Free Trade Agreements (FTAs) with the USA.)
- Market players assign long sea haulage (7100 Nautical miles and 30 days) as the reason for inability to service European markets, as quality on arrival is reported to be unfit for consumption. However, Ecuador and other South American countries which are equally far away (about 7200 NM) involving as long as 30 days of sailing, still are able to regularly supply large volumes to Europe. Therefore, Indian banana trade to identify supply chain practices and sea logistics solutions

Table 4: World import of fresh banana (2018-2020)

Leading Importing Countries	Global Ranking (Value)	Import (in MT & USD) 2020	Import (in MT & USD) 2019	Import (in MT & USD) 2018
USA	1	\$2549 Mn 4671 Mn MT	\$2529 Mn 4676 Mn MT	\$2592 Mn 4778 Mn MT
Russia	2	\$1116 Mn 1515 Mn MT	\$1119 Mn 1512 Mn MT	\$1154 Mn 1556 Mn MT
Japan	3	\$987 Mn 1067 Mn MT	\$958 Mn 1045 Mn MT	\$912 Mn 1002 Mn MT
Germany	4	\$956 Mn 1267 Mn MT	\$995 Mn 1331 Mn MT	\$944 Mn 1258 Mn MT
China	5	\$933 Mn 1746 Mn MT	\$1094 Mn 1939 Mn MT	\$896 Mn 1544 Mn MT
World	-	\$14460 Mn 22364 Mn MT	\$14401 Mn 21826 Mn MT	\$13926 Mn 20872 Mn MT

Source – ITC Trade MAP

1.5. Indian Trade Scenario

Total export of banana in 2019-20 stands at USD 101.4 Million and 0.21 MT. Since 2016, India's exports have grown at CAGR of 10.7% while during the same time the production has increased by only 5%. India has doubled the exports of banana since 2015-16 through concentrated measures at farm level and through development of market linkages. India has huge untapped potential for global trade in banana and can increase its exports many-fold.

Top global importers of Indian banana

Top buyers of Indian banana are Iran, UAE, and Oman. From 2016 to 20, top 10 importing countries consistently have been particularly Middle East nations, except for Nepal which keeps on featuring in Top 10 list and only once in top 5 list in 2019.

In terms of volume, Nepal is amongst the top 3 importers purely because cost of importing banana for Nepal is very less as India has a land border through LCS Maharajgunj in Uttar Pradesh. Iran has many sanctions imposed on it by USA due to which their normal trade is affected. India has been trading in bananas with Iran extensively during this period.

Table 5: Major buying of Indian banana (countries)

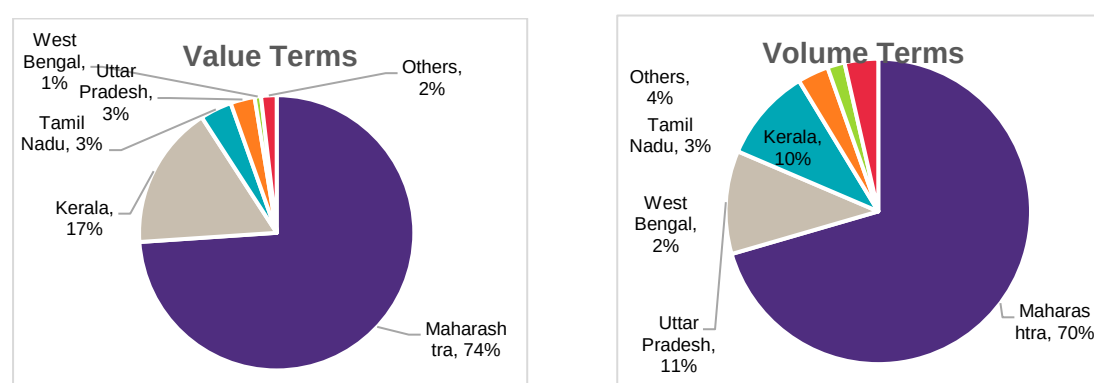
Country	2018		2019		2020		% share
	Volume	Value	Volume	Value	Volume	Value	
World	116741	55784	174668	76332	212831	101356	100%
Iran	18400	9791	34314	18882	73569	39403	21%
UAE	17416	13748	22574	16886	33041	21089	10%
Oman	20269	10287	24424	12007	22925	10434	7%
Iraq	8018	3766	12577	5264	18098	7423	6%
Saudi Arabia	7583	4914	8629	5173	12141	6449	4%
Nepal	30203	3469	54346	6104	33230	4093	3%
Qatar	3166	2564	3136	2701	5097	3535	3%
Afghanistan	3232	1477	4472	1856	5268	2743	2%
Kuwait	3020	2436	3404	2967	3084	1979	1%
Bahrain	1435	1060	2209	1685	2202	1439	1%

Source: ITC, Trade Map

Major banana exporting states of India

According to the latest data on APEDA, the top five banana exporters are Maharashtra followed by Kerala, Tamil Nadu, Uttar Pradesh, and West Bengal. In the given period the volume and value exported is 1.91 lakh MT and 82.8 Mn USD, respectively. Maharashtra exported the most and accounted for nearly 70% of banana trade. It exported 1.34 lakh MT banana valued at 61.2 Mn USD.

Figure 4: Top banana exporting states by value and volume (2020-21)



Source: APEDA Agri exchange

Uttar Pradesh stands second in volume exported and this is due to proximity to Nepal. Nepal receives its maximum imports from Uttar Pradesh via road which is cheaper than any other mode. In trade value, Kerala stands second but Kerala is not even top 5 producer of banana in the country. This is primarily due to higher trade volumes taking place through ports and air routes out of Kerala which is sourced mainly from Tamil nadu. Since Maharashtra is the leader in banana exports, Nhava Sheva/JNPT port of Maharashtra handled majority of banana exports followed by Cochin port and Cochin Airport. ICD Janori (Nashik) and Tuticorin port exported 7,286 MT and 6,095 MT, respectively. Cochin Airport is at 3rd in trade by value but is 5th in trade by volume solely because of expensive nature of air cargo.

Table 6: Major ports/airports by value and volume used in banana export (2020-21)

Port	Value (Mn USD)	Volume ('000 MT)	% share in value	Avg. Cost (USD/MT)
Nhava Sheva Sea	56.0	122.9	67.7%	455.9
Cochin Sea	5.7	9.7	6.9%	587.7
Cochin Airport	5.3	5.4	6.5%	992.1
ICD Janori	3.1	7.3	3.8%	426.6
Tuticorin Sea	2.9	6.1	3.6%	483.3
Trivandrum Airport	2.4	3.1	3.0%	790.1
Nautanwa (Sonauli) ICD	2.3	20.5	2.7%	110.4
ICD CFS Nashik	2.0	4.4	2.4%	442.3
Barhni Road/ Railway	0.6	4.4	0.7%	137.4
ACC Calicut Karipur Airport	0.5	0.7	0.6%	721.3
Others	6.6	1.8	2.2%	278.0
Total	82.8	191.2	100%	432.9

Source: APEDA, Agri exchange

Value of exports through air route is around 11% while volumes handled are around 5%. Value handled by road/railway is 1% while volume handled is around 4%. Road transport is cheaper, thereby leading to Nepal and Bhutan receiving banana from India through road. Air transport is expensive and Middle East nations import by means of both airports and seaports.

Table 7: Banana export by type of transportation (Between April '20 – Feb '21)

Port Type	Qty (MT)	Value (USD)	Price (USD/MT)	% Share
Sea Route	0.14	64.71	466.40	78%
Airport	0.01	9.08	906.40	11%
ICDs	0.03	7.84	227.70	9%
Road/Railway	0.01	1.12	141.00	1%
Total	0.19	82.76	432.90	100%

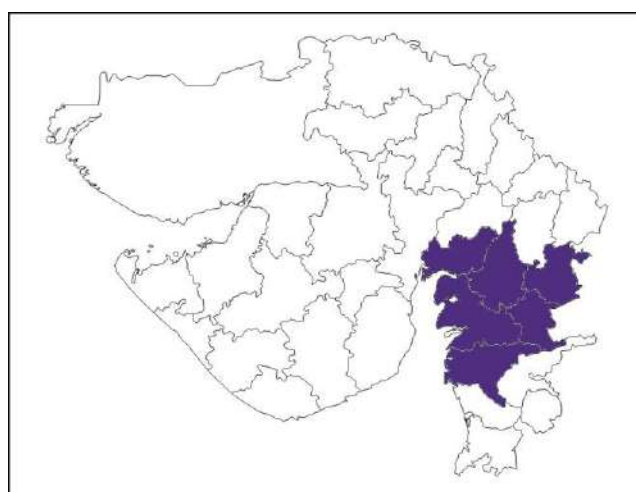
Source: APEDA Agri exchange

Key Cluster Wise Findings - Gujarat

Cluster-1 – Anand, Bharuch, Chhota Udepur, Narmada, Surat and Vadodara

Gujarat with its variety of soil, rainfall pattern, temperature regimes and irrigation and proximity to the port, made it as the hotspot for banana cultivation targeting export market. The state has a very high level of production and productivity in several agriculture and horticulture crops. The state leads not only in banana production, but also in Onion, Potato, Papaya, Groundnut, Cotton, Castor and Cumin. Over the years horticultural crops has provided good income to its growers. It has become a sustainable and viable venture for the small and marginal farmers.

Figure 5: Focal Cluster in Gujarat (6 districts)



6 Districts:

Anand
Bharuch
Chhota Udepur
Narmada
Surat
Vadodara

Source: GT Analysis

a. Total area and production under horticulture in the state (last 5 years)

Gujarat, the 5th largest state area wise, stands 5th in horticulture production as of 2018-19⁴. Last five-year horticulture production trend (2015-2020)⁵ shows that the production of horticultural crops has increased at a CAGR of 1.1% and in 2019-20 a total of 23.8 Mn MT of horticultural crops were produced. During the same period, the productivity has increased at a CAGR of 1.8% and as of 2020 it stands at 14.3 MT/Ha. Among horticultural crops, vegetables are grown most followed by fruits, spices and flowers. Among vegetables crops, potato is produced most followed by brinjal and cucurbits. Among the fruits, banana is produced in larger areas followed by mango and papaya.

Table 8: Total area and production under horticulture in Gujarat (2015-16 to 2019-20)

S.No.	Year	Area (in Lakh ha)	Production (in Mn MT)
1.	2019-20	18.30	23.78
2.	2018-19	15.84	22.78
3.	2017-18	16.87	23.43

⁴ NHB, Annual Horticulture Production, 2014-19 reports.

<http://nhb.gov.in/Statistics.aspx?enc=WkegdyuHokljEtehnJoq0KWL79sOQCy+W4MfOk01GFOWQSEvtp9tNHHoiv3p49g>

⁵ Horticultural cultivation of crops, area and production 2015-20, <https://doh.gujarat.gov.in/horticulture-census.htm>

4.	2016-17	16.11	23.23
5.	2015-16	15.32	22.20

Source: Director of Horticulture, Government of Gujarat

b. Production of banana in Gujarat

Gujarat stands 2nd in banana production in India with 4.6 Mn produced during 2019-20,. The state contributes 15% to the national banana production. Over the last 5 years, area under banana crop has increased from 64700 Ha to 69500 Ha, which means farmers are adopting the crop due to good earning obtained from it. In the same time production has increased from 4.2 Mn MT to 4.6 Mn MT. productivity has remained constant at 65 MT/Ha which is one of the highest in the whole world.

Table 9: Area, Production & Productivity of Banana in Gujarat (2015-16 to 2019-20)

S.No.	Year	Area (in '000 ha)	Production (in Mn MT)
1.	2019-20	69.54	4.63
2.	2018-19	70.18	4.61
3.	2017-18	68.15	4.47
4.	2016-17	66.31	4.29
5.	2015-16	64.69	4.18

Source: Director of Horticulture, Government of Gujarat

c. Production of banana in the cluster

The six districts of the study region produce 85% of the total banana of the state which is 3.9 Mn MT. In the last five years the banana production has increased at a CAGR of 2%. Bharuch is the leader in banana production followed by Anand and Narmada.

Table 10: Area, Production & Productivity of Banana in the cluster (Gujarat) (2015-16 to 2019-20)

S.No.	Year	Area (in '000 ha)	Production (in Mn MT)
1.	2019-20	56.22	3.91
2.	2018-19	55.42	3.82
3.	2017-18	54.66	3.78
4.	2016-17	53.19	3.61
5.	2015-16	51.96	3.53

Source: Director of Horticulture, Government of Gujarat

Extensive tissue culture, high density plantation, fertigation, bunch management along with water soluble fertilizers had contributed for extensive productivity in Gujarat

d. Marketable Surplus

Marketable surplus has been taken as banana being sent outside state. From the primary survey it was found that around 75-80% of the total produce of the state goes outside of the state. Majority of the trade happens in the months of April-August. The major destinations being Delhi, Rajasthan, Punjab and Haryana. Rajasthan being closest to Gujarat buys fulfils maximum requirement from Gujarat. Rest 20-25% of the produce is consumed within the state.

Table 11: Marketable Surplus of banana in the cluster (Gujarat)

S.No.	Varieties	Grade marketable (in Mn MT)	wise surplus	Total marketable surplus (in MT)	Shelf life
1	Grand Naine (95%)	2.85		3.00	7-10 days
2	Others – Mahalakshmi, Robusta (5%)	0.15			7-10 days

Source: Stakeholder consultations, GT Analysis

e. Percentage Losses of banana in the cluster

Losses in produce of banana occurs at different stages in the cluster. These include:

1. Pre-Production Stage: 10% of plants affected due to
 - abiotic stresses like moisture deficit stress
 - diseases - nematode, Sigatoka leaf spot, wilt, bunchy top
 - uprooted by animals (nilgai, wild boar)
 - extreme weather and/or flood
- Production and Value Addition Stage
 - Primary processing - Grading, Cleaning, Sorting and Packaging done at field itself.
 - all the operations are carried out manually yet losses are negligible at this stage
 - C-graded sorted and not bought
- Logistics, Marketing and Branding
 - Losses occurring are transit losses mainly. Can be reduced by adopting better packaging practices.
 - In ripening chamber few hands overripe or underripe, such can't be sold in market which is loss.
 - Unsold bananas are given away at throwaway prices.

During the stakeholder consultation it was found that total losses right from pre-production to marketing is 18-20% with around 10% losses in pre-production and around 7% losses during transportation.

f. Total No. of FPOs/Growers Association/Producer Groups (PGs) etc. involved in banana cultivation in cluster

Surat has one of the oldest cooperatives for banana. The Co-operative 'Kamrej Cooperative' was established in 1958. Besides, surat has 4 cooperatives for banana while Bharuch has three out of which 2 are unregistered. Three FPOs in the cluster actively involved in Banana trade and 5% of total banana growers are its members. Cooperative/FPO model has given the bargaining power to farmers. Their earnings have increased and farmer member of Cooperative/FPO is earning more than the regular farmers. The major areas of banana cultivation are Jhagadiya taluka in Bharuch, Anand taluka in Anand, Nandod taluka in Narmada, Vadodara taluka in Vadodara and Kamrej taluka in Surat.

g. Transport of banana from the cluster

Desai Fruits of Gujarat has been exporting banana since 2008. This company is one of the first organised players who started exporting banana from India. Exports of banana are taking place from JNPT. Export value and quantity cannot be ascertained from APEDA database as exports of Gujarat are considered as export from Maharashtra due to usage of JNPT port. However, from primary survey it was found that around 15000 MT of banana is exported annually from Gujarat whose value stands at 5.2 crore. Main export destinations of banana are middle east nations in which major ones are Iran, Oman and Saudi Arabia.

Table 12: Major domestic destinations for transfer of banana (from Gujarat cluster)

S.No.	States	State wise value (in INR)	State-wise mode of Transport (sea, air etc. to be specified)	Cost of transport (INR/kg)
1.	Delhi	277 Crore	Road – Trucks and Rail	Road – INR 4 Rail – INR 2 (after subsidy)
2.	Rajasthan	1130 Crore	Road – Trucks	2
3.	Punjab	457 Crore	Road – Trucks	4.5
4.	Haryana	418 Crore	Road - Trucks	3.5

Source - Stakeholder consultation with existing enterprises, respective central/state govt. department, GT Analysis

The main buyers of banana of Gujarat are Rajasthan, Delhi, Punjab and Haryana. Rajasthan being the neighbouring state buys maximum banana from Gujarat. Other minor destinations of banana of Gujarat are Jammu & Kashmir, Madhya Pradesh, Uttarakhand, Himachal Pradesh and Uttar Pradesh.

h. Prices with respect to grade/count/size/packaging etc.

Productivity from Grand Naine is high. When fresh tissue culture sapling is planted, each tree on an average produces a bunch of 28-30 kg which gives a productivity of 90 MT/Ha. Subsequent use of ratoon of mother plant, productivity first decreases to 23-25 kg per plant (75 MT/Ha) and further to 18-20 kg per plant (60 MT/Ha). In a period of 2.5 years, around 3 crop cycles of banana are executed.

According to AGMARK standards banana is classified into following classes:

Extra Class (commonly called as Grade A): Banana is of superior quality. Fingers are free of defects, with exception of very little superficial defects which do not affect general appearance and quality of the produce.

Class I (commonly called as Grade B): Banana is of good quality. Fingers have slight defects in shape and colour and other superficial defects which do not affect flesh. Area of defects does not exceed 2 sq. cm. of the total surface area.

Class II (commonly called as Grade B): These bananas satisfy minimum requirements. Area of defects does not exceed 4 sq. cm. of total surface area and the defects do not affect flesh.

Processable class (not part of AGMARK standards) (commonly called as Grade C): These are low quality banana which is not fit for human consumption but can be used for processing in which defects do not affect flesh. This class is extensively used in making banana chips.

Table 13: Average Prices with respect to grade/count/size/packaging etc

S.No.	Year	Based on grade	Market based (domestic/ export/ processing) – Price in Rs./kg	Price based on packaging (wooden/corrugated box/punnet etc.) in Rs./kg
1.	2020-21	Grade A – 10 Grade B – 8 Grade C – 5	Domestic – 10 Export – 13-15	Wooden – 42 Corrugated Box – 45
2.	2018-19	Grade A – 9 Grade B – 7 Grade C – 4	Domestic – 9 Export – 12-13	Wooden – 38 Corrugated Box – 42

Source - Stakeholder consultation with existing enterprises, central/state govt. departments, GT Analysis

2.1. Pre-Production and Production

a. Production of planting material

There are 11 laboratories in the cluster which have obtained subsidy under MIDH across the cluster. Average cost per plant sold through these laboratories is INR 12-13. State PSUs such as GSFC (Sardar brand) and GNFC (Narmada brand) are also providing services in terms of tissue culture plants and agri inputs through a predefined dealer network. Other players such as Jain Irrigation Limited, Rise and Shine and Nagarjuna Fertilizers also have a marketing network developed in the cluster through which they reach out to farmers.

Table 14: Tissue culture labs and material availability in the cluster (Gujarat)

Number of labs	Major varieties (including recently introduced varieties)	Capacity (number of plants/year)	Area coverage (in '000 Ha)	Average Selling Price/Plant	Capacity Utilisation %	Certified /Accredited	Annual demand of cluster (number of plants/year)	Constraints / Gaps with its reasons
11	check	2.2 Crore	14.67	15-17	100%	MIDH assisted	6.7 Cr	Private companies do aggressive marketing. Farmers buy from them in addition to GSFC, GNFC, etc.

Source: Stakeholder's consultation and government data, GT Analysis

b. Irrigation mechanism adopted

Table 15: Overview of micro and drip irrigation in banana cultivation in the cluster

District	Area of banana cultivation under drip/MIS (Ha)							
	<=1 Ha		1 Ha & <= 2Ha		2 Ha & <= 5Ha		>5 Ha	
	No	Ha	No	Ha	No	Ha	No	Ha
Anand	10	308.97	171	599.89	420	491	457	59.52
Bharuch	40	1420.56	580	2098.11	1496	1641.43	2072	289.64
C. Udepur	2	112.03	80	268.07	160	222.15	193	12.29
Narmada	41	2072.98	636	3126.49	2228	1739.19	3114	260.05
Surat	18	841.45	436	1913.3	1143	1164.33	1356	104.71
Vadodara	48	629.65	463	1381.51	913	1328.75	966	348.47
Total	159	5385.64	2366	9387.37	6360	6586.85	8158	1074.68

Source: Gujarat Green Revolution Company, GT Analysis

Gujarat Green Revolution Company (GGRC) is a state PSU registered under companies act of 1956. They have played a crucial role in making drip irrigation adoption popular in the cluster. GGRC has companies empanelled with it. When farmer applies through application then it's a tripartite agreement among farmer, supplier and GGRC. When drip is installed at farmers' field then he receives subsidy on installation cost.

Table 16: Irrigation scenario in banana cultivation (Gujarat)

Type of Irrigation method	% of farmers using the irrigation method	Cost with irrigation method / Ha)	Source of Water	Cluster requirement (in numbers/ area/ capacity etc.)	Gaps in irrigation mechanism
Micro	75-80%	8-10% of total cost incurred in cultivation	Ground Water/River water	No	—
Others (includes wells, borewells, renting from others)	20-25%	8-10% of total cost incurred in cultivation	Ground water/River water	No	-----

Source: Stakeholder consultations, GT Analysis

c. Good Agricultural Practices and fruit care activities

Since exports are primarily to Middle East nations, they do not warrant requirement of GAP certificate. Hence, there are practically no farmers with this certification. Slowly the number of farmers doing bunch care activities including bunch covering/ bagging/ skirting and other fruit care activities to grow quality produce is improving. Traders offer good price for good quality banana. There is subsidy on bagging, but it is not on plastic bags which is the prevalent practice but instead on non-woven bags which sticks to banana and results in damage to quality.

d. Harvesting practices

Currently, all the banana harvesting is done manually. When the bunch is cut, a labour carries the bunch on his shoulder to the vehicle. Adoption of portable cable conveyors was not seen in the cluster. The reason for low adoption is also that the conveyor installation requires land. Due to majority of the land holdings being small and marginal, adoption of this practice is still not that prevalent. Labour availability for haulage of banana is not a constraint as well.

2.2. Post-Harvest Management and Value Addition

a. Packhouse

Packhouses are not being used by traders for packaging since primary processing is being done at farm level/field itself by making makeshift packhouse. This saves time and reduces loading/unloading instances, prevents loss as well as economical and efficient. There was negligible interest seen in traders for usage of packhouse even if available. This is evident from the fact that a packhouse at Chhota Udepur district is currently nonfunctional.

Since Chhota Udepur does not have any presence of banana exporters as well, it's utility has been almost nil. Other exporters such as Reva Fruits do packaging at farm level itself, thereby eliminating the need of a packhouse. They are useful where fruit has long shelf life. Primary months of banana trading from Gujarat is April-July. Peak monsoon in July poses few constraints in handling banana but it is not deemed to be a major deterrent by the exporters and intermediaries.

b. Integrated Packhouse (APEDA Certified and independent)

Table 17: Integrated packhouse facilities in the cluster (Gujarat)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs./Kg	Single/ Multi commodity	Need for upgradation / Demand for new Infrastructure	Constraint s/ Gaps with its reason
Bharuch - 1 Narmada - 1 Surat - 1 CU - 1 (not in use) Cap of each >60MT	GRAND NAINE	Depending upon season (peak season of utilization is April to July)	120	APEDA packhouse on rent. Own packhouse no charges	Multi commodity	Yes	1. Chhota Udepur facility, not given on rent, no exporter ready 2. Farmers growing export quality and organic in Anand

Source: Stakeholder consultations, GT Analysis

c. Reefer vans

Reefer vans are not being used for banana transportation. They are expensive and moreover banana can withstand heat for 2 days. For protection of banana from heat, wet bagasse is spread on top of trucks loaded with bananas to maintain temperature and keep the fruit cooler.

d. Processing units

There is no organised player who is processing banana value added products in the cluster. Around 60 small scale units with a daily output capacity of 200 kg of chips are operating across the cluster. These are mostly run by large scale farmers themselves. Each unit provides employment to 6 people. Similarly, there are 10 such units in Bharuch district and 60 units in Narmada district.

Table 18: Processing scenario in the cluster (Gujarat)

Units	Input to output ratio	Total units	Production Capacity /unit	Setting-up cost/unit	Cluster requirement (no & capacity)	Constraints/ Gaps with its reason
Banana chips making units	5:1 (Raw banana used for making chips)	70	200 kg chips/day	5 lakhs	4 large scale units with daily input capacity of 10 MT	Need to be farmer run FPO/Cooperative should be promoted to take this up

Source: Stakeholder consultations, GT Analysis

e. Cold Storage

Current trade practices restrict farmer as fresh banana is preferred and purchased directly at farm level. In the cluster, 22 MIDH assisted cold storage facilities are present. These include Anand (8 nos.), Surat (8 nos.) and Vadodara (6 nos.)

Table 19: Cold storage presence in the cluster (Gujarat)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charge Rs./Kg	Single/ Multi commodity	Need for upgradation / Demand for new Infrastructure	Constraint s/ Gaps with its reason
22 in no (MIDH assisted) with 48427 MT capacity	Only used for export which is already in integrated packhouse	Depending upon season	150	Majorly privately owned	Multi commodity	No	Due to perishable nature of banana and current banana trading practices cold storage has not got much use

Source: Stakeholder consultations and government data

f. Ripening Chamber

Ripening Chambers are required for banana. A naturally ripened banana remains green in colour due to high temperature which may ultimately affect the market price. In the cluster, 21 ripening chambers are present which are MIDH assisted. These include Anand (7 nos.), Bharuch (2 nos.), Narmada (3 nos.) and Vadodara (9 nos.) Rest there are many ripening chambers who have not taken any subsidy. There are small scale chambers are at household level as well.

Table 20: Ripening Chamber in the cluster (Gujarat)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Usage Charge Rs./Kg	Single/ Multi commodity	Need for upgradation / Demand for new Infrastructure	Constraint s/ Gaps with its reason
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21 nos. with 1464 MT capacity	GRAND NAINA	As per prevailing demand	As per prevailing demand	INR 2	Multi commodity	Yes	Electricity requirement is high
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Source: Stakeholder consultations and government data

g. Quality

In the cluster, government quality testing facility is at Vadodara named as Food and Drugs Laboratory (FDL). At this facility very few samples of banana are received annually for quality check. Moreover samples from Vadodara are quality tested at the facility situated at Rajkot as FDLs have distributed districts among themselves from where they take samples for quality testing. System of cross checking wherein Vadodara lab checks samples of other districts and doesn't check samples of home district is followed. Currently fresh fruits are not being tested at FDL Rajkot. There is a requirement of Material Residue testing and Pesticide residue testing infrastructure in these facilities.

2.3. Logistics, Marketing and Branding

a. Mode of transport – Domestic market

Table 21: Mode of Transportation for domestic market (Gujarat)

Mode of Transport	Major market (location)	INR/Kg/ Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Small truck/ Vehicles	Within Gujarat and Mumbai	1-2	6-12	1-2% due to less distance	Good scope of rail
Truck	Delhi, Rajasthan, Haryana, Punjab	Rajasthan – 2.5-4 Haryana, Punjab, Delhi – 4.5 – 5.5	12-48	5-10%	Good scope of rail in every districts of the cluster
Train	Delhi	Delhi – 4 (excluding subsidy)	24	<5%	Seasonality is an issue

Source: Stakeholder consultations and government data, GT Analysis

Road transport is the prevalent mode of transit for transfer of banana produce. More than 99% of the banana trade is taking place via this route. Rail transport via 'Kisan Rail' is available at Vadodara Junction. It initiated in April 2021. Till June 2021, 850 MT had been transported. Subsidy is given on rail transport which is 50% of the freight cost. It is convenient to obtain this subsidy and the awareness and utilisation of Kisan Rail is expected to improve in future. When distance is short, small vehicles like Tata Ace, Tata pick up are used which is mostly the case when trader is Mumbai. For long distance 10 MT capacity of trucks are used.

b. Mode of transport – Export Market

For export packaging and transportation, there are variable practices. In the cluster, exporters are also doing primary processing at field level itself wherein bananas are packed in carton boxes. These boxes are transported to packhouse through vehicles and kept in pre-cooling chamber to remove field heat. These boxes are then transferred to a cold storage at a temperature of 13-15 degree Celsius. Meanwhile, a reefer container of 40 feet size of capacity of 20 MT is sourced from JNPT Port which on arrival at the integrated packhouse is loaded. The container is then dispatched to importing nations through JNPT.

Table 22: Mode of Transportation for exports market (Gujarat)

Mode of Transport	Major market (location)	Cost - INR/Kg/Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Sea	Iran, Oman, Saudi Arabia, UAE	3	10-14 days	<5%	Long distance not possible (Europe)
ICD, Sabarmati	Iran, Oman, Saudi Arabia, UAE	NA	NA	<5%	Far from cluster

Source: Primary Survey and Field Validation, GT Analysis

c. Marketing Channels

As described earlier, a major segment of the banana produce from the cluster is being sent to other states for consumption, primarily towards North India. Commission Agents play an important role in this chain of transfer wherein the farmers get to earn around INR 10/kg at farm gate.

Table 23: Marketing Channel Details in Gujarat cluster

Marketing Channel	Percentage of farmers covered	Volume of crop handled (in MT)	Selling Price Rs./Kg	Constraints/Gaps with its reason
Commission Agent	85-90%	31.45 Lakh	9-10	Traders high in number
FPO/Cooperative	5-10%	3.7 Lakh	10-11	Trader's control maximum trader
Trader/Wholesaler (Mother Dairy/Desai Fruits)	4%	1.48 Lakh	12-14	Very low organized trade
Exports	1%	~15000 MT	13-15	-

Source: Stakeholder consultations, GT Analysis

d. Current Branding & Marketing Initiatives

Currently, there are no existing GI tagging measures for banana in Gujarat. Typically, GRAND NAINIE is grown. A homegrown variety of brand for banana will help enhance exports but also leave with farmers options of growing many varieties and thereby enhance their income levels

Proposed Interventions and Recommendations

Table 24: Indicative proposed interventions for Gujarat cluster

Value Chain Activity	Gaps Identified	Recommended Component	Numbers (A)	Units (MT, Ha, Sqm, set, participants etc.)	Indicative Per unit cost in INR Lakh (B)	Indicative Cost of Interventions C=A*B (INR Cr.)
Pre-Production and Production	Limited number of Farmer Producer Organisations (FPOs)	FPOs formation and promotion	36	500 participants each	43	15.48
	Lack of GAP certification	Promotion and training of GAP	28,111	Hectares	0.1	28.11
	Limited quality testing at farm and post farm levels	MRL, pesticide residue testing lab	1	Nos.	400	4
		Basic testing facility at district level	6	Nos.	25	1.5
Post-harvest Management and Value Addition	Dependency on traders for ripening	Ripening Chambers*	10	10 MT each	10	1
	Lack of export promotion infrastructure	APEDA certified packhouse	1	At least 200 MT capacity	400	4
	Limited sorting, grading and packaging facilities	Collection centres	80	50 MT per day handling capacity	4	3.2
	Limited processing and value addition	Banana Chips/ Flour manufacturing unit	4	10 MT/day input capacity	75	3
		Banana Puree manufacturing unit**	1	96 MT/day input capacity	1500	15
Logistics, Marketing and Branding	Injury and loss due to improper packaging practices	Adoption of crates	700000	20 kg each	0.003	17.5
	Limited availability of refrigerated transport	Refrigerated Transport Vehicles	27	15 MT reefer vans	30	8.1
Total	Limited branding, no GI tag	Branding, Digital and Marketing Initiatives	1	Approx. 10% investment towards holistic development for development of cluster brand	10.09	10.09
	-	-	-	-	-	110.98

Source: GT Analysis

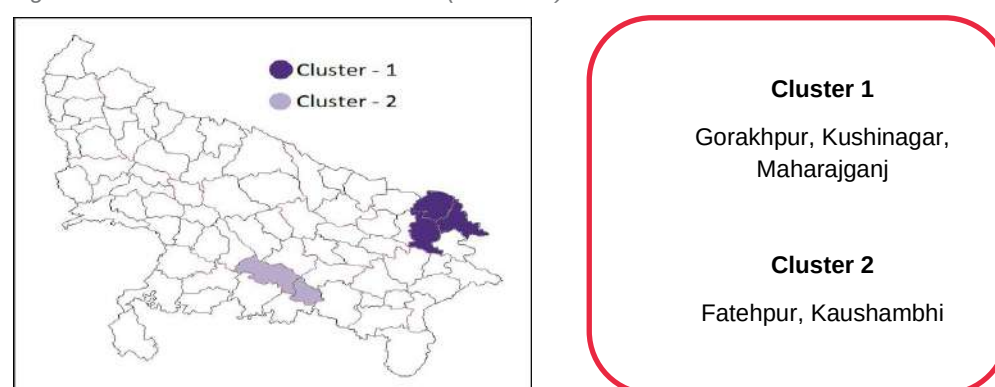
Key Cluster Wise Findings - Uttar Pradesh

Cluster 1 – Gorakhpur, Kushinagar and Maharajganj

3.1. Current Scenario

Uttar Pradesh has 75 districts in which variety of crops are grown. The agrarian landscape is dominated by sugarcane. Along with this, horticultural produce is also substantially produced. Amongst fruits, the major crops include banana and mango. For banana, the most cultivated varieties in the state are GRAND NAIN, Nendran, Monthan, Robusta, Dwarf Cavendish and Harichhal. Banana is majorly harvested in the months of September, October and November. The major areas in Uttar Pradesh where banana is cultivated on a large scale are Sidharth Nagar, Basti, Sant Kabirnagar, Maharajganj, Kushinagar, Faizabad, Barabanki, Sultanpur, Lucknow, Sitapur, Kaushambi and Allahabad.

Figure 6: Focal clusters in Uttar Pradesh (5 districts)



a. Total area and production under horticulture in the state (last 3 years)

The total area under horticulture production in the state has grown at a CAGR of 1.15% over the last 3 years while in terms of production the growth has been fairly constant. Aligarh, Fatehpur, Farrukhabad, Gorakhpur and Bulandshahr are the top 5 districts in fruit production while Agra, Farrukhabad, Hathras and Kannauj are the leading districts in vegetable production in the state.

Table 25: Area under horticulture cultivation in Uttar Pradesh (2017-18 to 2019-20)

S. No.	Year	Area (in Lakh Ha)	Production (in Mn MT)
1.	2019-20	17.72	38.19
2.	2018-19	17.38	38.35
3.	2017-18	17.33	38.35

Source: Department of Horticulture and Food Processing, Govt of Uttar Pradesh, GT Analysis

b. Production of banana in Uttar Pradesh

The overall production of banana in Uttar Pradesh for 2019-20 stood at 3.33 Million MT across a cultivated area of over 72000 ha. Faizabad, Bahraich, Farrukhabad, Fatehpur and Gorakhpur are the leading banana producing districts in the state accounting to over 44% of the state production while Fatehpur, Gorakhpur, Kaushambi and Kushinagar (which are a part of the study region) accounted for around 70% of the total cultivation area for banana in the state.

Table 26: Banana Production Uttar Pradesh (2017-18 to 2019-20)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
1.	2019-20	72.59	3.33
2.	2018-19	70.47	3.22
3.	2017-18	69.38	3.17

Source: Department of Horticulture and Food Processing, Govt of Uttar Pradesh, GT Analysis

c. Production of banana in cluster

The cluster comprising of 3 districts – Gorakhpur, Maharajganj and Kushinagar comprises of 55% of the cultivated land under banana in the state while at the same time amounts to 55% of the total state production as well (2019-20).

Table 27: Area and production under banana cultivation in cluster 1 (2017-18 to 2019-20)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
1.	2019-20	40.84	1.85
2.	2018-19	39.65	1.79
3.	2017-18	39.06	1.76

Source: Department of Horticulture and Food Processing, Govt of Uttar Pradesh, GT Analysis

d. Marketable surplus

From the stakeholder consultations, it was found that around 75-80% of the trade is taking place through movement of produce to outside the state. Around 20-25% of the produce is consumed within the state. The major destination geographies include Delhi, Punjab, Haryana and Madhya Pradesh.

Table 28: Marketable surplus of banana in Cluster 1 (Uttar Pradesh)

S.No.	Varieties	Grade wise marketable surplus (in Mn MT)	Total marketable surplus (in MT)	Shelf life
1	GRAND NAINE (95%)	1.222816	1.29	7-10 days
2	Others – Robusta, Harichal (5%)	0.06		7-10 days

Source: Stakeholder consultations and GT Analysis

e. Percentages losses in banana in cluster

The average losses in banana in the cluster amounted to around 22% as per stakeholder estimates. A major portion of these losses occur during transit procedures from farm level or aggregation point to consumption points. The overall losses at all stages is in the range of 20-25%. With major losses occurring during production (8-9%) and during transportation (9-10%).

f. Total no. of FPOs/Grower's association/Producer groups (PGs) etc. involved in banana cultivation in cluster

There are 15 FPOs in the cluster and majority are new which were formed in 2018-19. Out of these, 11 are in Gorakhpur and 4 are in Kushinagar district. FPOs here are acting as facilitator for farmers to sell their crops. Farmers first sell to FPOs and then FPOs sell it onwards to traders. The selling rate fixed by the FPO to farmer and the rate at which FPO sell to trader further usually differs (with the latter one being higher). The profit is used for day to day functioning of the FPO and net profit after deducting all expenditures is distributed among farmers.

g. Domestic transfer of banana from the cluster

A major segment of the cluster production of banana is seen making way to the neighbouring state of Bihar which is a large consumer of fresh banana fruit owing to its high population density and proximity

to the cluster. Other destinations domestically for sale of banana fruit from the cluster include Delhi, Rajasthan, Punjab and Haryana.

Table 29: Major domestic destinations for transfer of banana (from UP cluster 1)

S.No.	States	State wise value (in INR)	State-wise mode of Transport (sea, air etc. to be specified)	Cost of transport (INR/kg)
1.	Bihar	500 Cr	Road – Small trucks/trucks	<=2
2.	Delhi	80 Cr	Road – Trucks	3
3.	Rajasthan	330 Cr	Road – Trucks	4.5
4.	Punjab	133 Cr	Road – Trucks	4
5.	Haryana	122 Cr	Road - Trucks	3.5

Source: Stakeholder consultations and GT Analysis

h. Prices with respect to grade/count/size/packaging

S.No.	Year	Based on grade	Market based (domestic/ export/ processing) – Price in Rs./kg
1.	2020-21	Grade A – 11 Grade B – 8 Grade C – 5	Domestic – 10 Export – 13-15
2.	2018-19	Grade A – 10 Grade B – 8 Grade C – 4	Domestic – 9 Export – 12-13

Source: Stakeholder consultations and GT Analysis

3.2. Pre-Production and Production

a. Production of planting material

There is no tissue culture laboratory facility in any of the three districts of the cluster presently. The salespersons of private companies visit the farmers in the cluster and book required orders. Representatives of companies like Jain Irrigation and Rise and Shine to name a few visit the farmer to gauge their requirement. Similarly, few companies from Karnataka are also visiting the banana cultivators in this cluster for this purpose. The rates offered by these companies are highly competitive. Since there is no Tissue Culture Laboratory available for production of Banana TC Plants in the cluster, new Tissue Culture Units should be established in the cluster and nearby areas to reduce the cost of planting materials and also to avoid the damage during transportation.

Table 30: Tissue culture/nursery infrastructure availability in Cluster 1 (Uttar Pradesh)

Number Available	Major varieties (including recently introduced varieties)	Capacity (number of plants/year)	Area coverage (in Ha)	Average Selling Price/Plant (INR)	Capacity Utilisation %	Certified /Accredited	Annual demand of cluster (number of plants/year)	Constraints / Gaps with its reasons
14	GRAND NAIN	~20 lakhs	666	15-17	100%	Non-DBT approved	5 Cr	Private companies do aggressive marketing

Source: Stakeholder consultations and GT Analysis

b. Irrigation mechanism adopted

There is subsidy on adoption of drip irrigation under PMKSY (Pradhan Mantri Krishi Sinchai Yojana). But adoption of drip is very low in the cluster. One of the main reason is small land holding and another is surface water is being present in plenty. At present, there is only 20-22% of total banana cultivation under drip in the cluster.

Table 31: Irrigation Scenario in banana cultivation in Cluster 1 (Uttar Pradesh)

Type of Irrigation method	% of farmers using the irrigation method	Major blocks (with % coverage)	Cost with irrigation method / Ha)	Source of Water	Cluster requirement (in numbers/ area/ capacity etc.)	Gaps in irrigation mechanism
Micro	20-22%	Penetration of irrigation is less currently	8-10% of total cost incurred in cultivation	Ground Water/River water	Yes	Lack of private sector participation, awareness amongst farmers
Others (includes wells, borewells, renting from others)	80-85%	Right now, dependence on these sources is high	8-10% of total cost incurred in cultivation	Ground water/River water	No	

Source: Stakeholder consultations and GT Analysis

c. Good Agricultural Practices and fruit care activities

Irrespective of farm size, presently, adoption of GAP like mulching, INM, IPM and fruit care are minimal. No organisation involved for Global / India GAP certification. Nepal due to its vicinity is the major destination of export from this cluster. Compliance and GAP certification is seldom required/followed. However, the farmers are aware of the certification procedures to get higher price realization. Intermediaries are willing to pay higher rates for better quality produce.

d. Harvesting Practices

As like other places, harvesting is done manually. As the farm is fragmented, no mechanized practices are followed. Losses at the farm gate is very minimal.

3.3. Post-Harvest Management and Value Addition

a. Packhouse

Packhouses are being used in the cluster albeit only in Gorakhpur district. There are traders who do the primary processing at field itself. There are 8 packhouses in the district while other two districts have no packhouse. Highly primitive postharvest handling practices are followed with two plastic containers with alum and fungicidal treatments. Proper drying is not been done and then the dehandled fruits were loaded in the crates.

Table 32: Packhouse availability in Cluster 1 (Uttar Pradesh)

Type of infrastructure	Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs./Kg	Single/ Multi commodity	Need for upgradation/ Demand for new Infrastructure	Constraints/ Gaps with its reason

Packhouse	0	GRAND NAIN	Depend upon season (peak season of utilization is July, Aug-Sept)	90	Facility built by government, no usage charges	Multi commodity	No	Primary processing done at packhouse. On field primary processing practice can be adopted
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Source: Stakeholder consultations, Primary Survey, GT Analysis

b. Integrated Packhouse (APEDA certified)

Integrated packhouse facilities are absent in the cluster. The exports are currently happening only to Nepal via ICD Sonauli. Traders of Nepal come to the field and take the banana right from farm gate. Export norms are seldom followed due to cultural and geographical similarities.

c. Reefer vans

Reefer vans can be used only if exporters to countries other than Nepal are buying their banana from the state. The banana growing districts are far from JNPT from where primarily the exports of banana takes place (1700 km). Subsidy on transport to reefer vans should be given. The region has potential to grow export quality banana.

d. Processing units

There is only one processing unit in the cluster, Kushinagar, utilizing pseudostem for the extraction of banana fibre. From this fibre, various handloom and textile products can be made. The firm is also registered on e-commerce portals and websites for marketing the end products. Major destinations for sale of banana fibre are Patna, Delhi, Gorakhpur and Varanasi. Selling Price of 1 kg Banana fibre – A-grade at INR 300 to 400 per kg and B grade at INR 250 to 300 per kg.

Table 33: Production of banana fibre in cluster 1 (Uttar Pradesh)

Units	Input to output ratio	Total units	Production Capacity /unit	Setting-up cost/unit	Cluster requirement (no & capacity)	Constraints/ Gaps with its reason
Banana fibre making unit	8 to 12 stems required to extract 1 kg fibre	1	70 kg banana fiber/ day	2.65 lakh (machine cost)		FPO/Cooperative should be promoted in this sector

Source: Stakeholder consultations, DDH Office and GT Analysis

e. Cold Storage

Current banana trade practices do not require any cold storage for domestic purpose. For export purposes, the exporters need to store banana in cold environment, however, this is done at own packhouse or storage facility.

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs. /Kg	Single/ Multi commodity	Cluster requirement (in number/ area/ capacity etc.)	Need for upgradation/ Demand for new Infrastructure	Constraints/ Gaps with its reason
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17 in no (MIDH assisted) with 102078 MT capacity	Only used for export which is already in integrated packhouse	Depending upon season	15-180 days	Majorly private owned	Multi commodity	No	No	Due to perishable nature of banana and current banana trading practices cold storage has got negligible utility
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Source: Stakeholder consultations, DDH Office and GT Analysis

f. Ripening Chamber

Two ripening chamber facilities in the cluster with a combined capacity of around 69 MT. Ripened banana is preferred by the traders operating from near the farm level as there is minimal issue in transporting ripened banana. However, for longer distance, transportation of ripened banana is an issue which leads to transit loss. There are few small-scale ripening units at household level as well which banana is ripened either using liquid ethylene or through ice. These small household units are selling ripened banana in the nearby domestic markets. Traders use traditional method of ripening the bunches using smoke rooms.

Table 34: Ripening Chamber facilities in Cluster 1 (Uttar Pradesh)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Usage Charges Rs./Kg	Single/ Multi commodity	Cluster requirement (in number/ area/ capacity etc.)	Need for upgradation/ Demand for new Infrastructure	Constraints/ Gaps with its reason
2 in number (MIDH assisted) with total capacity of 69.13 MT)	GRAND NAINA	As per prevailing demand	As per prevailing demand	INR 2	Multi commodity	No	No	Electricity requirement is high

Source: Stakeholder consultations, DDH Office and GT Analysis

g. Quality assessment

For quality testing purposes, there are no testing infrastructure or laboratories in the cluster. Moreover, for domestic purpose no testing is done and required.

3.4. Logistics, Marketing and Branding

a. Mode of transport – Domestic market

Nearly 80% of total banana production of the cluster is distributed to outside the state while 15-20% of the produce is sold to domestic markets. The main domestic markets are Pratapgarh and East Uttar Pradesh. In addition, Bihar is a major destination. Gorakhpur district being close proximate to Bihar, the banana from the cluster, is being sent to all cities of Bihar through trucks and small trucks. Harvested banana bunches are transported from farm level by traders/wholesalers through light vehicle like Pick up, TATA ape, TATA ace, to avoid injuries to harvested produces.

Table 35: Mode of Transport for domestic market (Cluster 1 - Uttar Pradesh)

Mode of Transport	Major market (location)	INR/Kg/ Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Small truck/ Vehicles	For short distance domestic markets and Bihar	1-2	3-6	1-2% due to less distance	Good scope of rail
Truck	Delhi, Rajasthan, Haryana, Punjab	Delhi – 3 Raj – 4.5 Punjab – 4 Haryana -3.5	12-48	5-10%	Good scope of rail in every district of the cluster

Source: Stakeholder consultations, DDH Office and GT Analysis

b. Mode of transport – Export Market

In terms of exports, the primary outflow is into the neighbouring nation of Nepal through Land Custom Station facility of Sonauli located in Maharajganj district. Nepal is one of the largest importers of banana from India owing to geographical proximity and limited export related compliances required. The average value of exported banana from this LCD is around \$120 per MT which is considerably low than banana exported from other exit points from India.

Table 36: Exports of fresh banana from LCD Sonauli to Nepal (Maharajganj District - Uttar Pradesh)

Port	2018-19		2019-20		2020-21	
	Qty ('000 MT)	INR Crore	Qty ('000 MT)	INR Crore	Qty ('000 MT)	INR Crore
Nautanwa Sonauli LCD	26.48	15.94	36.57	25.97	26.23	21.99

Source: APEDA, HS Code - 08039010

c. Marketing channels

Banana business runs on market demand and prevalent prices. There are primarily 4 types of channels for marketing of banana from the cluster.

- Farmers sell their produce directly to commission agent (85%) dealing with banana crop. Small traders are selling these produces to wholesale traders in other cities in the state as well as the neighbouring state of Bihar. In turn, these produces are mobilized to retailers in sub urban, rural areas and finally to consumers.
- Farmers are selling their produce via Farmer producer organisations (5%) in limited cases. This is onward sold to districts within the cluster, primarily Gorakhpur as well as
- This type deals with export purpose. Farmers are selling the produces directly to whole sale traders and whole sale traders themselves export the produce or have tie ups with exporters for onward sale to Nepal. Reefer transport for exports is negligibly used.

Table 37: Marketing channels for fresh banana in Cluster 1 (Uttar Pradesh)

Marketing Channel	Percentage of farmers covered	Volume of crop handled (in MT)	Selling Price Rs./Kg	Constraints/Gaps with its reason
Commission Agent	85-90%	16.65 Lakh	9-10	Traders high in number, delay in payment, low price during glut
FPO/Cooperative	5-10%	.925 Lakh	10-11	Interference and control by trader lobbies
Trader/Wholesaler (Mother Dairy/Desai Fruits)	4%	.555 Lakh	12-14	Very limited organized trade

Exports	1%	.37 MT	13-15	Maximum exports to Nepal
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Source: Stakeholder consultations, DDH Office and GT Analysis

d. Current branding and marketing initiatives

Currently, there is no existing initiative or marketing exercises being undertaken for branding of produce from the cluster. There is no existing cluster brand either. Majority of the exports are also to the neighbouring Nepal market where stringent exports related requirements are not a priority (unlike EU and US markets).

3.5. Proposed Interventions and Recommendations

Table 38: Indicative proposed interventions for Cluster 1 - Uttar Pradesh

Value Chain Activity	Gaps Identified	Recommended Component	Numbers (A)	Units (MT, Ha, Sqm, set, participant etc.)	Indicative Per unit cost in INR Lakh (B)	Indicative Cost of Interventions C=A*B (INR Cr.)
Pre-Production and Production	Lack of Tissue Culture Planting Material	Tissue Culture Lab with Virus Indexing Facility & Hardening Center	10	2.3 Cr. plants every year	250	25.0
	Limited number of Farmer Producer Organisations (FPOs)	FPOs formation and promotion	18	500 participants each	43	7.7
	Lack of GAP certification	Promotion and training of GAP	20421	Hectare	0.1	20.4
	Limited quality testing at farm and post farm levels	MRL, pesticide residue testing lab	1	Number	400	4.0
		Basic testing facility at district level	3	Number	25	0.8
Post-harvest Management and Value Addition	Dependency on traders for ripening	Ripening Chambers	28	50 MT each	50	14.0
	Lack of export promotion infrastructure	APEDA certified packhouse	1	At least 200 MT storage	400	4.0
	Limited sorting, grading and packaging facilities	Collection centres	52	50 MT per day handling capacity	4	2.1
	Limited processing	Banana Chips/ Flour manufacturing unit	2	10 MT/day input capacity	75	1.5
		Banana Puree manufacturing unit	1	96 MT/day	1500	15.0
	Injury and loss due to improper packaging practices	Adoption of crates	320000	20 kg each	0.003	9.6
Logistics, Marketing and Branding	Limited availability of refrigerated transport	Refrigerated Transport Vehicles	10	15 MT reefer vans	30	3.0
	Limited branding, no GI tag	Branding, Digital and Marketing Initiatives	1	Approx. 10% investment towards holistic development of cluster brand	10.7091	10.7
Total			-	-	-	117.8

Source: GT Analysis

Cluster 2 – Kaushambhi and Fatehpur

4.1. Current Scenario

a. Production of banana in cluster

The cluster comprising of 2 districts – Kaushambhi and Fatehpur which accounts for 19% of the cultivated land under banana in the state while at the same time amounts to 21% of the total state production as well (2019-20).

Table 39: Area and Production of banana in Cluster 2 (Uttar Pradesh)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
1.	2019-20	14.97	0.71
2.	2018-19	14.54	0.69
3.	2017-18	14.32	0.68

Source: Department of Horticulture and Food Processing, Uttar Pradesh Govt, GT Analysis

b. Marketable surplus

Table 40: Marketable surplus of banana in Cluster 2 (Uttar Pradesh)

S.No.	Varieties	Grade wise marketable surplus (in Mn MT)	Total marketable surplus (in MT)	Shelf life
1	GRAND NAIN (95%)	0.51	0.54 lakh	7-10 days
2	Others – Robusta, Mahalakshmi (5%)	.03		7-10 days

Source: Department of Horticulture and Food Processing, Uttar Pradesh Govt, GT Analysis

c. Percentage losses of banana in cluster

The average losses in banana in the cluster amounted to around 22% as per stakeholder estimates. A major portion of these losses occur during transit procedures from farm level or aggregation point to consumption points. The overall losses at all stages are in the range of 20-25%. With major losses occurring during production (8-9%) and during transportation (9-10%).

d. Number of farmers involved in banana cultivation

S.No.	Type	Total No. of farmers	Average Farmers' Income (INR/Ha)	Blocks with higher concentration of farmers
1.	Marginal (< 1 Ha)	~3000	5,00,000	Campierganj in Gorakhpur Khadda in Kushinagar
2.	Small (1 – 2 Ha)	~7500		Nichnaul block in Maharajganj Bhitaura block in Fatehpur
4.	Large (> 2 Ha)	~2000		Kaushambi block in Kaushambi

Source: Stakeholder consultations, GT Analysis

e. Total no. of FPOs/Growers Association/Producer Groups (PGs) etc. involved in banana cultivation in cluster

There are 2 FPOs who are dealing in banana in the cluster. FPOs act as an intermediary in the trade of banana. They buy from farmers and sell it further to traders and in this process, profit is earned which is distributed among farmer members after deducting operational expenses for running the FPO. The FPOs also help farmers in advice and provision of agri inputs and market intelligence.

f. Domestic transfer of banana from the cluster

S. No.	States	State wise value (in INR)	State-wise mode of Transport (sea, air etc. to be specified)	Cost of transport (INR/kg)
1.	Bihar	220.6 Crore	Road – Small trucks/trucks	<=3
2.	Delhi	35.6 Crore	Road – Trucks	3
3.	Rajasthan	145.3 Crore	Road – Trucks	4
4.	Punjab	58.8 Crore	Road – Trucks	4
5.	Haryana	53.7 Crore	Road – Trucks	3.5

Source: Stakeholder consultations, GT Analysis

g. Prices with respect to grade/count/size/packaging

S.No.	Year	Based on grade	Market based (domestic/ export/ processing) – Price in Rs./kg
1.	2020-21	Grade A – 11 Grade B – 8 Grade C – 5	Domestic – 10 Export – 13-15
2.	2018-19	Grade A – 10 Grade B – 8 Grade C – 4	Domestic – 9 Export – 12-13

Source: Stakeholder consultations, GT Analysis

4.2. Pre-Production and Production

a. Adoption of new practices, innovation & technology in Pre-production and Production

Currently, tissue culture is being used for growing banana. Ratoons are used 2-3 times. Drip irrigation is slowly making inroads into the banana cultivation area in the two clusters of the state.

b. Production of planting material

There are no tissue culture labs in the cluster. Only hardening centres/nursery are available (3 in number). Their annual capacity is 14 lakh saplings. There is an evident need to introduce tissue culture lab where banana is grown in high quantity.

Table 41: TC Labs and nurseries scenario in the cluster (Cluster 1 - UP)

Number of nurseries	Major varieties (including recently introduced varieties)	Capacity (number of plants/year)	Area coverage (in Ha)	Average Selling Price/Plant	Capacity Utilisation %	Certified /Accredited	Annual demand of cluster (number of plants/year)	Constraints / Gaps with its reasons
3	GRAND NAINA	~3 lakhs	100	15-17	100%	Non-DBT approved	1.8 cr	There should be a tissue culture lab where banana plantation is in large area

Source: Stakeholder consultations, GT Analysis

c. Irrigation mechanism adopted

Currently, micro irrigation penetration is considerably low in the cluster when compared with other major banana growing regions in the country such as in Gujarat, Maharashtra or Andhra Pradesh. Only around 30-35% of the banana plantation is under drip irrigation.

Table 42: Irrigation Scenario in Cluster 2 (Uttar Pradesh)

Type of Irrigation method	% of farmers using the irrigation method	Major blocks (with % coverage)	Cost with irrigation method / Ha)	Source of Water	Cluster requirement (in numbers/ area/ capacity etc.)	Gaps in irrigation mechanism
Micro	30-35%	Right now, its adoption very less	8-10% of total cost incurred in cultivation	Ground Water/River water	Yes	More farmers need to adopt this
Others (includes wells, borewells, renting from others)	65-70%	Right now, dependence on these sources is high	8-10% of total cost incurred in cultivation	Ground water/River water	No	

Source: Stakeholder consultations, GT Analysis

d. Good Agricultural Practices and fruit care activities

In the cluster, negligible or no export is happening. Nepal procures its banana from districts of cluster-1 of Uttar Pradesh. GAP certificate is required for European countries. If farmers adopt good fruit care practices, then good crop will be obtained and for that they will definitely fetch higher price. Traders willingly pay good price for good quality of banana which are free of spots and bruises.

e. Harvesting Practices

In the cluster harvesting is taking place manually. However, harvesting losses are minimal. Portable Cable conveyor can be used but these are used more for export purpose. However, its adoption is negligible as it requires space at farm level for operation and deployment.

4.3. Post-Harvest Management and Value Addition

a. Packhouse

The cluster has total of 46 packhouses out of which 24 are in Kaushambi district and 22 are Fatehpur district. There are traders who use packhouse for primary processing. However, these packhouses are not solely for bananas, rather are multi commodity in nature.

Table 43: Packhouse availability in Cluster 2 (Uttar Pradesh)

Type of infrastructure	Capacity	Variety / Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs./Kg	Single/ Multi commodity	Cluster requirement (in number/ area/ capacity etc.)	Need for upgradation/ Demand for new Infrastructure	Constraints/ Gaps with its reason
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Packhouse	Kaushambi – 22 Fatehpur – 20 with total capacity of 4200 MT	GRAND NAINA	Depending upon season (peak season of utilization is July, Aug-Sept)	90	Facility built by government, no usage charges	Multi commodity	No	No	Primary processing done at packhouse. On field primary processing practice can be adopted
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Source: Stakeholder consultations, Primary Survey, GT Analysis

b. Integrated packhouse

Right now, there are no integrated packhouses in the cluster, as exports is negligible from the region.

c. Reefer Vans

Currently, reefer vans are not being used since exports are negligible from the cluster.

d. Processing units

There are no banana processing units in the cluster.

e. Cold Storage

There are 24 cold storages with cumulative capacity of 212069 MT operating in the cluster. These are multi-commodity storages, primarily involved in potato storage. Utilisation in case of banana storage is negligible due to the risk of chill injury. Instead of cold storage, ripening chamber is the only PHM infrastructure which is relevant for banana. So, the facilities available is not upto mark for banana management, because its standards are not fulfilling the exact requirement specific to banana.

Table 44: Cold Storage facility availability in cluster 2 (Uttar Pradesh)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs. /Kg	Single/ Multi commodity	Cluster requirement (in number/ area/ capacity etc.)	Need for upgradation/ Demand for new Infrastructure	Constraints/ Gaps with its reason
24 in no (MIDH assisted) with 212069 MT capacity	Only used for export which is already in integrated packhouse	Depending upon season	Depending upon season	Depending upon agreement	Multi commodity	Not for banana	No	Perishability of banana and banana trading practices, cold storage not got use

Source: Stakeholder consultations, GT Analysis

f. Ripening Chamber

In the cluster, there are 2 ripening chambers however, these are private in nature and are not MIDH assisted. Small scale ripening units at local level (home) are seen in a makeshift manner, wherein ethylene gas or ethylene liquid or ice is used to ripen banana and sell it in local market.

Table 45: Ripening Chamber availability in Cluster 2 (Uttar Pradesh)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Usage Charges Rs./Kg	Single/ Multi commodity	Cluster requirement (in number/ area/ capacity etc.)	Need for upgradation/ Demand for new Infrastructure	Constraints/ Gaps with its reason
2 nos. with 25 MT capacity each	GRAND NAINA	80%	300	INR 2	Multi commodity	No	No	Electricity requirement is high

Source: Stakeholder consultations, GT Analysis

g. Quality Assessment

There are no quality assessment laboratories in the cluster. Moreover, for domestic purpose testing of banana is not required.

h. Adoption of new practices, innovation and technology in Post-Harvest Management and Value Addition

Usage of crates should be promoted and primary processing at field itself should be promoted. This will help in reducing losses. Currently, only foliage and farm waste such as banana leaves and stems are used for packaging and for transit purposes

4.4. Logistics, Marketing and Branding

a. Mode of transport – Domestic market

Mode of Transport	Major market (location)	INR/Kg/ Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Small truck/ Vehicles	For short distance	1-2	3-6	1-2% due to less distance	Good scope of rail
Truck	Delhi, Rajasthan, Haryana, Punjab	Delhi – 3 Raj – 4 Punjab – 4 Haryana -3.5	12-48	5-10%	Good scope of rail in every district of the cluster

Source: Stakeholder consultations, GT Analysis

b. Mode of transport – Export Market

Right now, exports are negligible or not at all happening from the cluster. During the field work no exporter of banana from the cluster was found.

c. Marketing channels

Table 46: Marketing channels for fresh banana in Cluster 2 (Uttar Pradesh)

Marketing Channel	Percentage of farmers covered	Volume of crop handled (in MT)	Selling Price Rs./Kg	Constraints/Gaps with its reason
Commission Agent	85-90%	6.1 lakh	9-10	Traders high in number
FPO/Cooperative	5-8%	0.5 lakh	10-11	Trader's control maximum trader
Trader/Wholesaler (Mother Dairy/ Desai Fruits)	2-3%	0.2 lakh	12-14	Very low organized trade

Source: Stakeholder consultations, GT Analysis

e. Current branding and marketing initiatives

Currently, there is no available Geographical Indication (GI) tag for banana in the state. Typically, GRAND NAINA is cultivated, however there is potential and demand to grow newer varieties. Local varieties need to be promoted (Nendran, Red Banana etc.). This will help in enhancing exports which will lead to increase in income of farmers.

4.5. Proposed Interventions and Recommendations

Table 47: Indicative proposed interventions for Cluster 2 (Uttar Pradesh)

Value Chain Activity	Gaps Identified	Recommended Component	Numbers (A)	Units (MT, Ha, Sqm, set, participant etc.)	Indicative Per unit cost in INR Lakh (B)	Indicative Cost of Interventions C=A*B (INR Cr.)
Pre-Production and Production	Lack of Tissue Culture Planting Material	Tissue Culture Lab with Virus Indexing Facility & Hardening Center	4	With total capacity of 1 Cr. plants every year	250	10
	Limited number of Farmer Producer Organisations (FPOs)	FPOs formation and promotion	12	500 participants each	43	5.16
	Lack of GAP certification	Promotion and training of GAP	7487	Hectare	0.1	7.49
	Limited quality testing at farm and post farm levels	MRL, pesticide residue testing lab	1	Number	400	4
		Basic testing facility at district level	2	Number	25	0.5
Post-harvest Management and Value Addition	Dependency on traders for ripening	Ripening Chambers	11	50 MT each	50	5.5
	Lack of export promotion infrastructure	APEDA certified packhouse	1	At least 200 MT storage	400	4
	Limited sorting, grading and packaging facilities	Collection centres	8	50 MT per day handling capacity	4	0.32
	Limited processing	Banana Chips/ Flour manufacturing unit	2	10 MT/day input capacity	75	1.5
		Banana Puree manufacturing unit	1	96 MT/day	1500	15
	Injury and loss due to improper packaging practices	Adoption of crates	125000	20 kg each	0.003	3.75
Logistics, Marketing and Branding	Limited availability of refrigerated transport	Refrigerated Transport Vehicles	10	15 MT reefer vans	30	3
	Limited branding, no GI tag	Branding, Digital and Marketing Initiatives	1	Approx. 10% investment towards holistic development of cluster brand	6.02	6.02
Total	-	-	-	-	-	66.24

Source: GT Analysis

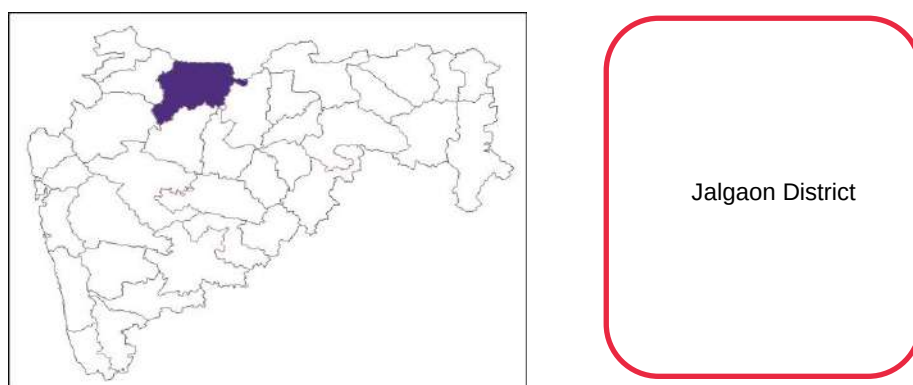
Key Cluster Wise Findings - Maharashtra

Cluster 1 - Jalgaon

5.1. Current Scenario

Maharashtra, the 3rd largest state in India, ranks 6th in horticulture production during 2018-19. Area under F&V cultivation is fairly constant at 1.41 Mn Ha (CAGR 1%) with It's contribution to total horticulture production at 7%. The state is famous for a fruits diversity such as Mango, Oranges, Banana and Grapes which are consumed far and wide within India and is exported as well to lucrative markets such as EU and USA. Besides, the fruits like Dragon fruit and date palm are being recently adopted by the farmers.

Figure 7: Focal Cluster in Maharashtra (1 district)



a. Total area and production under fruits in the state (last 5 years)

In terms of production of fruit, Maharashtra ranks 2nd in the country. Over the last 5 years, Maharashtra on an average has contributed 12% to the national fruit basket. The state is famous for its mangoes, orange, grapes and banana. These are supplied to all states and consumed across India. Banana is the most grown fruit in the state with 68,000 ha under cultivation yielding around 3.5 Mn MT of banana

Table 48: Total area and production under fruits in Maharashtra (2014-15 to 2018-19)

S. No.	Year	Area (in Lakh Ha)	Production (in Mn MT)
1.	2018-19	7.57	10.81
2.	2017-18	7.56	11.84
3.	2016-17	7.45	13.71
4.	2015-16	7.29	9.70
5.	2014-15	7.35	10.225876

Source: State Statistical Abstract, Department of Agriculture

b. Production of banana in Maharashtra

Maharashtra is the 3rd largest producer of Banana in India and contributes to 11% in the total national production. Over the last 5 years, in terms of area, the area under banana crop has reduced from 75,000 Ha to 68,000 Ha, while in the same period, the corresponding production has gone up to 3.73 Mn MT. The yield has also increased at CAGR of 3.4% and stands at 52 MT/Ha.

Table 49: Area, Production & Productivity of Banana in Maharashtra (2014-15 to 2018-19)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
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1.	2018-19	68.14	3.56
2.	2017-18	80.88	4.23
3.	2016-17	81.34	3.33
4.	2015-16	69.55	3.02
5.	2014-15	75.92	3.37

Source: State Statistical Abstract, Department of Agriculture

c. Production of banana in cluster

Jalgaon contributes to over 73% of the total banana production. The most cultivated varieties are Safed velchi (Ney Poovan), Basrai, Harichal (Robusta), GRAND NAINA, Lal velchi, Mutheli (Rasthali), Bankel and Bhurkel. The cultivation has prospered in the district owing to its locational advantage as well as widespread penetration of drip irrigation. Also, farm level interventions such as High-Density Planting (reducing impact of dry weather) and adoption of tissue culture banana plants positively impacted the spread of the industry. As a result, its yield of banana per ha (65 Mt/ha) is higher than that at state level (52 Mt/ha).

Table 50: Area, Production & Productivity of Banana in cluster (Maharashtra) (2014-14 to 2018-19)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
1.	2018-19	40.11	2.61
2.	2017-18	48.02	3.12
3.	2016-17	49.28	2.27
4.	2015-16	41.37	2.13
5.	2014-15	46.00	2.38

Source: State Statistical Abstract, Department of Agriculture

d. Marketable surplus

75-80% of the produce from this cluster is sold outside the state. The North Indian market is a major destination for banana from Jalgaon. Kisan Rail service is plying between Jalgaon and Delhi on a weekly basis. Around 2-3 consignments of banana from Jalgaon move to Delhi through train amounting to around 1100 MT weekly. The main harvesting season of the cluster is April to September every year. During this time maximum trade of banana happens from the cluster. The cluster also fulfils the requirement of the state. Local varieties are not grown much in the cluster. GRAND NAINA is the dominant variety in the cluster.

Table 51: Marketable Surplus of banana in the cluster (Maharashtra)

S.No.	Varieties	Grade wise marketable surplus (in Mn MT)	Total marketable surplus (in Mn MT)	Shelf life
1	GRAND NAINA (95%)	1.95	2.00	7-10 days
2	Others – Robusta, Harichhal (5%)	0.05		7-10 days

Source: Stakeholder consultations, DDH Office and GT Analysis

e. Percentage losses of banana in the cluster

From pre-production to retail the level of losses in the state of Maharashtra has been observed to be 30.5%. Maximum losses are observed at retail level wherein percentage losses are to the tune of 14.5% followed by losses at field level (6.8%), losses during transport (4.0%), losses during storage and ripening (3.6%) and losses during assembly market/ wholesale market (1.6%)⁶.

⁶ All India Coordinated Research Project Trial, from NRCB report

f. Total no. of FPOs/Growers Association/Producer Groups (PGs) etc. involved in banana cultivation in cluster

There are 165 FPOs in the cluster (Jalgaon district) out of which few are dealing in banana along with other fruits. FPOs act as an intermediary in the trade of banana. They buy from farmers and sell it further to traders and in this process, profit is earned which is distributed among farmer members after deducting operational expenses for running the FPO. Tapti valley Banana Processing & Products Co-op society, is a successful banana FPO set up at the cost of INR 7.50 Cr. They are making use of waste pseudo stem to make banana fibre. Muktainagar block is one of the major banana producing block of the district.

g. Domestic transfer of banana from the cluster

The major buyer of the Jalgaon banana are the north Indian states of Delhi, Uttar Pradesh, Haryana and Punjab. Kisan rail plies between Bhusawal Junction of Jalgaon and Adarsh Nagar station of Delhi. 23 MT capacity wagons are used. During one trip at least 15 and maximum 25 wagons are sent to Delhi depending upon demand in Delhi.

Table 52: Major domestic destinations for transfer of banana (from Maharashtra cluster)

S. No.	States	State wise value (in INR)	State-wise mode of Transport (sea, air etc. to be specified)	Cost of transport (INR/kg)
1.	Delhi	370 Crore	Road – Trucks	Delhi- INR 4
2.	Uttar Pradesh	385 Crore	Road – Trucks	Haryana- INR 4
3.	Punjab	535 Crore	Road – Trucks	Punjab- INR 5
4.	Haryana	488 Crore	Road – Trucks	Uttar Pradesh - INR 4-4.5

Source: Stakeholder consultations and GT Analysis

h. Prices with respect to grade/count/size/packaging

For growing good quality of banana farmers are offered good prices by traders. It becomes utmost importance that farmers adopt fruit care practices and grow quality banana. Though this will incur additional expenditure but price obtained will be more by farmers. For growing export quality a farmer easily obtains 13-15 INR/kg. For good quality for domestic a farmer easily obtains 10-12 INR/kg.

Table 53: Average Prices with respect to grade/count/size/packaging etc.

S.No.	Year	Based on grade	Market based (domestic/ export/ processing) – Price in Rs./kg
1.	2020-21	Grade A – 11 Grade B – 7 Grade C – 5	Domestic – 10 Export – 13-15
2.	2018-19	Grade A – 9 Grade B – 7 Grade C – 4	Domestic – 9 Export – 12-13

Source: Stakeholder consultations and GT Analysis

i. Status of protection and promotion initiatives for GI, Trademarks, etc.

Currently, there is a Geographical Indication (GI) tag for banana of Jalgaon and its banana is called 'Jalgaon Banana'. After receiving GI tag, exports have increased and the price realised by farmers for growing export quality banana (INR 15/kg) is around 50% more when compared to a regular farmer (INR 10/kg).

5.2. Pre-Production and Production

a. Production of planting material

There are 25 tissue culture labs in the cluster, which includes a facility run by Jain Irrigation Limited as well. This facility has a capacity of producing 10 crore saplings annually. There are other tissue culture labs as well who also coordinate by means of an association.

Table 54: Tissue culture labs and material availability in the cluster (Maharashtra)

Number of labs	Major varieties (including recently introduced varieties)	Capacity (number of plants/year)	Area coverage (in Ha)	Average Selling Price/Plant	Capacity Utilisation %	Certified /Accredited	Annual demand of cluster (number of plants/year)	Constraints / Gaps with its reasons
25	GRAND NAINA	>10 Cr. (Only for Jain Irrigation)	25000	13-17	80%	DBT approved	7.2 Crore	Nil

Source: Stakeholder consultations, GT Analysis

b. Irrigation mechanism adopted

Currently, micro irrigation penetration is high in the cluster. More than 90% of the banana plantation is under drip irrigation.

Table 55: Overview of micro and drip irrigation in banana cultivation in the cluster

Type of Irrigation method	% of farmers using the irrigation method	Cost with irrigation method / Ha)	Source of Water	Cluster requirement (in numbers/ area/ capacity etc.)	Gaps in irrigation mechanism
Micro	90-100%	8-10% of total cost incurred in cultivation	Ground Water/ River water	Yes	More farmers need to adopt this
Others (includes wells, borewells, renting from others)	5-10%	8-10% of total cost incurred in cultivation	Ground water/River water	No	

Source: Stakeholder consultations, GT Analysis

c. Good Agricultural Practices and fruit care activities

GAP certificate is required for exports of banana to EU markets. There are farmers who are following GAP and are producing export quality banana. But majority of them are exported to Gulf countries. Traders willingly pay good price for good quality of banana which are free of spots and bruises.

d. Harvesting Practices

In the cluster, primarily, manual harvesting is being practiced. Cutting of bunch is done manually and once detached, the bunch is carried by labour on shoulder to outside the farm where it undergoes primary processing at farm level or is directly loaded onto a vehicle.

5.3. Post-Harvest Management and Value Addition

a. Packhouse

There are 2 packhouses in the cluster. One is at Raver and another one at Savda. They are used for primary processing of banana.

b. Integrated packhouse

There are 3 integrated packhouses in the cluster. They are all privately owned with a combined capacity of 1,125 MT.

c. Reefer vans

Right now, reefer vans are not being used since exports as percentage of total production is very less ~2% from the cluster.

d. Processing units

Jain Farm Fresh has a banana processing unit. They produce banana puree, concentrate and juice. Apart from that they make blended juices like apple- banana blend, mango- banana juice and strawberry - banana juice. chips are being made using Grand Naine banana. Major market of banana chips is Nagpur, Nashik and Mumbai.

e. Cold Storage

There are multiple privately owned cold storages operating in the district numbering over 50 nos. These are multi commodity in nature. Banana storage is minimal in this regard.

f. Ripening Chamber

There are 2 ripening chambers in the cluster which has a cumulative capacity of 100 MT.

g. Quality Assessment

There is one quality testing laboratory in the cluster known as district public health laboratory located at Jalgaon. It is FSSAI registered. However, the number of samples being received here for fresh or processed banana products is negligible. Negligible samples of banana are received in lab for quality testing. The lab present in the cluster does the testing of manufactured products.

h. Adoption of new practices, innovation and technology in Post-Harvest Management and Value Addition

Usage of crates should be promoted and primary processing at field itself should be promoted. This will help in reducing losses.

5.4. Logistics, Marketing and Branding

a. Mode of transport – Domestic market

Harvested banana bunches are transported from farmers field to storage godowns of whole sale Agents/ whole sale banana traders through light vehicle like Pick up, TATA ape, TATA ace, just to avoid injuries to harvested produces. Price of banana is decided by an APMC in Jalgaon. At the price decided the whole Jalgaon follows that price. After buying of banana bunches are separated into hands, then hands are graded and sorted out. Either hands are loaded as it is in the vehicle or they are placed in crates and then transported further to the ultimate destination.

Table 56: Mode of Transportation for domestic market (Maharashtra)

Mode of Transport	Major market (location)	INR/Kg/ Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Small truck/ Vehicles	For short distance	1-2	3-6	1-2% due to less distance	Nil
Truck	Delhi, Haryana, Punjab and Uttar Pradesh	Delhi-4 Haryana-4 Punjab-5 UP – 4-4.5	12-48	5-10%	Usage of Kisan Rail needs to be promoted further
Train	Delhi	4.5 (excluding subsidy)	24 hrs. of travelling time, next 24 hrs. for loading and unloading	<5%	Nil

Source: Stakeholder consultations, GT Analysis

b. Mode of transport – Export Market

Jalgaon is a major exporter of banana in India. Jalgaon has GI tag for banana. Desai Fruits and INI Farms and other small, organized players are active in the cluster. Reefer vans are not used. Reefer container is booked at JNPT which on arrival to integrated packhouse is loaded with carton boxes of 13 kg or 7 kg capacities.

Table 57: Mode of Transportation for exports market (Maharashtra)

Mode of Transport	Major market (location)	Cost - INR/Kg/Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Sea	Iran, Oman, Saudi Arabia, UAE	3-4	9-10 hours	<5%	Long distance not possible (Europe)

Source: Stakeholder consultations, GT Analysis

c. Marketing channels

Farmer sell their produce directly to commission agents dealing with banana crop only. Small traders are selling this produce to wholesalers in other cities in states such as Delhi, Haryana, Punjab and Rajasthan. In turn, this produce is mobilized to retailers in sub urban, rural areas and finally to consumers. Maximum trade is happening through commission agents. FPOs are coming up as an optional trade channel. FPOs need marketing support for selling fruit in the market. This will increase their income and then the income of farmers will increase.

Table 58: Marketing Channel Details of the cluster (Maharashtra)

Marketing Channel	Percentage of farmers covered	Volume of crop handled (in MT)	Selling Price Rs./Kg	Constraints/Gaps with its reason
Commission Agent	85-90%	18 lakh	10-11	Traders high in number
FPO/Cooperative	5-8%	1 lakh	11-13	Trader's control maximum trader
Trader/Wholesaler (Mother Dairy/Desai)	2-3%	0.6 lakh	12-13	Low organized trade
Exports	1-2%	0.4 lakh	13-15	

Source: Stakeholder consultations, GT Analysis

5.5. Proposed Interventions and Recommendations

Table 59: Indicative proposed interventions for Jalgaon Cluster (Maharashtra)

Value Chain Activity	Gaps Identified	Recommended Component	Numbers (A)	Units (MT, Ha, Sqm, set, participants etc.)	Indicative Per unit cost in INR Lakh (B)	Indicative Cost of Interventions $C=A*B$ (INR Cr.)
Pre-Production and Production	Limited number of Farmer Producer Organisations (FPOs)	FPOs formation and promotion	6	500 participants each	43	2.58
	Lack of GAP certification	Promotion and training of GAP	20,056	Hectares	0.1	20.06
	Limited quality testing at farm and post farm levels	MRL, pesticide residue testing lab	1	Nos.	400	4
		Basic testing facility at district level	1	Nos.	25	0.25
Post-harvest Management and Value Addition	Dependency on traders for ripening	Ripening Chambers	22	50 MT each	50	11
	Lack of export promotion infrastructure	APEDA certified packhouse	0	At least 200 MT storage	400	0
	Limited sorting, grading and packaging facilities	Collection centres	28	50 MT per day handling capacity	4	1.12
	Limited processing and value addition	Banana Flour manufacturing unit	2	10 MT/day input capacity	75	1.5
		Banana Puree manufacturing unit	1	96 MT/day	1500	15
	Injury and loss due to improper packaging practices	Adoption of crates	300000	20 kg each	0.003	7.5
Logistics, Marketing and Branding	Limited availability of refrigerated transport	Refrigerated Transport Vehicles	49	15 MT reefer vans	30	14.7
	Limited branding, no GI tag	Branding, Digital and Marketing Initiatives	1	Approx. 2% investment towards holistic development of cluster brand	1.55	1.55
Total	-	-	-	-	-	79.26

Source: GT Analysis

Key Cluster Wise Findings - Andhra Pradesh

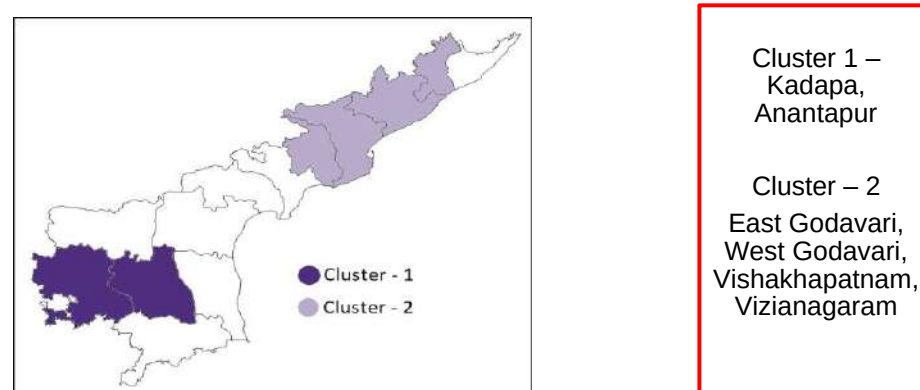
Cluster 1 – Anantapur and Kadapa

This cluster consists of Anantapur and Kadapa Districts

6.1. Current Scenario

Andhra Pradesh, the 7th largest state in India, stands 4th in horticulture production in 2018-19. Key metrics with regards to horticulture production in the state for the last 5 years. In terms of fruit production, Andhra Pradesh stands first nationally. In last five years Andhra Pradesh, on an average has contributed 14% to fruit basket of India. The state majorly grows banana and mango. These are supplied to all states and consumed across India. Banana is the most grown fruit in the state with 1.04 Lakh Ha area under banana cultivation yielding around 6.2 Mn MT.

Figure 8: Focal Cluster in Andhra Pradesh (6 districts)



a. Total area and production under horticulture in the state (last 5 years)

Total area under horticulture crop in the state increased at a CAGR of 3.16% during the period of 2015-20. During the same time the production increased at a CAGR of 9.49%. The change is phenomenal. According to horticulture department of the state the shift is voluntary as return from horticulture is more compared returns from cultivation of food grains. Horticulture contributes 16.07% to the state GVA (Gross Value Added).

Table 60: Total Area and Production under horticulture in Andhra Pradesh (2015-16 to 2019-20)

S. No.	Year	Area (in Lakh Ha)	Production (in Mn MT)
1.	2019-20	17.84	31.23
2.	2018-19	17.40	30.52
3.	2017-18	16.68	28.81
4.	2016-17	15.93	23.86
5.	2015-16	15.27	19.84

Source: AP Horticulture Department, GT Analysis

b. Production of banana in the state

Andhra Pradesh is the top producer of banana in the country. Andhra Pradesh alone contributes 20% to the banana basket of India. Over the last five years, area under banana cultivation has increased from

79000 Ha to 103000 Ha. Due to this the production has also increased from 3.5 Mn MT to 6.2 Mn MT. In Andhra Pradesh, PPP model was adopted by the state government to increase the yield which has given positive results. The yield of banana in the state is 60 MT/Ha which has increased from 44 MT/Ha at CAGR of 6.4%.

Table 61: Area, Production & Productivity of Banana in Cluster 1 (Andhra Pradesh)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
1.	2019-20	97.70	5.86
2.	2018-19	103.73	6.22
3.	2017-18	99.37	5.96
4.	2016-17	88.16	4.67
5.	2015-16	75.72	3.57

Source: AP Horticulture Department, GT Analysis

c. Production of banana in cluster

Area under banana cultivation in the cluster increased at a CAGR of 10% almost doubling in five-year period of 2015-20. In the same time production increased at a CAGR of 8.76%. Area under banana cultivation in Anantapur district in 2015 was 11712 Ha which in 2020 became 15116 Ha. In 2015, Kadapa district area under banana cultivation was 12800 Ha which in 2020 became 25887 Ha an increase of more than 100%.

Table 62: Area, Production & Productivity of banana in Cluster 1 (Andhra Pradesh)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
1.	2019-20	41.00	2.46
2.	2018-19	39.35	2.36
3.	2017-18	41.54	2.49
4.	2016-17	28.70	1.52
5.	2015-16	24.51	1.62

Source: AP Horticulture Department, GT Analysis

d. Marketable surplus which is sent out of state

Table 63: Marketable Surplus of banana in the cluster (Andhra Pradesh Cluster-1)

S.No.	Varieties	Grade wise marketable surplus (in M MT)	Total marketable surplus (in Mn MT)	Shelf life
1	GRAND NAINA (95%)	1.8	1.87	7-10 days
2	Others – Chakrakeli, Karpura (5%)	0.09		7-10days

Source: Stakeholder Consultations, GT Analysis

75-80% of the produce is sold outside the state. Telangana being close proximate to the cluster is the major buyer of banana of the cluster of Andhra Pradesh. Other major buyer is Maharashtra (cities like Pune, Nagpur and Mumbai). Traders of Delhi also buy banana from the cluster. The main harvesting season of the cluster is November to April. During this time maximum trade of banana happens from the cluster. The cluster also fulfils the requirement of the state. Local varieties are not grown much in the cluster. Grand Naine is the dominant variety in the cluster.

e. Losses of banana in cluster

From pre-production to retail the level of losses observed in the state of Andhra Pradesh has been observed to be around 23%. Maximum losses are observed at retail level wherein percentage losses are to the tune of 9.4% followed by losses during storage and ripening (4.8%), losses during assembly market/ wholesale market (3.9%), losses at field level (2.9%) and losses during transport (1.6%). Total no. of FPOs/Growers Association/Producer Groups (PGs) etc. involved in banana cultivation in cluster.

There are 13 FPOs in the cluster which are dealing in banana. They provide inputs to farmers at cheaper cost and after the formation of FPOs farmers income has increased and FPO members earn more than regular farmers. Out of these 13, 8 are in Anantapur and rest are in Kadapa. Gokulavedam FPO is one of the major FPO dealing in banana in Anantapur. The farmer member of the FPO are producing 1300 MT of banana. Other major ones are Sun Rise FPC and Turmeric and Banana FPO in Kadapa. Mandal which are major banana producers are Putluru, Yellanur, P. Pappur and Narpala in Anantapur.

f. Domestic transit of banana from the cluster

Around 80% of the produce of the cluster is going out of the state. Major ones being Telangana, Maharashtra and Delhi. Other states which are not major buy of the banana of the cluster are Odisha, Tamil Nadu, Uttar Pradesh and others. Small trucks and vehicles are used for short distance while for long distance large trucks are used.

Table 64: Major domestic destinations for transfer of banana (from Andhra Pradesh Cluster-1)

S. No.	States	State wise value (in INR)	State-wise mode of Transport (sea, air etc. to be specified)	Cost of transport (INR/kg)
1.	Maharashtra	842 Crore	Road	3
2.	Telangana	266 Crore	Road	1.5
3.	Tamil Nadu	541 Crore	Road	1.5
4.	Delhi	126 Crore	Road	6.5

Source: Stakeholder Consultations, GT Analysis

g. Prices with respect to grade/count/size/packaging

Primarily, based on shape, size and colour, the banana produced is graded into 3 variants – A, B and C with A grade being of the highest quality and fetching maximum farm level price of INR 10-11/kg. For better quality fruits traders offer good price to farmers.

Table 65: Average prices with respect to grade/count/size/packaging etc.

S.No.	Year	Based on grade	Market based (domestic/ export/ processing) – Price in Rs./kg
1.	2020-21	Grade A – 12 Grade B – 9 Grade C – 5-6	Domestic – 11 Export – 13-15
2.	2018-19	Grade A – 9 Grade B – 7 Grade C - 4	Domestic – 10 Export – 12-13

Source: Stakeholder Consultations, GT Analysis

6.2. Pre-Production and Production

a. Production of planting material

There are two private players tissue culture labs in Anantapur district. Their cumulative capacity is producing 50 lakh saplings per year. Rest of the demand of the cluster of tissue cluster saplings is met through companies like Jain, Nagarjuna among other companies. In the cluster there are many companies whose salesmen come and advertise their products and convince farmers to use their products.

Table 66: Tissue culture labs and planting material availability in Cluster 1 (Andhra Pradesh)

Number of nurseries	Major varieties (including recently introduced varieties)	Capacity (number of plants/year)	Area coverage (in Ha)	Average Selling Price/Plant	Capacity Utilisation %	Certified /Accredited	Annual demand of cluster (number of plants/year)	Constraints / Gaps with its reasons
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2	Grand Naine	~50 lakhs	1666	15-17	100%	Non-DBT approved	4.9 Crore	Private companies do aggressive marketing. Farmers buy from them.
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Source: Stakeholder Consultations, GT Analysis

b. Irrigation mechanism adopted

There is high adoption of drip irrigation in the cluster. It is more than in the 90% area of banana cultivation. This is due to state government incentives and push to make farmers adopt drip irrigation. (Andhra Pradesh Micro Irrigation Project - APMIP)

Table 67: Overview of micro and drip Irrigation in banana cultivation in the Cluster

Type of Irrigation method	% of farmers using the irrigation method	Cost with irrigation method / Ha)	Source of Water	Cluster requirement (in numbers/ area/ capacity etc.)	Gaps in irrigation mechanism
Micro	85-90%	8-10% of total cost incurred in cultivation	Ground Water/River water	No	_____
Others (includes wells, borewells, renting from others)	5-10%	8-10% of total cost incurred in cultivation	Ground water/River water	No	-----

Source: Stakeholder consultations, GT Analysis

c. GAP and fruit care activities

The PPP model executed through an MoU between the state's Horticulture Department and private players (INI Farms and Desai Fruits) with an objective to encourage and support productivity improvement, post-harvest infrastructure, capacity building and value chain development in Bananas has led to an incremental rise in the volumes of banana exported from the cluster. Farmers are trained on tissue culture adoption, irrigation techniques, fruit care practices etc. An area of 1500 acres is being targeted to be covered under this initiative.

d. Harvesting practices

As is the case in whole of India harvesting is done manually in the cluster. Being a robust fruit, this is carried on shoulders to the outside of field and then either primary processing is done or bunch is directly loaded into the vehicle. Major months of banana trading are winter months of November to April.

6.3. Post-Harvest Management and Value Addition

a. Packhouse

There are many packhouses in the cluster. Many are at field level also. Those packhouses are for storage of coconuts and mangoes. Out of total 2002 packhouses, 1156 are in Kadapa alone and rest in Anantapur. The districts are major producer of coconut and mango and for its primary processing farmers with the subsidy from NHM established under PPP model got built farm level packhouses.

Table 68: Details of packhouses in Cluster 1 (Andhra Pradesh)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs./Kg	Single/ Multi commodity	Need for upgradation/ Demand for new Infrastructure	Constraints/ Gaps with its reason
2002 packhouses having total capacity of 60060 MT	GRAND NAINA	Depending upon season (peak season of utilization is November to April)	180	Government packhouses can be used free of charge	Multi commodity	Yes	Primary Processing is done at field itself which has reduced reliance on packhouses

Source: Stakeholders consultation, GT analysis

b. Integrated Packhouse

There are 7 integrated packhouses out of which 6 are in Anantapur district. They have a capacity of 112 MT and are also utilised for cashew as well. These integrated packhouses help in exports.

Table 69: Integrated pack house facilities in cluster 1 (Andhra Pradesh)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs./Kg	Single/ Multi commodity	Need for upgradation/ Demand for new Infrastructure	Constraints/ Gaps with its reason
7 integrated packhouses in the cluster, 6 at Anantapur and 1 at Kadapa	GRAND NAINA and other local varieties	Peak season November to April	180	NA	Multi commodity	No	Primary Processing is done at field itself which has reduced reliance on packhouses

Source: Stakeholder consultations, DDH Office and GT Analysis

c. Reefer vans

Under the aegis of MIDH and initiatives of the state government, till date, around 2 nos. reefer vans have been given on subsidy to banana farmers in the cluster. Those reefer vans are used by farmers to transfer their fresh produce from the field to be sold in the market. The private players like Desai Fruits and INI Farms seldom use reefer vans. Reefer vans are costly and hence not preferred for exports or other uses.

d. Processing units

Right now, there are no banana processing units in the cluster. There are about 41 processing units but they are either for mango processing or for cashew processing or others. But none for banana processing.

e. Cold storage

Cold storages are not required for banana. If banana is kept in cold storage, then it will begin ripening. Moreover, a farmer cannot pre-cut and keep it in cold storage and then sell later when price is high. Current trade practices restrict farmer from doing that. Fresh banana right from field are bought. In the cluster 16 cold storages are MIDH assisted.

Table 70: Cold storages in Cluster 1 (Andhra Pradesh)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charge Rs./Kg	Single/ Multi commodity	Need for upgradation / Demand for new Infrastructure	Constraints/ Gaps with its reason
16 nos (MIDH assisted) with 97,500 MT capacity	GRAND NAINA	Depending upon season	180	Majorly privately owned	Multi commodity	No	Due to perishable nature of banana & current banana trading practices cold storage has not got much use

Source: Stakeholder consultations, DDH Office and GT Analysis

f. Ripening Chamber

In the cluster, 41 MIDH assisted ripening chambers are available. There are multiple ripening chambers as well who have not taken any subsidy as well as at household level. These include Anantapur (12 nos. – 779 MT) and Kadapa (29 nos.-985 MT).

Table 71: Ripening Chambers in Cluster 1 (Andhra Pradesh)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Usage Charge Rs./Kg	Single/ Multi commodity	Need for upgradation / Demand for new Infrastructure	Constraints/ Gaps with its reason
41 in no with 1764 MT capacity	GRAND NAINA	85%	As per prevailing demand	INR 2	Multi commodity	Yes	Electricity requirement is high

Source: Stakeholder consultations, DDH Office and GT Analysis

g. Quality Assessment

There are no quality testing labs in the cluster. For domestic selling it is not required. Soil testing labs should be there in the cluster.

6.4. Logistics, Marketing and Branding

a. Mode of transport – Domestic market

Table 72: Mode of Transportation for domestic market Cluster 1 (Andhra Pradesh)

Mode of Transport	Major market (location)	INR/Kg/ Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Small truck/ Vehicles	For short distances	1-2	3-6	1-2% due to less distance	Good scope of rail
Truck	Maharashtra Tamil Nadu Delhi	Maharashtra - 3	12-48	5-10%	Good scope of rail in every districts of the cluster

		Telangana and Tamil Nadu – 2 Delhi – 7			
Train	Delhi	Delhi – 5.5 (excluding subsidy) Mumbai – 2.5	24-48	<5%	Seasonality is an issue

Source: Stakeholder consultations, GT Analysis

More than 99% of banana trade from the cluster is taking place through roadways. Rail transport is newer and available through Kisan Rail service via Tadipatri Junction in Anantapur district. Through this service, a volume of 45000 MT has been transported to Delhi (35000 MT) and Mumbai (10000 MT) over the past year.

b. Mode of transport – Export Market

Table 73: Mode of Transportation for export market Cluster 1 (Andhra Pradesh)

Mode of Transport	Major market (location)	Cost - INR/Kg/Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Sea	Iran, Oman, Saudi Arabia, UAE	20	10-14 days	<5%	Long distance not possible (Europe)

Source: Stakeholder Consultations, GT Analysis

In order to export bananas, JNPT port in Maharashtra is the major exit point. Visakhapatnam port though in close proximity is seldom used. The major banana importing nations from India are in the Middle East for which dispatch through JNPT port is more viable. The current practice is of booking reefer containers at JNPT Port, which on arrival is filled with banana in carton boxes. The reefer container then is shipped back to JNPT port and from there the container is exported to the importing nation.

c. Marketing Channels

Table 74: Mode of Transportation for export market Cluster 1 (Andhra Pradesh)

Marketing Channel	Percentage of farmers covered	Volume of crop handled (in MT)	Selling Price Rs./Kg	Constraints/Gaps with its reason
Commission Agent	85-90%	21 Lakh	9-10	Traders high in number
FPO/Cooperative	5-10%	1.17 Lakh	10-11	Trader's control maximum trader
Organised trade (Mother Dairy/Desai/ etc.)	3-4%	0.77 Lakh	12-14	Very low organized trade
Exports	1-2%	42935 MT	13-15	-

Source: Stakeholder consultations

d. Current Branding & Marketing Initiatives

Currently, Andhra Pradesh has no GI tag credentials. Branding initiatives are also non-existent. There are other companies of other states who buy banana from the cluster and brand as their own banana. The companies like Desai Fruits ('Happy' banana brand) and INI Farms ('Kimaye' brand) are procuring banana from the cluster. Banana of the cluster are being exported through JNPT. State government runs special reefer wagon trains which plies between Tadipatri railway station, Anantapur and JNPT port. This initiative

has supported exporters as transportation price decreased. This will encourage exporters to export more from the region and farmers income will increase

6.5. Proposed Interventions and Recommendations

Table 75: Indicative proposed interventions for Andhra Pradesh Cluster 1 (Anantapur and Kadapa)

Value Chain Activity	Gaps Identified	Recommended Component	Numbers (A)	Units (MT, Ha, Sqm, set, participant etc.)	Indicative Per unit cost in INR Lakh (B)	Indicative Cost of Interventions C=A*B (INR Cr.)
Pre-Production and Production	Lack of Tissue Culture Planting Material	Tissue Culture Lab with Virus Indexing Facility & Hardening Center	8	With total capacity of 2 Cr. plants every year	250	20
	Limited number of Farmer Producer Organisations (FPOs)	FPOs formation and promotion	12	500 participants each	43	5.16
	Lack of GAP certification	Promotion and training of GAP	20502	Hectare	0.1	20.50
	Limited quality testing at farm and post farm levels	MRL, pesticide residue testing lab	1	Number	400	4
		Basic testing facility at district level	2	Number	25	0.5
Post-harvest Management and Value Addition	Dependency on traders for ripening	Ripening Chambers	13	10 MT each	10	1.3
	Lack of export promotion infrastructure	APEDA certified packhouse	1	At least 200 MT storage	400	4
	Limited sorting, grading and packaging facilities	Collection centres	0	50 MT per day handling capacity	4	0
	Limited processing	Banana Chips/ Flour manufacturing unit	2	10 MT/day input capacity	75	1.5
		Banana Puree manufacturing unit	1	96 MT/day	1500	15
	Injury and loss due to improper packaging practices	Adoption of crates	300000	20 kg each	0.003	9
Logistics, Marketing and Branding	Limited availability of refrigerated transport	Refrigerated Transport Vehicles	42	15 MT reefer vans	30	12.6
	Limited branding, no GI tag	Branding, Digital and Marketing Initiatives	1	Approx. 10% investment towards holistic development of cluster brand	9.36	9.36
Total	-	-		-	-	102.92

Source: GT Analysis

Cluster 2 – East Godavari, West Godavari, Vizianagaram, Vishakhapatnam

7.1. Current Scenario

a. Production of banana in the cluster (EG, WG, Vizianagaram, Vishakhapatnam)

The area under banana cultivation in the cluster increased at CAGR of 1.26% in the period of 2015-20. During the same time the production of banana in the cluster increased at CAGR of 11%. The reason can be attributed to tissue culture adoption.

Table 76: Area, Production & Productivity of Banana (Andhra Pradesh Cluster-2) (2015-16 – 2019-20)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
1.	2019-20	36.08	2.16
2.	2018-19	42.12	2.53
3.	2017-18	37.16	2.23
4.	2016-17	40.06	2.12
5.	2015-16	33.88	1.28

Source: AP Horticulture Department, GT Analysis

b. Marketable surplus

This is the cluster where local varieties are more prevalent. These include Chakrakeli and Karpura which have received GI tags as well. Prevalence of Amrutpani and Grand Naine variety was found to be limited in the cluster. Due to this most of the produce of the cluster remains and consumed within the state. From stakeholder consultation it was found that 60% of the produce remains within the state and only 40% of the produce goes outside the state. Here, APMC based trade is still prevalent and significant. Farmers are getting better price at APMC rather than from local traders. This is understood as local traders are selling the banana within the state itself where they won't be getting good price. At APMC auctioning takes place due to which farmer earns better price.

Table 77: Marketable surplus of banana in the cluster (Andhra Pradesh Cluster-2)

S.No.	Varieties	Grade wise marketable surplus (in Mn MT)	Total marketable surplus (in Mn MT)	Shelf life
1	GRAND NAINE (33%)	0.27	0.82 lakhs	7-10 days
2	Karpura (38%)	0.31		7-10 days
3	Others (29%) – Chakrakeli, Red Chakrakeli, Amruthpani, and other local varieties	0.24		7-10 days
Figures in % mean out of total production of banana in the cluster				

Source: Stakeholder consultations, GT Analysis

c. % losses of banana in cluster

From pre-production to retail the level of losses observed in the state of Andhra Pradesh has been observed to be 23.6%. Maximum losses are observed at retail level wherein percentage losses are to the tune of 9.4% followed by losses during storage and ripening (4.8%), losses during assembly market/ wholesale market (3.9%), losses at field level (2.9%) and losses during transport (1.6%)⁷.

d. FPOs/Growers Association/Producer Groups (PGs) etc. involved in banana cultivation in cluster

There are 4 FPOs in the cluster which are dealing in banana of which 2 each are in East and West Godavari districts. They help farmers in terms of providing inputs at cheaper cost. Mandals in the cluster

⁷ All India coordinate research project trial, derived from NRCB report

which are major banana producers are Payakaraopeta, Golugonda and S. Rayavaram in Visakhapatnam district. In East Godavari district, the major blocks are Atreyapuram, Ravulapalem and Kothapeta.

e. Domestic transfer of banana from the cluster

The major states receiving banana from the cluster are East Indian states such as Odisha and, West Bengal which form a large portion of the offtake of banana moving out of the state. Amongst others, Karnataka is a major consumer as well along with Tamil Nadu and the neighbouring state of Telangana. Local varieties of the cluster are also being sold to another states. These include Amrutpani, Red Chakrakeli and Karpura.

Table 78: Major domestic destinations for transfer of banana (Andhra Pradesh Cluster-2)

S. No.	States	State wise value (in INR)	State-wise mode of Transport (sea, air etc. to be specified)	Cost of transport (INR/kg)
1.	Odisha	108 Crore	Road	3
2.	West Bengal	236 Crore	Road	5.25
3.	Karnataka	158 Crore	Road	3.5
4.	Tamil Nadu	187 Crore	Road	2.25
5.	Telangana	92 Crore	Road	2.25

Source: Stakeholder consultations, GT Analysis

f. Prices with respect to grade/count/size/packaging

Table 79: Average prices with respect to grade/count/size/packaging/etc.

S.No.	Year	Based on grade	Market based (domestic/ export/ processing) – Price in Rs./kg
1.	2020-21	Grade A – 12 Grade B – 9 Grade C – 5-6	Domestic – 11 Export – 13-15
2.	2018-19	Grade A – 9 Grade B – 7 Grade C - 4	Domestic – 10 Export – 12-13

Source: Stakeholder consultations, GT Analysis

7.2. Pre-Production and Production

a. Production of planting material

There are no tissue culture laboratories in the cluster. The demand of tissue culture plants is being fulfilled by laboratories in other districts of Anantapur, Chittoor and Guntur as well as through private sector suppliers such as Jain Irrigation and Nagarjuna Fertilizers.

b. Irrigation mechanism adopted

There is low adoption of drip irrigation in the cluster. Farmer are dependent on surface water available in the cluster.

Table 80: Irrigation scenario in banana cultivation (Andhra Pradesh Cluster-2)

Type of Irrigation method	% of farmers using the irrigation method	Cost with irrigation method / Ha)	Source of Water	Cluster requirement (in numbers/ area/ capacity etc.)	Gaps in irrigation mechanism
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Micro	15-20%	8-10% of total cost incurred in cultivation	Ground Water/River water	No	_____
Others (includes wells, borewells, renting from others)	85-90%	8-10% of total cost incurred in cultivation	Ground water/River water	No	-----

Source: Stakeholder consultations, GT Analysis

c. GAP and fruit care activities

Limited exports are taking place from the cluster. Adoption is low as exports as proportion of total production is negligible. Also, availing such certifications and training is deemed expensive by the grower which is a major reason for low adoption. Small number of farmers are involved in GAP and fruit care practices.

d. Harvesting practices

As is the case in whole of India harvesting is done manually in the cluster. Being a robust fruit, this is carried on shoulders to the outside of field and then either primary processing is done, or bunch is directly loaded into the vehicle.

7.3. Post-Harvest Management and Value Addition

a. Packhouse

Packhouses in the cluster are multi commodity in nature. These are primarily utilised for storage of mango and/or coconut. The state government is actively promoting on field storage. Out of total 1036 packhouses, East Godavari has the highest number with 422 nos, followed by West Godavari and Vizianagaram with 279 and 206 nos. respectively. The details of packhouses are as follows:

Table 81: Details of packhouses in Cluster 2 (Andhra Pradesh)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs./Kg	Single/ Multi commodity	Need for upgradation/ Demand for new Infrastructure	Constraints/ Gaps with its reason
1036 packhouses having total capacity of 31080 MT	GRAND NAINA and all other local varieties	Depending upon season (peak season of utilization is November to April)	180	Government packhouse can be used free of charge	Multi commodity	Yes	Primary Processing is done at field itself which has reduced reliance on packhouses

Source: Stakeholder consultations, GT Analysis

b. Integrated Packhouse

There are 3 integrated packhouses in the cluster. Except Vizianagaram, all the other districts have 1 integrated packhouse each. These integrated packhouses are used for exports purpose.

Table 82: Details of integrated packhouse in Cluster 2 (Andhra Pradesh)

Capacity	Variety/ Grade process ed	Capacit y Utilisati on	No. of days of operati on	Lease Charges Rs./Kg	Single/ Multi commodi ty	Need for upgradatio n/ Demand for new Infrastruct ure	Constrain ts/ Gaps with its reason
3 integrated packhouses in the cluster, 1 at each district except Vizag	GRAND NAINA and other local varieties	Depending upon season (peak season of utilization is November to April)	180	Government built integrate packhouse are given at rent to interested party	Multi commodity	No	Used mostly by exporter

Source: Stakeholder consultations, GT Analysis

c. Reefer vans

Under the state government program, reefer vans are given on subsidy to farmers. Till date 3 has been given on subsidy to farmers of the cluster. The capacity of those vans is 20 MT each. These vans are utilised by farmers to sell their fresh produce in nearby markets. Moreover, exports are minimal from this cluster.

d. Processing units

Processing of fresh banana is not taking place throughout the cluster. There are no chips/flour making units of banana. Processing of banana need to be promoted as during the time of pandemic bananas were selling at as low price as INR 1 per kg. Value addition will lead to increase in shelf life.

e. Cold Storage

Cold storages are not required for banana. If banana is kept in cold storage, then it will begin ripening. Moreover, a farmer cannot pre-cut and keep it in cold storage and then sell later when price is high. Current trade practices restrict farmer from doing that. Fresh banana right from field are bought. In the cluster 32 cold storages are MIDH assisted. Rest there are many cold storages who have not taken any subsidy. Number of such cold storages can be just estimated like 10 more in each district can be assumed of capacity more than 100 MT.

Table 83: Cold storages in Cluster 2 (Andhra Pradesh)

Capacit y	Variety/ Grade processe d	Capacity Utilisatio n	No. of days of operatio n	Lease Charge s Rs. /Kg	Single/ Multi commodit y	Need for upgradation / Demand for new Infrastructu re	Constraint s/ Gaps with its reason
32 in no (MIDH assisted) with 125850 MT capacity	Only used for export which is already in integrated packhouse	Dependin g upon season	180	Majorly private ly owned	Multi commodity	No	Due to perishable nature of banana & current banana trading

							practices cold storages have limited utility
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Source: Stakeholder consultations, GT Analysis

f. Ripening Chamber

There are 19 ripening chambers in the cluster which include Vizianagaram (6 nos.), Vishakhapatnam (6 nos), East Godavari (4 nos.) and West Godavari (3 nos.) with a combined capacity of 5816 MT.

Table 84: Ripening Chambers in the cluster (Andhra Pradesh Cluster-2)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Usage Charges Rs./Kg	Single/ Multi commodity	Need for upgradation / Demand for new Infrastructure	Constraints/ Gaps with its reason
19 in no with 5816 MT capacity	GRAND NAINA	As per prevailing demand	As per prevailing demand	INR 2	Multi commodity	No	Electricity requirement is high

Source: Stakeholder consultations, GT Analysis

g. Quality assessment

There are no quality assessment labs in the cluster. For domestic trade testing is not required. But in each cluster, there should be soil testing lab and other relevant labs which are required for making sure quality of banana being grown is fulfilling the all the requirements.

7.4. Logistics, Marketing and Branding

a. Mode of transport – Domestic market

Road transport is the prevalent transport. More than 99% of banana trade happening through road only. Rail transport is not available in the cluster. Primary processing is either done at field itself or directly bunches are loaded in the vehicle. For small distance small vehicles like Tata Ace, Tata Ape or others are used. For large distance large vehicle of 10 MT capacity are used.

Table 85: Mode of transportation for domestic market (Andhra Pradesh Cluster-2)

Mode of Transport	Major market (location)	INR/Kg/ Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Small truck/ Vehicles	For short distances	1-2	3-6	1-2% due to less distance	Good scope of rail
Truck	Odisha West Bengal Karnataka Tamil Nadu Telangana			5-10%	Good scope of rail in every districts of the cluster

Source: Stakeholder consultations and government data

b. Mode of transport – Export Market

Exports are minimal from the cluster. Instead, the maximum exports is taking place from Anantapur cluster.

c. Marketing Channels

Maximum trading is happening through local traders. Organized trade is limited. FPO and cooperatives are acting as a trader. Direct selling to market is not being practiced by FPOs. They just act as an intermediary.

Table 86: Marketing channel details of the Cluster 2 (Andhra Pradesh)

Marketing Channel	Percentage of farmers covered	Volume of crop handled (in MT)	Selling Price INR/ Kg	Constraints/Gaps with its reason
Commission Agent	85-90%	19.44 Lakh	9-10	Traders high in number
FPO/Cooperative	5-10%	1.51 Lakh	10-11	Trader's control maximum trader
Organise trade (Mother Dairy/Desai)	3-4%	0.64 Lakh	12-14	Very low organized trade

Source: Stakeholder consultations, GT Analysis

d. Current Branding & Marketing Initiatives

Currently, Andhra Pradesh has no GI tag for its banana. Branding initiative also is also not existent. There are other companies of other states who buy banana from here and brand as their own banana. Private companies like Desai Fruits and INI Farms are less active in the cluster. They operate more from Anantapur cluster. This branding awareness will lead to brand recall and demand of banana from the cluster will increase. This will lead to increase in farmers of income.

7.5. Proposed Interventions and Recommendations

Table 87: Indicative proposed interventions for Andhra Pradesh Cluster 2 (EG, WG, Vizianagaram, Vishakhapatnam)

Value Chain Activity	Gaps Identified	Recommended Component	Numbers (A)	Units (MT, Ha, Sqm, set, participant etc.)	Indicative Per unit cost in INR Lakh (B)	Indicative Cost of Interventions C=A*B (INR Cr.)
Pre-Production and Production	Lack of Tissue Culture Planting Material	Tissue Culture Lab with Virus Indexing Facility & Hardening Center	9	With total capacity of 2.25 Cr. plants every year	250	22.5
	Limited number of Farmer Producer Organisations (FPOs)	FPOs formation and promotion	18	500 participants each	43	7.74
	Lack of GAP certification	Promotion and training of GAP	18039	Hectare	0.1	18.04
	Limited quality testing at farm and post farm levels	MRL, pesticide residue testing lab	1	Number	400	4
		Basic testing facility at district level	4	Number	25	1
Post-harvest Management and Value Addition	Dependency on traders for ripening	Ripening Chambers	3	10 MT each	10	0.3
	Lack of export promotion infrastructure	APEDA certified packhouse	1	At least 200 MT storage	400	4
	Limited sorting, grading and packaging facilities	Collection centres	0	50 MT per day handling capacity	4	0
	Limited processing	Banana Chips/ Flour manufacturing unit	3	10 MT/day input capacity	75	2.25
		Banana Puree manufacturing unit	1	96 MT/day	1500	15
	Injury and loss due to improper packaging practices	Adoption of crates	250000	20 kg each	0.003	7.5
Logistics, Marketing and Branding	Limited availability of refrigerated transport	Refrigerated Transport Vehicles	10	15 MT reefer vans	30	3
	Limited branding, no GI tag	Branding, Digital and Marketing Initiatives	1	Approx. 10% investment towards holistic development of cluster brand	8.53	8.53
Total	-	-	-	-	-	93.86

Source: GT Analysis, Note: Bonta variety may be promoted for processing purposes.

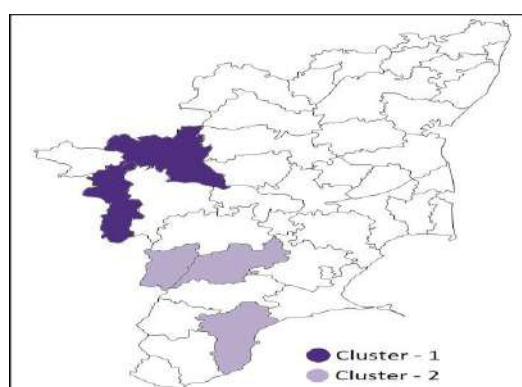
Key Cluster Wise Findings - Tamil Nadu

Cluster 1 – Erode and Coimbatore

8.1. Current Scenario

Tamil Nadu Horticulture Development Agency was set up in Tamil Nadu to look after development of horticulture in the region. TANHODA has been registered as a Society under Tamil Nadu Societies Registration Act, 1975. TANHODA is a "Special Purpose Vehicle" for the purpose of implementing schemes like MIDH - National Horticulture Mission, PMKSY - Micro Irrigation, TN - IAMWARM, National Bamboo Mission and National Mission on Medicinal plants.

Figure 9: Focal Cluster in Tamil Nadu (5 districts)



Cluster 1 – Erode
and Coimbatore

Cluster – 2
Madurai, Theni
and Thoothukudi

a. Total area and production under horticulture in the state (last 5 years)

Table 88: Total area and production under horticulture in Tamil Nadu (2015-16 to 2019-20)

S. No.	Year	Area (in Lakh Ha)	Production (in Mn MT)
1.	2019-20	14.60	19.32
2.	2018-19	13.62	16.69
3.	2017-18	12.81	14.01
4.	2016-17	13.15	13.59
5.	2015-16	13.33	16.68

Source: Horticulture Department, Government of Tamil Nadu

The area under horticulture crop has increased at a CAGR of 1.84% during the period of 2015-20. During the same period the production has increased at a CAGR of 2.97%.

Table 89: Area, Production & Productivity of Banana in Tamil Nadu (2015-16 to 2019-20)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
1.	2019-20	92.41	3.99
2.	2018-19	87.51	3.48
3.	2017-18	78.48	3.04
4.	2016-17	90.99	3.03
5.	2015-16	94.21	3.6

Source: Horticulture Department, Government of Tamil Nadu

The area under banana cultivation has decreased at a CAGR of 1.39% during the period of 2015-20. During the same period the production has increased at a CAGR of 2%. This is due to increased adoption of Grand Naine. The productivity of Grand Naine is higher than local varieties. Moreover, under tissue culture a single bunch weighing 30 kg is obtained of Grand Naine. Moreover, spacing requirement of Grand Naine is low and 3000 plants can be planted in 1 hectare land. Local varieties require more spacing this again reduces productivity.

b. Production of banana in the cluster

Table 90: Area, Production & Productivity of Banana in Tamil Nadu Cluster-1 (2015-16 to 2019-20)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
1.	2019-20	21.19	0.85
2.	2018-19	18.69	0.85
3.	2017-18	16.90	0.65
4.	2016-17	19.56	0.59
5.	2015-16	20.53	0.87

Source: Horticulture Department, Tamil Nadu

The production of banana in the cluster has increased at a CAGR of 0.63% during the period of 2015-20. During the same time the production has decreased at a CAGR of 1.57%. In this cluster the Grand Naine production is low. In 2015, in Erode, area under banana cultivation was 12335 Ha and area under banana cultivation in Coimbatore was 8199 Ha. In 2020, in Erode the area under banana cultivation increased to 14116 Ha and in Coimbatore it decreased to 7071 Ha. However, the increase in area in Erode was higher than decrease in area in Coimbatore.

c. Marketable Surplus which is sent out of state

Table 91: Marketable Surplus of banana in the cluster (Tamil Nadu Cluster-1)

S. No.		Varieties	Grade wise marketable surplus (in Mn MT)	Total marketable surplus (in Mn MT)	Shelf life
1	Grand Naine (50%)	0.24 lakh	0.48 lakh	10-14 days	
2	Others (50%)	0.24 lakh		10-14 days	
Figures in % mean out of total production of banana in the cluster					

Source: Stakeholder consultation and GT analysis

Out of total production Grand Naine constitutes 50% and other local varieties constitutes another 50%. Around 60% of the production is sent out of state. The major buying states being Karnataka and Kerala. The banana of the cluster is also being sold to Mumbai. In Karnataka the major city buying banana of Tamil Nadu cluster-1 is Mysore and Bangalore and in Kerala the major city buying banana of Tamil Nadu cluster-1 is Cochin. **Losses of banana in the cluster**

From pre-production to retail the level of losses observed in the state of Tamil Nadu has been observed to be 9%. Maximum losses are observed at retail level wherein percentage losses are to the tune of 3.9% followed by losses during storage and ripening (2.1%), losses during assembly market/ wholesale market (1.9%), losses at field level (0.8%) and losses during transport (0.3%)⁸. The losses are comparatively lower as very less production of Tamil Nadu is sent too far to other states. Moreover inter district transport is also less as every district is significant producer of banana which can cater to the local demand.

d. Total no. of FPOs/Growers Association/Producer Groups (PGs) etc. involved in banana cultivation in cluster

There are 6 FPOs in the cluster which are dealing in banana. They provide inputs to farmers at cheaper cost and after the arrival of FPO ecosystem, farmer income has improved. FPOs can make investment in

⁸ All India coordinate research project trial, derived from NRCB report

ripening chamber so that they are able to sell ripened banana themselves. Thangamam FPO is one of the major FPO dealing in banana in Coimbatore. Another are Ranganathan FPO, Vellangiri FPO and Aurnamalai FPO in Coimbatore. Major blocks which are producing banana are Karamadai, Chondumuthur and Annur block in Coimbatore and Sathyamangalam, Gobichettipalayam, Bhavanisagar, Bhavani, TN Palayam and Anthiyur in Erode district.

e. Export value and quantity of the focus crop (last 5 years)

Table 92: Major domestic destinations for transfer of banana (from Tamil Nadu Cluster-1)

S. No.	States	State wise value (in INR)	State-wise mode of Transport (sea, air etc. to be specified)	Cost of transport (INR/kg)
1.	Mysore, Karnataka	296 Crore.	Road	1-1.5
2.	Cochin, Kerala	162 Crore	Road	1-1.5

Source: Stakeholder consultations at DDH office, GT Analysis

f. Prices with respect to grade/count/size/packaging

Table 93: Average Prices with respect to grade/count/size/packaging etc.

S.No.	Year	Based on grade	Market based (domestic/ export/ processing) – Price in Rs./ kg	Price based on packaging (wooden/corrugated box/punnet etc.) in Rs./kg
1.	2020-21	Grade A – 10 Grade B – 8 Grade C – 5	Domestic – 10 Export – 13-15	Wooden – 42 Corrugated Box – 45
2.	2018-19	Grade A – 9 Grade B – 7 Grade C - 4	Domestic – 9 Export – 12-13	Wooden – 38 Corrugated Box - 42

Source: Stakeholder Consultations, GT Analysis

For good quality of banana farmer obtain good price. For that farmer need to adopt fruit care practices and good agricultural practices. This will improve income of farmers. There should be a good focus of government on making fruit care practices adoption a norm in India. Large farmers are able to bear the cost, small farmers need to be supported for that. The prices mentioned in the table are of GRAND NAIN variety. For other varieties the price obtained by farmer are different. For Red banana, it is 15-18 INR/kg and for Nendran Nadu it is 18-20 INR/kg.

g. Status of protection and promotion initiatives for GI, Trademarks, etc.

Presently, Tamil Nadu has 2 GI tags for its banana. These are for varieties of Virupakshi Hill and Sirumalai Hill. However, there is negligible branding initiatives for banana in the cluster.

8.2. Pre-Production and Production

a. Production of planting material

There are 2 tissue culture labs in the cluster. Their cumulative capacity is to produce 23 lakh saplings in a year. Among the 2 companies SPIC Agro Tech is major supplier of tissue cultured plants in the cluster.

Table 94: Tissue culture labs and material availability in the cluster (Tamil Nadu Cluster-1)

Number of labs	Major varieties (including recently introduced varieties)	Capacity (number of plants/year)	Area coverage (in Ha)	Average Selling Price/Plant	Capacity Utilisation %	Certified /Accredited	Annual demand of cluster (number of plants/year)	Constraints / Gaps with its reasons
2	GRAND NAINA	~23 lakhs	766	15-17	100%	SPIC company is an old company and is DBT approved	2.5 cr	Tissue culture saplings of local varieties need to be also produced

Source: Stakeholder Consultations, GT Analysis

b. Irrigation mechanism adopted

There is good penetration adoption of drip irrigation in the cluster. Around 75-80% of the banana cultivation area is under drip irrigation. Farmers are adopting micro irrigation system to conserve water, to reduce labour cost, to get quality produce. Farmers receive subsidy under PMKSY (Pradhan Mantri Krishi Sinchai Yojana). Department of horticulture is implementing the scheme to render subsidies to farmers to lay out drip system.

Table 95: Overview of micro and drip Irrigation in banana cultivation in Cluster-1 (Tamil Nadu)

Type of Irrigation method	% of farmers using the irrigation method	Cost with irrigation method / Ha)	Source of Water	Cluster requirement (in numbers/ area/ capacity etc.)	Gaps in irrigation mechanism
Micro	75-80%	8-10% of total cost incurred in cultivation	Ground Water/River water	No	_____
Others (includes wells, borewells, renting from others)	20-25%	8-10% of total cost incurred in cultivation	Ground water/River water	No	-----

Source: Stakeholder consultations with farmers and DDH office

c. GAP and fruit care activities

Negligible exports are happening from the districts of the cluster. An FPO named Kazhani Farmer Producer Organisation is exporting banana through Madurai airport to the Middle East market. This is confirmed by the department of horticulture officials of the cluster. GAP and fruit care practices adoption is limited in the cluster.

d. Harvesting practices

As is the case in whole of India harvesting is done manually in the cluster. Being a robust fruit, this is carried on shoulders to the outside of field and then either primary processing is done, or bunch is directly loaded into the vehicle. Normally banana bunches are harvested manually by using sharp knife or nylon rope cutting method. This technique helps in prevention of injuries to fruits during harvest.

8.3. Post-Harvest Management and Value Addition

a. Packhouse

There is a total of 120-130 packhouses in the cluster out of which 60 packhouses are present in the district of Coimbatore and 60-70 nos are in the district of Erode. These packhouses are of 600 sq ft area and the capacity of these packhouses is 1 MT each. They can be used for various horticultural crops for the purpose of preliminary grading, sorting and packaging. These facilities have been built under subsidy from NHM and by farmers as Public-Private-Partnership model.

b. Integrated Packhouse

Since exports are minimal and only one FPO is present in the cluster, which is involved in exports, there is no presence of an integrated packhouse facility in the cluster. The FPO is utilising Madurai airport for exporting agri produce to the Middle East (6 MT/week on average). The FPO has it's own godown facility where produce is stored.

c. Reefer vans

Since exports are minimal, reefer vans are not being used. This is as confirmed by officials of horticulture and agriculture marketing departments at the district level in the cluster.

d. Processing units

In terms of value addition, presently, banana chips are being produced in the cluster by means of numerous household units. There is one such unit in Erode which is being run by an FPO and several such units in Coimbatore which are being run by individuals. Large scale processing is not happening in the cluster.

e. Cold Storage

Six cold storage facilities are available in Erode district with a combined capacity of 1125 MT while there are around 3 facilities in Coimbatore district with a combined capacity of 1500 MT.

Table 96: Cold Storage presence in the Cluster-1 (Tamil Nadu)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs. /Kg	Single/ Multi commodity	Need for upgradation / Demand for new Infrastructure	Constraints/ Gaps with its reason
9 in number with total capacity of 2625 MT	Only used for export which is already in integrated packhouse	80%	300	-	Multi commodity	No	Due to perishable nature of banana & current banana trading practices cold storage has limited utility

Source: Stakeholder consultations, GT Analysis

f. Ripening Chamber

In the cluster, only 1 ripening chamber of 25 MT capacity is available Coimbatore (not assisted by MIDH). Erode has no ripening chamber facilities. The banana is being ripened at household level itself. This is as confirmed by district horticulture department officials.

Table 97: Ripening Chamber availability in Cluster-1 (Tamil Nadu)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Usage Charge Rs./Kg	Single/ Multi commodity	Need for upgradation / Demand for new Infrastructure	Constraint s/ Gaps with its reason
1 in no with 25 MT capacity	GRAND NAINA and other local varieties	90%	300	INR 2	Multi commodity	Yes	Electricity requirement is high

Source: Stakeholder consultations, DDH Office and GT Analysis

g. Quality assessment

In Erode there is one government lab of agriculture department which can test pesticide residue level in banana. In Coimbatore, small level testing is done by Tamil Nadu Agriculture University. This is as confirmed by horticulture officials of the cluster.

8.4. Logistics, Marketing and Branding

a. Mode of transport – Domestic market

Table 98: Mode of Transportation for domestic market (Tamil Nadu Cluster-1)

Mode of Transport	Major market (location)	INR/Kg/ Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Small truck/ Vehicles and Trucks	Supply outside state to Mysore and Cochin at short distance from cluster	1-1.5	3-6	1-2% due to less distance	Good scope for Kisan Rail

Source: Stakeholder consultations and government data, GT Analysis

b. Mode of transport – Export Market

Minimal exports are happening from the cluster. For exports Madurai airport is being used. (Approximately 6 MT per week)

c. Marketing Channels

Table 99: Marketing Channels details from Cluster 1 (Tamil Nadu)

Marketing Channel	Percentage of farmers covered	Volume of crop handled (in MT)	Selling Price Rs./Kg	Constraints/Gaps with its reason
Commission Agent	85-90%	7.6 Lakh	9-10	Traders high in number
FPO/Cooperative	8-9%	0.7 Lakh	10-11	Trader's control maximum trade
Organised trade (Mother Dairy/Desai)	1-2%	0.1 Lakh	12-14	Very low organized trade
Exports	<0.1%	~250 MT	13-15	-

Source: Stakeholder consultations, GT Analysis

d. Current Branding & Marketing Initiatives

Presently Tamil Nadu has 2 GI tag for its bananas. The 2 GI tags are for the varieties of Virupakshi Hill Banana and Sirumalai Hill Banana. However, there is no known brand of banana from the cluster.

8.5. Proposed Interventions and Recommendations

Table 100: Indicative proposed interventions for Tamil Nadu Cluster 1 (Erode and Coimbatore)

Value Chain Activity	Gaps Identified	Recommended Component	Numbers (A)	Units (MT, Ha, Sqm, set, participant etc.)	Indicative Per unit cost in INR Lakh (B)	Indicative Cost of Interventions C=A*B (INR Cr.)
Pre-Production and Production	Lack of Tissue Culture Planting Material	Tissue Culture Lab with Virus Indexing Facility & Hardening Center	5	With total capacity of 1.25 Cr. plants every year	250	12.5
	Limited number of Farmer Producer Organisations (FPOs)	FPOs formation and promotion	12	500 participants each	43	5.16
	Lack of GAP certification	Promotion and training of GAP	10594	Hectare	0.1	10.59
	Limited quality testing at farm and post farm levels	MRL, pesticide residue testing lab	1	Number	400	4
		Basic testing facility at district level	2	Number	25	0.5
Post-harvest Management and Value Addition	Dependency on traders for ripening	Ripening Chambers	24	50 MT each	50	12
	Lack of export promotion infrastructure	APEDA certified packhouse	1	At least 200 MT storage	400	4
	Limited sorting, grading and packaging facilities	Collection centres	38	50 MT per day handling capacity	4	1.52
	Limited processing	Banana Chips/ Flour manufacturing unit	2	10 MT/day input capacity	75	1.5
		Banana Puree manufacturing unit	1	96 MT/day	1500	15
	Injury and loss due to improper packaging practices	Adoption of crates	100000	20 kg each	0.003	3
Logistics, Marketing and Branding	Limited availability of refrigerated transport	Refrigerated Transport Vehicles	10	15 MT reefer vans	30	3
	Limited branding, no GI tag	Branding, Digital and Marketing Initiatives	1	Approx. 2% investment towards holistic development of cluster brand	1.46	1.46
Total	-	-		-	-	74.23

Source: GT Analysis. Note: In Tamil Nadu, traditional varieties may be preferred for banana chips processing. (Virupaksha, Sirumalai and Nendran)

Cluster 2 – Theni, Tuticorin and Madurai

9.1. Current Scenario

a. Production of banana in the cluster

The area under banana cultivation in the cluster has remained almost constant at 17000 Ha during the period of 2015-20. But the production has increased at a CAGR of 5.4%. Theni is developed as a major cluster for banana in Tamil Nadu. In 2015 the area under banana cultivation was 9379 Ha in Thootukudi, 5914 Ha in Theni and 2512 Ha in Madurai. In 2020 the area under banana cultivation decreased in Thootukudi to 8237 Ha, in Theni it increased to 5966 Ha and in Madurai it increased to 3003 Ha.

In Tamil Nadu the crop is available throughout the year. During some time, the prices are high and during some time prices are low. This gap needs to be addressed. This is unlike in other states where there is a particular season of maximum trading of banana and fluctuation is comparatively less compared to Tamil Nadu. The state need to sync the banana production with market. Excessive usage of chemical and fertilizer is increasing cost of farmers and damaging quality of banana. GAP adoption, fruit care practices adoption is the need of the hour in the cluster.

Table 101: Area, Production & Productivity of Banana in Cluster-2 (Tamil Nadu) (2015-16 to 2019-20)

S. No.	Year	Area (in '000 Ha)	Production (in Mn MT)
1.	2019-20	17.216	0.97
2.	2018-19	16.97	0.89
3.	2017-18	15.71	0.79
4.	2016-17	18.07	0.74
5.	2015-16	17.80	0.75

Source: Horticulture Department, Government of Tamil Nadu

b. Marketable Surplus

Table 102: Marketable Surplus of banana in Cluster-2 (Tamil Nadu)

S.No.		Varieties	Grade wise marketable surplus (in Mn MT)	Total marketable surplus (in Mn MT)	Shelf life
1.	Grand Naine (33%)		0.12 lakh	0.35 lakh	10-14 days
2.	Others (67%) – Nendran, Red banana, Karpuravalli, Poovan, Rasthali		0.23 lakh		10-14 days
Figures in % mean out of total production of banana in the cluster					

Source: Primary survey

Cluster-2 being in close proximate to Kerala state, the banana of the cluster is being sent to all cities of Kerala state through lorries and small trucks. Harvested banana bunches are transported from farmers field to storage godowns of wholesale Agents/ wholesale banana traders through light vehicle like Pick up, TATA Ape, TATA ace, just to avoid injuries to harvested produces. After auction for produce take place, the banana bunches are separated into hands, then hands are graded and sorted out. 13 kg of graded banana hands are packed in plastic grades covering with newspaper on bottom and top and loaded in trucks like Eicher, 407 lorry to transport to Chennai, other cities of the Tamil Nadu and Kerala.

c. Losses of banana in the cluster

From pre-production to retail the level of losses observed in the state of Tamil Nadu has been observed to be 9%. Maximum losses are observed at retail level wherein percentage losses are to the tune of 3.9% followed by losses during storage and ripening (2.1%), losses during assembly market/ wholesale market (1.9%), losses at field level (0.8%) and losses during transport (0.3%)⁹. The losses are comparatively

⁹ All India coordinate research project trial, derived from NRCB report

lower as very less production of Tamil Nadu is sent too far to other states. Moreover inter district transport is also less as every district is significant producer of banana which can cater to the local demand.

d. Total no. of FPOs/Growers Association/Producer Groups (PGs) etc. involved in banana cultivation in cluster

There are 7 FPOs in the cluster which are dealing in banana out of which 5 nos. are in Theni and 2 nos. are in Madurai. Among these FPOs, the major ones include Banana Farmers Association in Theni which has 150 banana growers as its members and the other major FPO is Tamil Nadu Banana Producers Federation in Theni which has around 60 members. 2 FPCs of Madurai which dealing in banana are Thirumangalam and Azhagarmalaiyan FPC. These FPCs provide inputs to farmers at cheaper cost and after the coming of FPOs farmers income has increased and FPO members earn more than regular farmers. Major block producing banana in the cluster are Cumbum, Chinnamanu, Uthamapalayam and Periyakulam in Theni.

e. Domestic transfer of banana from the cluster

Table 103: Major domestic destinations for transfer of banana (from Tamil Nadu Cluster-2)

S. No.	States	State wise value (in INR)	State-wise mode of Transport (sea, air etc. to be specified)	Cost of transport (INR/kg)
1.	Cochin, Kerala	351 Crore	Road	1-1.5

Source: Stakeholder consultations and GT Analysis

The state mentioned in the table is major banana buyer of the cluster. This cluster is geographically situated in southern Tamil Nadu and far from other states. Only Kerala is the major buyer of the banana of this cluster.

f. Prices with respect to grade/count/size/packaging

Table 104: Average Prices with respect to grade/count/size/packaging etc.

S.No.	Year	Based on grade	Market based (domestic/export/ processing) – Price in Rs./ kg	Price based on packaging (wooden/corrugated box/punnet etc.) in Rs./kg
1.	2020-21	Grade A – 10 Grade B – 8 Grade C – 5	Domestic – 10 Export – 13-15	Wooden – 42 Corrugated Box – 45
2.	2018-19	Grade A – 9 Grade B – 7 Grade C - 4	Domestic – 9 Export – 12-13	Wooden – 38 Corrugated Box - 42

Source: Stakeholder consultations and GT Analysis

For good quality of banana farmer obtain good price. For that farmer need to adopt fruit care practices and good agricultural practices. This will improve income of farmers. There should be a good focus of government on making fruit care practices adoption a norm in India. Large farmers are able to bear the cost, small farmers need to be supported for that. The prices mentioned in the table are of Grand Naine variety. For other varieties the price obtained by farmer are different. For Red banana it is 15-18 INR/kg and for Nendran Nadu it is 18-20 INR/kg. Other varieties also trade in range of 12-20 INR/kg.

g. Status of protection and promotion initiatives for GI, Trademarks, etc.

Presently, Tamil Nadu has 2 GI tags for its banana. The 2 GI tags are for the varieties of Virupakshi Hill Banana and Sirumalai Hill Banana . However, there is no known brand of banana from the cluster.

9.2. Pre-Production and Production

a. Production of planting material

In the cluster there are no tissue culture labs and the requirement is being fulfilled through neighbouring districts. There are 6 private nurseries which buy saplings from the companies of Hosur or districts of cluster-1 of Tamil Nadu. In House three are 7 DBT approved tissue culture labs. The annual requirement of tissue culture plants comes to be around 2 crores but of different varieties (not only of Grand Naine).

b. Irrigation mechanism adopted

Table 105: Overview of micro and drip irrigation in banana cultivation in the cluster-2 (Tamil Nadu)

Type of Irrigation method	% of farmers using the irrigation method	Cost with irrigation method / Ha)	Source of Water	Cluster requirement (in numbers/ area/ capacity etc.)	Gaps in irrigation mechanism
Micro	75-80%	8-10% of total cost incurred in cultivation	Ground Water/River water	No	_____
Others (includes wells, borewells, renting from others)	15-20%	8-10% of total cost incurred in cultivation	Ground water/River water	No	-----

Source: Stakeholder Consultations, GT Analysis

c. GAP and fruit care activities

Significant exports are happening from the cluster. There are merchants in the cluster who send 1 MT of perishable commodities on daily basis to Kerala for onward export to GCC nations. Theni cultivates good volume of Grand Naine which is export friendly. For exports to these nations, GAP certificate is also not necessary which is an enabling factor. From Tamil Nadu, the exports of banana in 2020-21 stood at approximately 7000 MT. Kerala seaports and airports are also used by Tamil Nadu for exports.

d. Harvesting practices

As is the case in whole of India harvesting is done manually in the cluster. Being a robust fruit, this is carried on shoulders to the outside of field and then either primary processing is done, or bunch is directly loaded into the vehicle.

9.3. Post-Harvest Management and Value Addition

a. Packhouse

At farmers field level, 5 small size pack houses with capacity of below one MT for an area of 600 sq.feet each are available in Chinnamanur, Cumbum and Uthamapalayam blocks to meet the preliminary grading and sorting purpose of harvested banana bunches¹⁰. These structures were established with subsidies from NHM by farmers as Public-Private Partnership.

Table 106: Details of packhouses in the Cluster-2 (Tamil Nadu)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs./ kg	Single/ Multi commodity	Need for upgradation/ Demand for new	Constraints/ Gaps with its reason
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¹⁰ Data only available for Theni district. Data from Madurai and Thoothukudi not received.

						Infrastructure	
5 packhouses	Grand Naine and all other local varieties	Depending upon season	360	Government packhouse can be used free of charge	Multi commodity	Yes	Primary Processing is done at field itself which has reduced reliance on packhouses

Source: Stakeholder consultation and GT Analysis

b. Integrated Packhouse

Presently, 2 private Integrated Banana Pack houses and 3 Government owned Integrated banana pack house is in operation to run export business¹¹. The Primary Processing Centres set up by Dept of Agriculture Marketing at Chinnamanur (1 nos), Cumbum (2 nos) are utilized by private exporters on annual lease basis. At the time of peak period of international banana trade, the present capacity available to process banana is not adequate. Presently, Public owned Integrated Banana Pack house is in construction at Chinnamanur at an outlay of INR 537.50 lakhs through MIDH- NHM scheme with capacity of 50 MT daily.

Table 107: Integrated packhouse facilities in the Cluster-2 (Tamil Nadu)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs./Kg	Single/ Multi commodity	Need for upgradation/ Demand for new Infrastructure	Constraints/ Gaps with its reason
5 integrated packhouses in the cluster	Grand Naine and other local varieties	Depending upon season	360	Government built integrate packhouse are given at rent to interested party	Multi commodity	No	Used mostly by exporter

Source: Stakeholder consultation, DDH office and GT Analysis

c. Reefer vans

Currently, reefer vans are not being used. However, looking at the quantum of exports the vans can be provided at subsidy to exporters. This will help in enhancing exports and maintaining the quality.

d. Processing units

There is no organised player making chips in the cluster. In the state there are many household units which makes chips and sell it to the customer on demand. Customers in Tamil Nadu prefer freshly baked banana chips. Nendran variety is used for making chips.

e. Cold Storage

Cold storages are not required for banana. If banana is kept in cold storage, then it will begin ripening and will suffer from chilling injuries also. Moreover, a farmer cannot pre-cut and keep it in cold storage and

¹¹ Data available only for Theni. Data for Madurai and Thoothukudi not received

then sell later when price is high. Current trade practices restrict farmer from doing that. Fresh banana right from field are bought. In the cluster 14 cold storages are in Theni¹².

Table 108: Cold storage presence in the Cluster-2 (Tamil Nadu)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Lease Charges Rs./Kg	Single/ Multi commodity	Need for upgradation / Demand for new Infrastructure	Constraints/ Gaps with its reason
14 in number with capacity of 13180 MT	Only used for export which is already in integrated packhouse	Depending upon season	180	Majorly privately owned	Multi commodity	No	Due to perishable nature of banana & current banana trading practices cold storage not got any use

Source: Stakeholder consultation, DDH Office and GT analysis

f. Ripening Chamber

Ripening Chambers are required for banana. If banana is not ripened in Ripening Chamber, then it does not develop yellow color. A naturally ripened banana remains green in color which may remain unsold in market, as a layman will think that it is unripe. In the cluster 5 ripening chamber which are MIDH assisted in Theni. There are chambers which are at household level.

Table 109: Ripening Chamber in Cluster-2 (Tamil Nadu)

Capacity	Variety/ Grade processed	Capacity Utilisation	No. of days of operation	Usage Charges Rs./Kg	Single/ Multi commodity	Need for upgradation / Demand for new Infrastructure	Constraints/ Gaps with its reason
5 in no with 895 MT capacity	GRAND NAINA and other local varieties	As per prevailing demand	As per prevailing demand	INR 2	Multi commodity	Yes	Electricity requirement is high

Source: Stakeholder Consultation, DDH Office and GT analysis

g. Quality assessment

There are no quality assessment labs in the cluster. For domestic trade testing is not required. But in each cluster, there should be soil testing lab and other relevant labs which are required for making sure quality of banana being grown is fulfilling the all the requirements.

9.4. Logistics, Marketing and Branding

a. Mode of transport – Domestic market

¹² Data available only for Theni district. Data for Madurai and Thootukudi not available.

Table 110: Mode of Transportation for domestic market Cluster 2 (Tamil Nadu)

Mode of Transport	Major market (location)	INR/Kg/ Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Small truck/ Vehicles and Trucks	Supply outside state to Cochin at short distance from cluster	1-1.5	3-6	1-2% due to less distance	Good scope of rail Good scope of rail in every district of the cluster

Source: Stakeholder consultations and government data

The cluster is close proximate to Kerala state. The banana of the cluster is sent to all cities of Kerala state through lorries, small trucks. Harvested banana bunches are transported from farmers field to storage godowns of whole sale Agents/ whole sale banana traders through light vehicle like Pick up, TATA ape, TATA ace, just to avoid injuries to harvested produces. After auction for produces take place, the banana bunches are separated into hands, then hands are graded and sorted out. 13 kg of graded banana hands are packed in plastic crates covering with newspaper on bottom and top and loaded in trucks like Eicher, 407 lorries to transport to Chennai, other cities of the Tamil Nadu and Kerala. Rail connectivity is not available in Theni Banana cluster as Madurai to Bodi railway connection is under broad gauge conversion at present.

b. Mode of transport – Export Market

Table 111: Mode of Transportation for exports market - Cluster 2 (Tamil Nadu)

Mode of Transport	Major market (location)	Cost - INR/Kg/Km	Transit time (in hrs.)	% losses incurred during transportation	Constraints and gaps
Sea	Iran, Oman, Saudi Arabia, UAE	2-3	10-14 days	<5%	Long distance not possible (Europe)
Airport	Malaysia, Maldives	2	24 hours	<5%	Exports to European nations need to be increased

Source: Primary Survey and Field Validation

Main export destinations for banana from the cluster is GCC and Middle East markets such as Dubai, Saudi Arabia, Iran, Bahrain and Qatar. Limited quantity of banana was shipped to Malaysia, Singapore, Maldives, South Korea and even to London.

In order to harvest banana meant for exports purpose, different methods are deployed. The bunches are harvested in field and hands are separated by nylon rope cutting in field itself. These banana hands are graded and sorted out. The damaged, injured and poor matured fruits are removed, and good quality hands are packed in plastic grades and transported to Integrated Banana Pack house / Export house for processing.

After cleaning, washing in water, the banana hands weighing 13 kg is packed in poly sheets, unaerated by vacuum cleaner, placed in card board boxes, thus packaging is completed. The packed cardboard boxes are loaded in refer containers and 20.85 MT refer containers are transported by container lorries from Pack house to ports like Tuticorin, Cochin and Chennai.

In Theni district, Private Transport vehicle owners are operating trucks like Eicher, 407, Pick up, TATA ace, TATA ape for transport of banana from field to Godowns of wholesale commission agents, from godowns to Retail sellers. TATA ace, Mobile vending carts, pulling carts, tow wheeler are used to sale banana fruits to consumers by street vendors. For exports, both seaports and airports of Kerala and Tamil Nadu are utilised, primarily, Cochin seaport and Tuticorin seaport. Cochin and Trivandrum airport from Kerala and Chennai airport from Tamil Nadu are used as well.

c. Marketing Channels

Table 112: Details of marketing channels in the Cluster-2 (Tamil Nadu)

Marketing Channel	Percentage of farmers covered	Volume of crop handled (in MT)	Selling Price Rs./ Kg	Constraints/Gaps with its reason
Commission Agent	85-90%	8.74 lakhs	9-10	Traders high in number
FPO/Cooperative	5-10%	0.5 lakhs	10-11	Trader's control maximum trader
Organised trade (Mother Dairy/Desai/ etc.)	3-4%	0.4 lakhs	12-14	Very low organized trade
Exports	1-2%	~10000 MT (including exports happening through Kerala)	13-15	-

Source: Stakeholder consultations, GT Analysis

The numbers related to exports above are taking into account the volumes being exported from the cluster but using the seaports and airports from Kerala.

d. Current Branding & Marketing Initiatives

Presently Tamil Nadu has 2 GI tag for its bananas. The 2 GI tags are for the varieties of Virupakshi Hill Banana and Sirumalai Hill Banana . However, there is no known brand of banana from the cluster.

9.5. Proposed Interventions and Recommendations

Table 113: Indicative proposed interventions for Tamil Nadu Cluster 2 (Theni, Tuticorin, Madurai)

Value Chain Activity	Gaps Identified	Recommended Component	Numbers (A)	Units (MT, Ha, Sqm, set, participant etc.)	Indicative Per unit cost in INR Lakh (B)	Indicative Cost of Interventions C=A*B (INR Cr.)
Pre-Production and Production	Lack of Tissue Culture Planting Material	Tissue Culture Lab with Virus Indexing Facility & Hardening Centre	5	With total capacity of 1.25 Cr. plants every year	250	12.5
	Limited number of Farmer Producer Organisations (FPOs)	FPOs formation and promotion	24	500 participants each	43	10.32
	Lack of GAP certification	Promotion and training of GAP	8603	Hectare	0.1	8.6
	Limited quality testing at farm and post farm levels	MRL, pesticide residue testing lab	1	Number	400	4
		Basic testing facility at district level	3	Number	25	0.75
Post-harvest Management and Value Addition	Dependency on traders for ripening	Ripening Chambers	26	50 MT each	50	13
	Lack of export promotion infrastructure	APEDA certified packhouse	1	At least 100 MT storage	200	2
	Limited sorting, grading and packaging facilities	Collection centres	46	50 MT per day handling capacity	4	1.84
	Limited processing	Banana Chips/ Flour manufacturing unit	3	10 MT/day input capacity	75	2.25
		Banana Puree manufacturing unit	1	96 MT/day	1500	15
	Injury and loss due to improper packaging practices	Adoption of crates	100000	20 kg each	0.003	3
Logistics, Marketing and Branding	Limited availability of refrigerated transport	Refrigerated Transport Vehicles	13	15 MT reefer vans	30	3.9
	Limited branding, no GI tag	Branding, Digital and Marketing Initiatives	1	Approx. 2% investment towards holistic development of cluster brand	1.54	1.54
Total	-	-	-	-	-	78.71

Source: GT Analysis

Opportunities in Creation of Circular Economy: Waste to Wealth

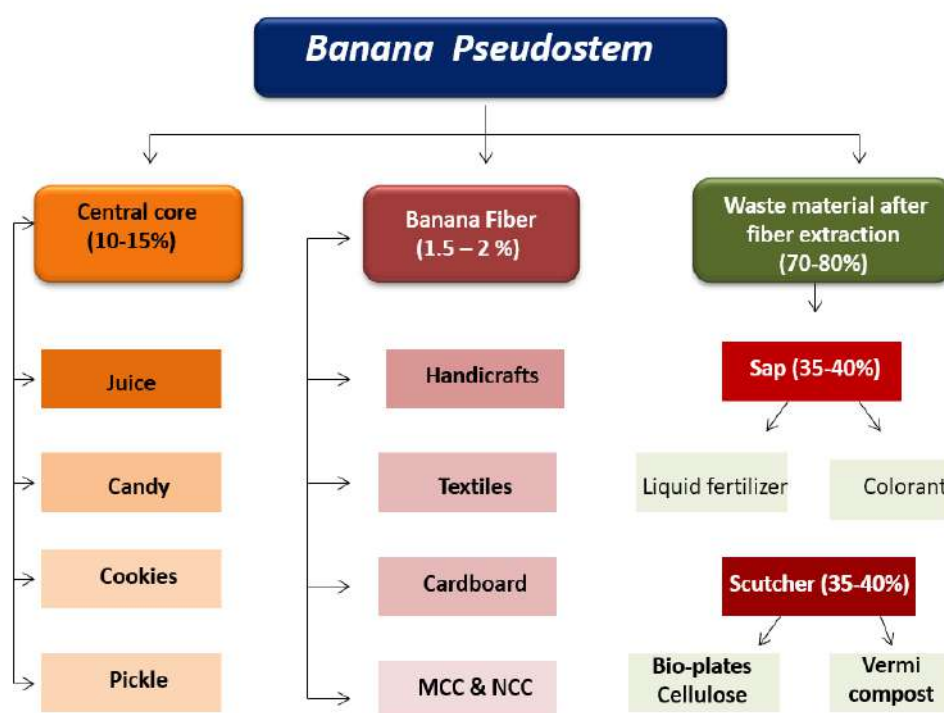
Banana is a pack house of nutrients and bio-active compounds with global acceptance. The changes in consumer behaviour and food consumption pattern have increased the potential of exploring new, innovative and greener products in the spectrum of food technology. Management of crop residues after harvest of the economic part is one of the major issues in agriculture. Though in some of the crops like sugarcane, wheat, rice and maize, it has been achieved to some extent, yet many of the horticultural wastes are left unutilized. Banana generates huge volume of the biomass including pseudostem (30-34%), leaves, flower & bracts (5%) and rhizome (12-14%) after the harvest of fruits. Currently less than 2% of these wastes is used for human consumption and the remaining are incinerated and wasted. The growing importance on the concept of circular economy has a way to attain sustainable development in the fruit processing sector. Large numbers of valorised by-products from banana plant, including bio ethanol production are gaining attention. Utilization of biomass from various banana waste (i.e. pseudostem, banana peel, peduncle, flower and reject bananas) conversions into novel and innovative high value products with emphasis on secondary and tertiary processing and products like xylitol, cellulose, MCC, hydrogel needs to be discussed. On the whole, banana can be a suitable candidate for development of future food due to its inherent availability of bioactive components. Fine chemistry, innovation and development of niche markets coupled with the circular economy approach can manifold enhance the value of products. The application of new methods, processes and product achieves global competitiveness and sustainable growth through creation of new business models and employment opportunities.

1. Waste Generated from Banana Crop

'Living well within the limits of our planet', is the tagline to understand the concept of circular economy. With rapid increases in extraction and exploitation of our non-renewable natural resources, those are having a wide range of negative environmental impacts. Offering 380-degree solutions will be much more appreciated by the world leaders and the policy makers, given the current scenario of climate change. There is a need to be in symbiotic relationship with our planet to save its resources for our future generation. The concept of circular economy presents great relevance in the global economy's evolution as an alternative to the present-day linear economy which represents make-consume-dispose economic model. The concept of linear economy exerts great pressure on the environment and human health, and also reduces opportunities for increasing the competitiveness of several sectors. In contrast, circular economy could increase the efficiency of primary resource consumption that aims to maintain the utility of products, components and materials and retain their value. Its adaptation will minimise the need for new inputs of materials and energy, while reducing environmental pressures linked to resource extraction, emissions and waste. This alternative model could create economic and environmental co-benefits, as the dependency on extraction and imports declines in parallel with a reduction in the emissions in the environment.

Apart from fruit, banana crop also generate huge quantity of biomass in the form of pseudostem, leaves, suckers etc. The average quantum of pseudostem yielded in India per hectare is 60-70 MT. This poses multiple challenges to the farmer in terms of disposal and management as the waste occupies space at farm level, is a recurring process, is not easily disposable and is labour intensive. Also, decomposing waste leads to disease, foul smell and pests.

Figure 10: Waste components generated from harvest of banana crop



Source: National Research Centre for Banana

For disposing purpose, farmers spending INR 8,000 to 10,000/ha used as dumping on field bunds and burning, disposing in natural drains which causing environmental problems. Currently, the practice is such that the waste is disposed off in canals, landfills and dumping grounds through makeshift means.

One of the central pillars of a circular economy is feeding materials back into the economy and avoiding waste being sent to landfill or incinerated, thereby capturing the value of the materials as far as possible and reducing losses. It is a known fact that nowadays, consumers demand food products, which are nutritious as well as convenient to use. With high banana production projected in the upcoming years, there is an imperative need to balance this production with proper utilization. Large amount of organic residues and wastes will be available in banana farm after harvesting its fruit bunch (stem, leaves, flower, etc) which are store house of nutraceuticals, have limited application as fertilizers, livestock feed and others. Technical application in the field of waste utilization can promote sustainable economy for banana cultivating farmers. Novel value-added products (starch, antioxidants, and pigments) have huge application in food, pharmaceutical and cosmetic industry. There have been a number of broad research studies about recovery/production of pectin, enzymes, antioxidants and phenolic compounds through eco-friendly methods, which shows the importance of the recycling and re-valourising of banana by-products as great resources for future developments in the prevention of current economic and environmental problem.

Potential Raw Material Availability

Banana is grown in the regions from the humid tropics to humid subtropics and semi-arid tropics like Tamil Nadu, Maharashtra, Gujarat, Andhra Pradesh, Karnataka, Uttar Pradesh, Madhya Pradesh, Bihar, West Bengal, Assam and Odisha. The Cavendish clones occupies 52% of the total area under banana cultivation and contributes 64% of the total banana production. The factors like production function, purchasing power, distribution system, end point delivery greatly influences the supply chain of banana trade. The disruption of supply chain has geared up again a regular vociferous noise of utilizing banana in processing industry. In spite of a sunrise sector, processing industry in India is still under-performing and hardly processing less than 5% of agricultural produces. Nonetheless of agricultural and allied sectors sharing major chunk of labour force of over 50% and providing employment to over 400 million dependent population, the processing industry gives employment to less than 2 million population with the share of

0.5%. Due to mishandling of the produce about 25-40% is being wasted and only 2% is processed into value added products, the remaining being used in the raw form. This leads to price imbalance and large price variations both spatial and temporal, which disheartens farmers. Besides, it can be easily appended with the major processed produce from India i.e. Mango for its pulp. As mango is seasonal, the machines run only for four months 24 x 7 and for the remaining period hugely kept idle with single shift running of the unit just to leverage the operational cost. The time has ripened for food industry to look into the alternative sources to sustain their business with the utilization of low-cost high volume produces and convert them into high cost- low volume processed products.

Circular economy aims to develop a system that has no net effect on the environment, ensuring that there is a reduction in waste production by processing and designing their utilisation into the systems as valuable co-products. The concept of circular economy when adopted for banana, it should not be loosely based on a fragmented collection of ideas derived from some scientific and semi scientific concepts. The scientific knowledge should be strongly coupled with economic, environmental and social spectrum for a sustainable development. In the transition towards a circular economy for banana cultivation and processing, there is a need to collect and share data, produce exemplars, make innovation investments and facilitate business collaborations. On the whole, creating a circular economy requires fundamental changes throughout the value chain, from product design and technology to new business models, new ways of preserving natural resources (extending product lifetimes) and turning waste into a resource (recycling), new modes of consumer behaviour, new norms and practices, and education and finance.

Starch, pectin, and cellulose are used in the food industry as gelling agent, thickening agent and stabilizers. Starch, a group of carbohydrates, is commercially available and produced from plants such as corn, potato, rice, wheat, and cassava. Banana by-products that can be processed into edible starches includes the pith of pseudostem and the green culled banana which are removed during fruit selection and processing. Similarly, industrial products like pectin and cellulose can be extracted from the lingo-cellulosic wastes of banana. An innovative approach on the utilization of banana is required through seamless blending of modern scientific knowledge with traditional wisdom. Banana flour has been incorporated into various products such as flakes, high-fibre bread, slowly digestible cookies, pasta and noodles. The blossoms contain a class of photochemical known as saponins. Saponins lower LDL or bad cholesterol, boost our immunity against infection and are thought to inhibit the growth of cancer cells. They also have antioxidant activity and so can reduce our risk of chronic diseases including cardiovascular disease.

Assuming one cropping cycle on an annual basis for the widely grown GRAND NAIN variety, the total potential quantum of banana pseudostem getting generated annually at a national level is estimated to be around 61.6 Mn MT across an area of 8.78 lakh hectares. With regards to the states under study region, the broad breakup of potential pseudostem availability is tabulated below.

Table 114: Potential pseudostem generated from banana in states under study region

Sr. No.	Name of State	Area under banana cultivation ('000 ha)	Annual Pseudostem generation (Mn MT) ¹³
1	Gujarat	69.54	4.90
2	Uttar Pradesh	72.59	5.10
3	Maharashtra	68.14	4.80
4	Andhra Pradesh	97.70	6.80
5	Tamil Nadu	92.413	6.50
	TOTAL	400.37	28.10

Source: Stakeholder consultations, State Horticulture Departments, NHB, GT Analysis

2. Scope for Value addition and key value-added products

The scope for deriving value added products from banana stem is manifold. This includes utilization of banana leaves for making paper plates and disposable items as well as utilizing the bulk of pseudostem to derive multiple products such as fertilizer, fibres and papers which has applications in textile and handloom industry as well.

Management of crop residues after harvest of the economic part is one of the major issues in agriculture. Global food production has a trend to increase by at least 60% by 2050 (EEA, 2016). Several approaches

¹³ Assuming 1 cropping cycle per year

are used to tap the efficient use of agricultural biomass for improving the overall of food systems, from producer to consumer, to make a cascade effect and to ignite the transition from linear to circular economy. Banana farming generates huge quantities of biomass most of which goes as waste due to non-availability of suitable technology for its commercial utilization. After harvesting the fruit bunch, huge plant bio-waste is generated, out of which pseudostem (30-34%), flower & bracts (5%) and rhizome (12-14%) together contribute to about 50 % of the banana plant biomass. Currently less than 1% of these wastes production is used for human consumption and the remaining are incinerated and wasted. Hence, banana plant bio-waste (pseudostem and rhizome) endowed with rich nutrient and medicinal properties are worth pursuing.

Competitive landscape

Natural fibres from agro-waste exhibit excellent characteristics like good mechanical strength, stiffness, low density, non-abrasiveness, high disposability, renewability and are considered to be eco-friendly over synthetic fibres. Among the natural fibres, 90 % are of plant origin and classified as seed fibres such as cotton, bast/skin fibres like flax, ramie, hemp, banana and jute. Among them, 80 % of fibre was constituted by cotton and remaining by other long fibres such as flax, jute, pineapple, hemp, ramie, coir, sisal and banana.

15000 litres

Organic Fertilizer From 70 MT pseudostem (1 ha)

- Fresh sap of pseudostem can be enriched with essential plant promoting substances viz., gibberellic acid (GA) and cytokinin which can be used for plant liquid fertilizer for different crops like banana, papaya, sugarcane, etc.
- Leads to saving of about 20-40% and reduction in carbon load in soil
- NAU, Navsari developed the innovative product called as NAUROJI Novel Organic Liquid Fertilizer
- Rich in N, P, K, Zn, B, Fe, Mn, Mg, Ca, S, Cu etc. & PGH such as Gibberellic acid and Cytokines
- Can reduce use of chemical fertilizer to the tune of 20%

1 MT

Banana fibre/paper from 70 MT pseudostem (1 ha)

- Banana fibre can be extracted from the sheath of pseudostem either by hand or machine
- The natural fibre has multifaceted uses in making various value-added products
- Banana fibres can be blended with cotton and jute fibres to make eco-friendly fabrics.
- It is used as an alternative for garment which is commonly made from silk, so that when it comes to accessories banana fibre is frequently used for making scarves, hats and gloves as sustainable organic products become increasingly popular
- Can be promoted for use in government documents, currency notes, academic documents, land documents etc on a large-scale basis due to its longevity.

Potential in Study Region

Table 115: Annual potential to generate novel organic fertilizer and banana fibre in study region

Sr. No.	Name of State	Organic Novel Fertilizer (Mn MT)	Banana fibre (Mn MT)
1	Gujarat	1.04	0.06
2	Uttar Pradesh	1.08	0.07
3	Maharashtra	1.02	0.06
4	Andhra Pradesh	1.46	0.09
5	Tamil Nadu	1.38	0.09
	TOTAL	5.98	0.37

Source: GT Analysis, NAU, Gujarat

Other potential products include preparation of candies and pickles from the tender central core of pseudostem. Also, scutcher based products such as vermicompost and fish feed have potential to be derived from pseudostem.

Banana is a crop which produces huge quantity of fresh biomass and so far not much attention has been given towards the waste utilization. Every year millions of tonnes of banana pseudostem is dumped as waste, is either burnt or left to decompose, thereby increasing pollution. In order to utilize this residue in an effective and profitable way, many agricultural universities and national institutions have undertaken research initiatives. As highlighted above, using pseudostem many products can be developed which has the capacity to generate additional income and also provides eco-friendly raw materials to the industries like pharmaceuticals, textile, paper, confectionary, etc. It can also help to enhance employment levels in rural areas thereby improving social and economic conditions of the farming population.

The male flower (bell) and bracts are hardly being used by people for kitchen purposes, in spite of the fact that they have higher medicinal and nutritional value. Several beneficial properties of these parts were reported earlier. Since ancient times, these parts of banana plants have been an exemplary source of medicine in Ayurveda and other Indian literatures. In the last decade, there was prolific growth in pharmaceutical and nutraceutical industry interested in natural bioactive principles from cheap and renewable agricultural sources. This is mainly due to safety concerns, contradictory toxicological data about synthetic source, increasing microbial resistance and the consumer preferences for natural additives. Thus, these points offer exciting opportunities to exploit bio-waste from banana with nutraceutical importance. Additionally, flower bract could be used as a bio-colourant which could be used as a nutraceutical compound through encapsulation methods. XoS and FoS could be extracted from other parts of banana, which have a high potential in the future nutraceutical and pharmaceutical markets.

Non-food applications

Good quality fibres can be extracted along with numerous other uses (juice) with bioconversion potential. Currently less than 2% of these wastes production is used for human consumption and for production of fibre and the remaining are incinerated and wasted. These residues should be used effectively not only to avoid the environmental pollution but to create wealth from the waste. The stem of banana, commonly known as pseudostem is an aggregation of leaf stalk bases in cylindrical form. A normal banana plant has 15-18 leaf sheaths making the pseudo-stem. The outer sheaths are composed of coarse fibres and the soft fibres are obtained from inner sheath. Pseudo-stem is the trunk part of the plant which consists of a central core (pith) wrapped by leaf sheaths. The central core contributes to 10-15% of the pseudo-stem, fibre is 1.5-2% and the rest is non-fibrous material. About 30-40% of the nonfibrous material is sap and 60-70% is scutcher. Pseudo-stem of Banana could be used in pulp and paper industries due to its cellulosic content. Banana pseudo-stem has been known as potential cellulose source. During mechanical extraction of banana fibre from pseudostem, scutching waste (30 to 35 MT/ha), sap (15000 to 20000 l/ha) and central core (10 to 12 MT/ha) are also obtained simultaneously. Banana wastes could be effectively utilized in variety of products and industries namely textiles, making of woven fabrics, non-woven fabrics, hand-made papers, composite boards, automobile etc. This versatile use of the wastes makes Banana as a viable complementary material to over- exploited natural fibres like cotton and Jute. Utilization of biomass from various banana waste (i.e. pseudostem, sheath fibres, banana peel, peduncle and spoiled banana) conversions into novel and innovative high value products with emphasis on secondary and tertiary processing and products like bioethanol, xylitol, cellulose, MCC and hydrogel is explored by many researchers. Development of non-edible products from banana fibre and sap are also area of attraction since valorisations of waste have a positive impact on environment. Improved mechanical, thermal and chemical properties of this residual waste from banana farm will give a good scope for its reach.

The sap obtained during banana fibre extraction could be used to make organic liquid fertilizer and to prepare mordant for the paint industry. The Liquid portion Sap obtained along with scutcher during Fibre extraction from Banana Pseudo-stem is good source of plant nutrients such as N, P, K, Micronutrients along with growth promoting hormones like Cytokinin & Gibberellic Acid GA3 etc. If used as liquid fertilizer either through drip system or drenching to crops, it can save 20-40% dose of chemical fertilizers. Scutcher material after fibre extraction can be useful as manure after composting or as raw material for the preparation of vermicompost. As a part of banana fibre extraction, high value products viz., mordant from sap, microcrystalline cellulose powder from fibre are obtained.

3. Current Initiatives in India

Various research institutes and State Agriculture Universities (SAUs) are currently undertaking varying levels of research initiatives to generate value added products from banana waste. One of these is the Navsari Agriculture University from Gujarat which has undertaken farmer survey at ground level to understand pseudostem disposal methods and garnered farmer feedback under the aegis of NAIP. Based on this work, they have derived protocols and technical specifications to manufacture high quality organic fertilizer as well as banana fibre from pseudostem waste generated from banana plant.

The NAU has developed a method and process to manufacture organic fertilizer from banana pseudostem. It manufactures the same in house as well and sells it to the farmers through a network of outlets at a rate of INR 200 per litre. It has also provided technology transfer services to private sector players for adoption of technology on a royalty paid basis. Through this initiative, to date, 33 units are functional across the country. These primarily include Gujarat (12 nos.), Maharashtra (12 nos.), Uttar Pradesh (1 nos.), Bihar (1 nos.), Madhya Pradesh (1 nos.) and West Bengal (1 nos.). All of these units are currently manufacturing around 55 lakh liters of novel fertilizer on an annual basis which is less than 1% of the total potential for manufacturing this product based on volume of pseudostem generated annually. The NAU has also distributed this Novel Fertilizer to 411 Krishi Vigyan Kendras (KVKs) across India for demonstration and distribution to farmers across India.

4. Potential Interventions at cluster level

The area under banana in 8 clusters across 5 states and 23 districts accounts to around 37% of the national area under banana cultivation. This has huge potential to generate pseudostem, thereby opening up avenues for lucrative value addition opportunities at cluster level. Integrated facilities which include sub-units fibre production unit and novel organic fertilizer production unit may be proposed at cluster level (not limited to). Indicative cost components for each sub-units is summarized below.

Table 116: Indicative Project cost for novel organic fertilizer extraction unit (10 MT month capacity)

Sr. No.	Items	Quantity (nos.)	Cost per unit (INR)	Total Cost (INR)
1	Raspador Fiber Extraction Machine	40	80,000	32,00,000
2	Pseudostem cutter Machine	8	45,000	3,60,000
3	5000 litre bio-digesters	10	7,00,000	70,00,000
4	7 stage filter unit	3	2,50,000	7,50,000
5	Storage plastic tanks	15	7000	1,05,000
6	Hydraulic press	4	70,000	2,80,000
7	Induction sealer	4	30,000	1,20,000
8	Other Utilities	1	2,00,000	2,00,000
9	Technology Transfer fee to NAU, Navsari	1	3,50,000	3,50,000
	TOTAL			1,23,65,000

Source: NAU, Gujarat, project cost excludes land and lease costs

Table 117: Indicative Project cost for fibre extraction unit (1 MT per month capacity)

Sr. No.	Items	Quantity (nos.)	Cost per unit (INR)	Total Cost (INR)
1	Raspador Fiber Extraction Machine	40	80000	32,00,000
2	Pseudostem cutter Machine	8	45000	3,60,000
3	Other recurring utilities	1	50000	50000
	Total			36,10,000

Source: NAU, Gujarat, Stakeholder Consultations

As per discussions with local stakeholders as well as Navsari Agriculture University, it is prudent to set up cottage level fibre extraction units at district level and organic fertilizer extraction units as well as paper manufacturing unit at cluster level to utilize the available banana waste, thereby enhancing income levels of the banana farmers and boost value addition.

- Fibre extraction unit capacity: 1 MT per day
- Fertilizer extraction unit capacity: 1 lakh litres per month
- Paper Manufacturing unit capacity: 1 MT per day

Table 118: Proposed Investment in banana waste value addition at cluster level

S.No	Name of cluster	Fibre extraction unit			Fertilizer extraction unit			Paper Manufacturing Unit		
		No. of units	INR per unit (Lakh)	Total Indicative Cost (INR Cr.)	No. of units	INR per unit	Total Indicative Cost (INR Cr.)	No. of units	INR per unit (Lakh)	Total Indicative Cost (INR Cr.)
1	Gujarat	30	36.1	10.83	4	123.65	4.95	1	100	1
2	Uttar Pradesh 1	10	36.1	3.61	2	123.65	2.47	1	100	1
3	Uttar Pradesh 2	10	36.1	3.61	2	123.65	2.47	1	100	1
4	Maharashtra	10	36.1	3.61	5	123.65	6.18	1	100	1
5	Andhra Pradesh 1	10	36.1	3.61	2	123.65	2.47	1	100	1
6	Andhra Pradesh 2	15	36.1	5.415	4	123.65	4.95	1	100	1
7	Tamil Nadu 1	10	36.1	3.61	2	123.65	2.47	1	100	1
8	Tamil Nadu 2	15	36.1	5.415	3	123.65	3.71	1	100	1
		110	-	39.71	24	-	29.7	-	-	8

Source: GT Analysis, Navsari Agriculture University, Gujarat. Cost estimates are exclusive of land, lease and labour costs. Costs may vary from state to state. The cost estimates are indicative in nature.

5. Strategies for Promoting Banana Based Industries

- Sustenance of any processed product in the market and thriving of industry in the longer period required proper planning, road map and razor-sharp execution. Unlike developed countries, the food processing sector in India is dominated by the unorganized sector, contributing more than 80% share. Bringing them into the MSME bracket is pre-requisite to boost the processing of horticultural produces especially the crops like banana. Creation of cross functional convergence with development agencies, cluster formation in production points by converging Farmer producer organizations (FPOs), market linkages and assessment of the impact through technological incubation and cross cutting initiatives are primary step for the successful establishment of processing industry. In the end, larger industry players should come up with the plan to utilize the banana flour, fruits and juices in their product portfolio to scale up the industrial use and to market it effectively.
- The share of processed industries in agricultural employment has to be enhanced from the present 0.5% to at least 10% with the enhancement of productivity of workers by retaining the youth in agriculture by giving better remuneration and allotting them a higher productivity works. With that we can eradicate the dependency of workers in low productivity agricultural activities and shift them to higher remuneration agri-based industrial activities.
- Developing methodologies based on input-output models which provide information about the cluster wise production of banana and could identify the major catchments for supplying the fresh produces. This will allow in calculating the impact of any change in the supply chain. Using AI/ IoT/ App based platforms to provide and generate real time data points.
- In order to sustain the production and growth potential, it is essential to make value added products based on banana, so that farmers get an assured price for their produce all the time. India is the land of more than 30 ethnic varieties in different states with different taste, flavor and nutritional advantages. For example, Nendran is rich in vitamin A and thus could be used for making functional powders to alleviate malnutrition among the rural masses. Likewise, the

enormous processing potential could be tapped by developing and commercializing variety dependent technologies.

- Building right infrastructure for converting raw banana into flour, ripe banana into health powder and making banana puree are need to be prioritized. All these products have a greater nutritional potential and could be used in functional food markets with its pre-biotic characters. The low glycaemic, sugar controlling capability of banana flour has to be rightly marketed to enhance the demand for banana-based products.

6. Potential Impact Envisaged

a. Enhancement in farmer income:

- The banana cultivators, who earlier had to resort to traditional practices such as dumping banana waste in canals or burning, thereby leading to pollution and diseases, will get an additional income source by means of supplying pseudostem to processors at district/cluster level. The average selling rate for 1 MT of banana pseudostem is INR 400. Considering the volume of pseudostem generation potential based on the area under banana cultivation in the study region, there is a potential to generate an additional cumulative quantum of INR 1100 Crore approximately which will directly benefit the farmer, bolstering his profit margin and overall well-being considerably.

b. Reduction in consumption of NPK fertilizers:

- As per research conducted by Navsari Agriculture University, Gujarat, who are pioneers in development of Novel organic fertilizer from pseudostem, efficient utilisation of the same can help reduce consumption of chemical fertilizers (NPK) by almost 20% on a per hectare basis. Gross NPK consumption spread the 5 states in the study region (across crops) stood at approximately 26.3 Million MT. Substituting 20% of this will help in reduction of NPK consumption to the tune of almost 2.31 Million MT on an annual basis. This will also help enhance soil quality, reduce carbon content and reduce subsidy load on the government as NPK fertilizers are heavily subsidised.

c. Additional Employment Generation:

- With the proposed banana fibre, fertilizer extraction and paper manufacturing units, it is estimated to generate an additional employment of 22000 nos. This is of vital as most of this will be at district or taluka/block level, thereby providing a boost to rural employment including an impetus to women employment

Key Recommendations

The key recommendations in this section are focusing on the entire banana value chain, ensuring a holistic view of its constituent stages and concurrently focusing on various factors so as to enable a congruent development of the sector. These are broadly divided into 3 segments namely:

1. Pre-Production and Production Stage

- a. **Production of quality planting material:** Banana is a commodity requires intensive cultivation. ; Traditionally banana has been propagated by suckers but realizing the advantages of in vitro propagation, mass multiplication through shoot tip culture has gained popularity for the production of disease-free plants. High multiplication rate per unit time, better establishment with hardening, uniformity at flowering and harvesting, higher returns with increased yield are some of the benefits of micropropagation. Underutilization of tissue culture laboratory and practice of utilizing ratoons from the subsequent plantation cycle reduces productivity and quality of produce. Presence of tissue culture laboratories at cluster level catering to the overall demand of planting material (earlier compensated through sources within and outside state) will bring down transportation costs as well.
- b. **Higher impetus on GAP, Fruit Care and Climate resilient practices:** Training and capacity building being undertaken by farmers was found to be minimal across the clusters. Incessant application of fertilizers and inputs is another challenge which has been highlighted by district level government representatives (DoH, KVK, ATMA etc). Presently, less than 5% of the farmers are aware or practice GAP and fruit care practices. If we take the example of the global leader in banana exports, viz. Ecuador, this figure stands at over 60%. Hence, a holistic strategy to bring area under banana cultivation under GAP and certification compliances like global gap, fair trade, organic gap, HACCP and related certification systems should be developed in coordination with state departments, KVKs, ATMA and SAUs. Influencers and innovative farmers may be approached, success stories to be highlighted and better income generation avenues to be promoted and showcased so as to motivate cultivators to undertake such practices. Crop selection and changes in crop calendars will help farmers adapt to new temperatures and rainfall patterns. Farmers will also need to adopt more systematically measures to respond to increased frequency of floods and more intensive rainfalls. A combination of erosion control actions and better drainage capacities will be needed.
- c. **Promotion of FPOs/ FPCs:** During the primary survey with FPOs in the study region, it has emerged that the formation of FPOs/Cooperative led to increase in farm gate prices to the tune of 15-20% due to better bargaining power, assured offtake and periodicity. FPO membership ensures better input availability, access to market intelligence as well as avenues for capacity building when compared to conventional farmers. It is recommended to have a larger participation of farmers under FPO ecosystem in the case of banana for sustainability and for the efficient use of inputs.
- d. **Promotion of organic farming & conservation agriculture:** Clusters could be identified within the special export zones, to give more impetus on promoting organic banana cultivation. Special varieties other than Grand Naine may also be included in the scheme of things when to promote organic farming in banana. The exclusivity of traditional bananas may be promoted in the European and US market as premium varieties and special vehicles may be mooted for better delivery systems.

2. Post-Harvest handling and Value Addition

- a. **Primary Processing:** Currently, a majority of processing in banana happens at field level itself adopting traditional methods. This reduces quality of banana and may also aid in spread of diseases and insect pests. Hence, improving availability and access to primary processing facilities such as packhouses and collection centres closer to production clusters is vital. Presence of packhouses at taluk/block level will also help in establishing value addition facilities for other horticultural commodities other than banana. Establishment of ripening chambers, packhouses (incl. mobile), collection centres etc. along with suitable sensitisation of farmers and market intermediaries will help in effective utilisation of such facilities and aid in migration from traditional practices to efficient and hygienic handling of banana post farm gate.

- b. Secondary Value Addition:** Traditionally, banana in India has been majorly consumed as dessert fruit and <5% is consumed in processed form. Secondary and tertiary processing is almost negligible and thus led to poor adoption and profit realization. Banana puree, green banana flour, ripe banana powder, central core stem powders are gaining popularity in international market due to its versatility, higher resistant starch, low glycaemic properties. There is immense potential domestically as well as in exports markets, hence, integrated pulping lines and manufacturing lines at cluster level should be established. The private sector interest in this regard is evident and the same should be tapped through concentrated efforts and incentives. Subsequently efforts in branding and marketing space, the export can also be enhanced considerably catering to existing and new untapped markets.
- c. Leverage Waste to Wealth Opportunities:** Banana is one of the topmost 'high volume' fruit crops in India generating mammoth volumes of resultant waste at farm level. The potential to leverage this waste is immense, in terms of farm level use as well as at customer level through value addition. Banana fibre extraction and organic fertilizer generation through banana pseudostem is an untapped opportunity currently waiting to be exploited. District and cluster level interventions elaborated in this regard can help in generating a cumulative increase in farmer income (INR 1100 crore+), reduction in use of NPK fertilizers (2.3 Mn MT) as well as additional employment generation potential (20000+) in the study region.

3. Marketing, Branding and Logistics

- a. Improved Availability of Kisan Rail:** Through Kisan Rail, the transportation cost is reduced by INR 1.5-2/kg. However, it's current penetration is limited and pocketed to limited clusters. Also, the awareness with regards to its availability, schedule, costs involved and process to be followed needs to be highlighted among the stakeholders. There is high potential for utilisation of Kisan Rail as the cost of transit and resultant losses will get substantially reduced when compared to road transport through a mix of adoption of crates, reefer-based transit etc. Overall, better promotion of Kisan Rail availability, new stops at key production junctions and clusters, cost structures and process to be followed needs to be undertaken suitably. Financial assistance on reefer wagons from transport from cluster to distant markets may also be explored.
- b. Changes in banana production and market driven high through-put cultivation:** Unlike other countries where the banana is grown in large area and monitored by big firms, Indian banana industry is highly fragmented. Cluster formation or co-operative farming for procurement of all the inputs on collective basis, better implementation of the technologies, credit approval and better price realization need to be strengthened and encouraged. Traceability of the farm should be given importance with the stricter adherence to good agricultural and handling practices. Farmers need to be educated about export requirements and international quality standards. It is essential that the farmers work together to reduce the input cost and enhance uniform maintenance of plantations and thus able to get uniform quality, exportable bunches.
- c. Branding and marketing Initiatives:** Though banana is one of the mostly widely consumed fruits in India, there is still immense potential to leverage an already strong foothold through improved value addition, product diversification and accessibility to newer markets. Creation of specific demand and branding to Indian banana and promotion of traditional banana are essential to have a foot print in the international market. Value added products such as banana flour can be substituted to at least 30% with conventional wheat flour to cater to the diabetic population due to its low glycaemic index. Integration through Mid-day Meal schemes and PDS system may be promoted to boost the utilization of banana-based products. Registration and acquiring of GI tag and focussed marketing in exports markets, especially EU and USA should also be focused upon other than traditional markets like west Asia. Specific impetus on branding initiatives as a separate component should be laid upon in subsequent policy and subsidy regimes.
- d. Enabling IoT/ AI based ecosystems along with traceability:** Developing methodologies based on input-output models which provide information about the cluster wise production of banana and could identify the major catchments for supplying the fresh produces. This will allow in calculating the impact of any change in the supply chain. Enabling start-ups, skill development, identifying innovations through value chain approach and developing IoT/ AI based ecosystems, based on the demand of present and future markets. Leveraging Blockchain enabled traceability solution which will help in

ascertaining source of production, variety of produce, farming practices followed, organic certification as well as help in mapping entire chain of transfer of produce. This will help in ascertaining price trends, peak and lean seasons of a crop, anticipating glut/scarcity of crop in an area and hence in inventory management on basis of dashboards and reports developed. Farm level blockchain technology is the need of the hour to compete globally for exports and becoming a leader in exports.

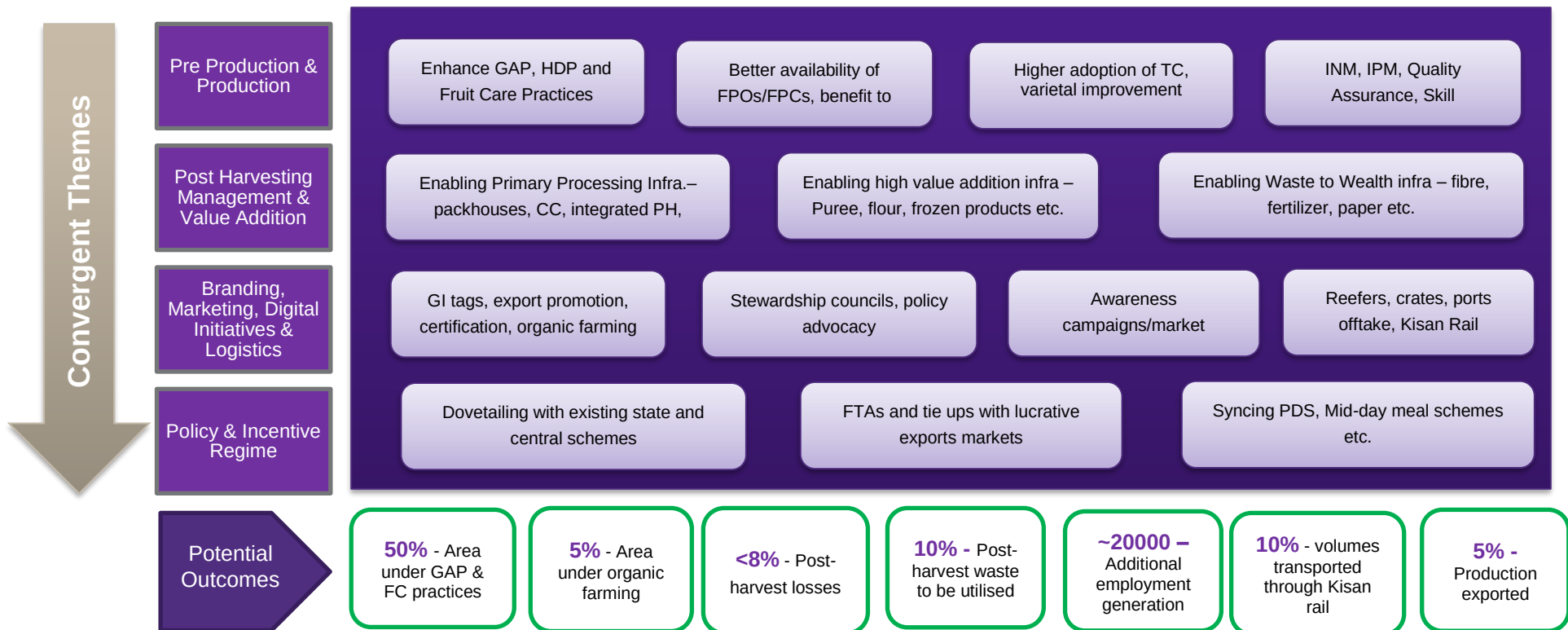
4. Others

- a. **Exports Facilitation and Sea Protocols:** India lags behind in fresh and value-added banana exports due to underdeveloped trade routes, practices and penetration in lucrative markets. Through all of the farm level initiatives summarised above, as well as focussed marketing efforts, over time, availability of exports compliant produce can be achieved. Development of sea protocols and favourable access to distribution networks and sea routes should be undertaken. Orientation of Free Trade Agreements (FTAs) and trade policy in favour of banana exports should be evolved (for eg. Ecuador, Columbia and Mozambique have trade agreements with EU). Banana could also be suggested as a tradable commodity in bilateral agreements to reduce the customs duty upon arrival. India should get the Most favoured Nations (MFN) when trading banana with other countries with zero or reduced tariff to be competitive in the market. The entrepreneurs need to be encouraged with incentives such as tax holidays, reduction in GST, providing special green channel at ports, reduction in handling time, etc. Being a perishable commodity, fast clearances at port level through a single point of inspection should be prioritised.
- b. **Promotion and Branding of traditional bananas and creation of export window:** Specification and grade standards should be made available for traditional banana varieties. Protocol for postharvest handling of Nendran, Ney Poovan and Red banana need to be perfected for shipping by sea. Export window may be created in the nodal place like ICAR- NRC Banana, to cater to need of exporters. Indian bananas should be marketed with specific brand name to get renowned global visibility. Phytosanitary manual, SOPs, GAPs should be prepared, involving all the stakeholders and exports standards for traditional varieties may also be developed.
- c. **Twining Value Chain Actors for Synergy:** Synergizing with the large number of value chain stakeholders such as the MoFPI, MoA, APEDA, NHB, FIEO, EIC, FSSAI as well as food processing industry associations across relevant clusters is key. This will enable in leveraging mutual strengths and mitigating challenges which may be faced across the value chain. These include exports related certification, best practices and compliances, quality testing and assurance as well as visibility in target markets. Policy advocacy through chamber engagement or common commodity specific committee/stewardship council may also be explored. Support from NRCB and IFFPT may be sought across all value chain interventions to access various prototypes, models, access to market intelligence and latest sectoral trends.
- d. **IT infrastructure/ Dashboard:** It is suggested that a real time dashboard highlighting details of post-harvest management infrastructure pertaining to banana is maintained. The portal can capture the beneficiaries of schemes where number of marginal, small and large farmer beneficiaries is maintained separately. The portal will help a farmer in gaining market intelligence. An IT portal will help in identifying the lacunae and address the same simultaneously for SML farmers. The portal can also maintain information on market intelligence, Kisan Rail availability and timetable as well as track movement of produce through such channels. It will help in identifying demand and supply from particular clusters/states. Also, such a portal will provide access to plant and machinery suppliers, maintenance experts as well as available technologies for primary and secondary processing. Stakeholders widely suggested that an e-NAM like portal may be developed for better marketing of banana to benefit in terms of price discovery; payment guarantee and access to nationwide market and traders' benefits from reduced transaction costs and delivery guarantee. This becomes even more imperative since negligible banana trade takes place through APMC ecosystem.

Banana - Vision 2030

Banana is one of the most widely grown and consumed fruits in India, almost ensuring a year-round availability. It is a focal commodity for the Ministry of Food Processing Industries, Govt of India in order to improve its post-harvest management profile and reducing losses, concurrently benefiting the cultivator as well as consumer. Albeit, well established, a focussed vision for the sector is imperative to lay down a strategic roadmap going ahead, which will guide and benefit the banana ecosystem in years to come.

There is a need for a concentrated effort on part of state governments in association with all central government ministries and line departments to devise a holistic vision so as to boost the wellbeing of the banana sector in the country. Cross expertise synergies need to be leveraged along with regional presence and acumen. Since banana is a commodity which is cultivated, almost pan India, this is of vital importance. Also, it's awareness and positioning amongst the cultivator stakeholders as a 'high value, high volume' crop rather than 'low value, high volume crop' needs to be developed. Focussing in silos on only one or individual aspects of the value chain will not be of intended success. There is a need for holistic focus on all factors right from farm level to consumer level, encompassing value chain actors such as market intermediaries, government machinery, exporters, logistics players, academicians as well as the business fraternity.





Assessment Study for Identifying Gaps in Infrastructure & Processing Facilities for Development of Banana Value Chain Under Operation Greens Scheme

Annexures

October 2021

Content

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Meetings with Industry Anchor (Keventer Agro Limited)



Meeting 1

The Ministry of Food Processing Industries organized a kickoff meeting with the team of Grant Thornton Bharat LLP as well as representatives from the industry (Keventer Agro) to discuss on the particulars of the study in hand and key particulars related to the banana sector in India. A brief summary of the meeting is provided below.

Key Particulars	Details
Date and Time of Meeting	08/06/2021 at 15:00 PM
Key Participants	Ministry of Food Processing Industries, Govt of India - Honorable Joint Secretary - Ms. Reema Prakash - Deputy Secretary – Shri S.K. Singh
	Keventer Agro Limited (Industry Anchor) - Shri Mayank Jalan – CMD, Keventer Agro Limited - Shri Sunil Kajaria – CEO, Keventer Agro Limited
	Grant Thornton Bharat LLP - Chirag Jain – Director - Karan Sharma – Associate Director - Khushrow Chinoy – Manager - Tushar Doshi – Assistant Manager - Shashank Shukla – Assistant Manager
Key Discussion Points	- At the beginning of the meeting Honorable Joint Secretary introduced participants from Grant Thornton team and Keventer Agro Limited followed by a brief context about the proposed study being taken up. - Team GTBL presented on the objectives and understanding of the proposed study, key preliminary findings, and roadmap for the study. This included: - o Background and understanding of the study. - o Findings and learnings from previous assessments and experience - o Survey Tools to be used for the study viz. surveys, questionnaires, etc. - o Broad timeline for the deliverables of the study. - Mr. Jalan and Mr. Kajaria invited Team GT to hold a detailed discussion with their team to understand their operations and gauge an overall view of the banana sector in India.

A detailed follow up discussion was also scheduled with the team of Keventer Agro Limited by GT.

Key Particulars	Details
Date and Time of Meeting	25/06/2021 at 12 PM
Key Participants	Grant Thornton Bharat LLP - Khushrow Chinoy – Manager - Tushar Doshi – Assistant Manager - Shashank Shukla – Assistant Manager Keventer Agro Limited (Industry Anchor) - Shri Sunil Kajaria – CEO, Keventer Agro Limited - Shri SP Chaudhuri – General Manager, F&V
Key Discussion Points	Challenges in Banana cultivation and procurement - The traditional model of sourcing bananas from aggregators and vendors suffers from the following challenges - o Inconsistent quality - o Seasonal variations in output - o Wild fluctuations in prices - o Excessive bruising due to poor post-harvest management - o Outsized commission to vendors & aggregators - o Large scale rejection at pack-house/ripening facility - o Delays in payments Current Operations - Keventers source 80% of their requirement of raw bananas from the 4 main banana growing districts of West Bengal in Nadia, Murshidabad, Hooghly, and 24 Parganas - Ripening facilities at – Kolkata, Durgapur and Siliguri - In addition to WB, catering to Jharkhand markets – Ranchi & Jamshedpur too - Over 300,000 farmers are involved in banana farming, with 41,000 hectares under cultivation (average landholding less than 0.3 acres) - Expected Annual volumes proposed to be handled – 40,000 MT (FY22) Branding Initiatives - Bananas are stickered extensively with a collection of stickers for various occasions / festivals - Mascot “Keven” is a clear favourite of kids and personifies the active lifestyle - Utilisation of various ATL & BTL platforms for enhanced visibility and taking the health aspect directly to consumers in an effort to highlight the positive health aspects of ethylene-ripened bananas

Snapshots of Meeting held with Keventer Agro Limited

One of the largest organized players of branded bananas in India

Operations across 4 states

Engaging with farmers at grassroot level through efficient aggregation

Efficient marketing through FMCG model of sale

Better Remuneration to Producer

- All grades of banana are bought at an average of INR 10/kg and premium is also given for A Grade bananas, contrary to competitors who insist only on A grade.
- Generally, G-9 variety is being procured and sold in market.
- 30,000 MT of it is being procured annually out of which 80% is from WB and rest from other states

Continued Investment in infrastructure

- In the 13 years of banana trade, setup 7 Ripening Chambers across India with total output of more than 100 MT/day.
- 4 in WB across 4 districts with 70 MT capacity, 1 each of 15 MT capacity in Ranchi, Patna and Roorkee
- Shade net facilities for plantlets demo farms for GAP training to farmers in West Bengal

Good Practices

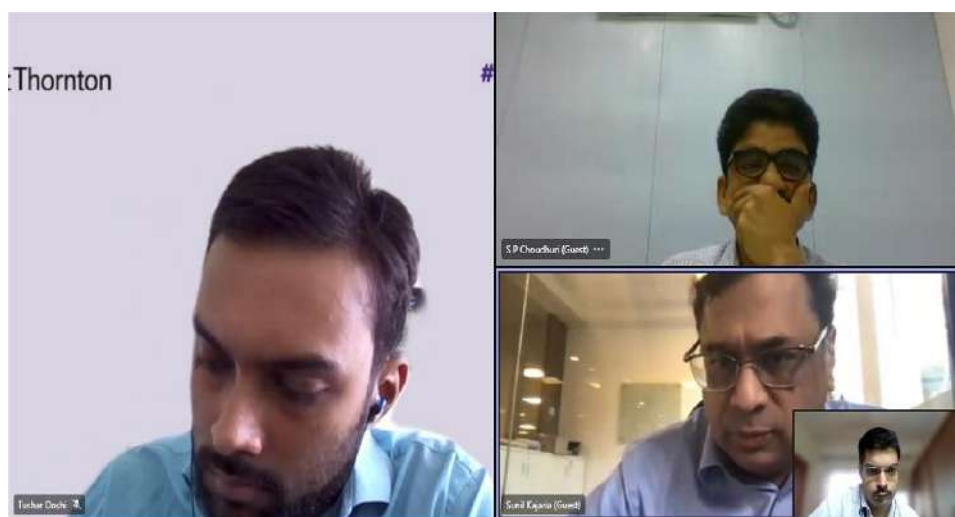
- Ripening through ethylene (no carcinogenic calcium carbide)
- Pioneer in East India for ethylene-based ripening (FSSAI approved).
- Bananas not transported in looms due to which transit losses <.5%.
- Introduced use of crates @15 kg + foam.
- Insistence on Farm gate cleaning to reduce losses (sap clearance)

Extension Support to farmers

- Making farmers aware of Good Agricultural Practices under their Keventer Assisted Farming program
- Currently 10% of their procurement coming from such farmers.
- Farmers are also being made aware of Tissue culture and its benefit to counter widespread use of sucker method of banana cultivation.

Branding & Marketing

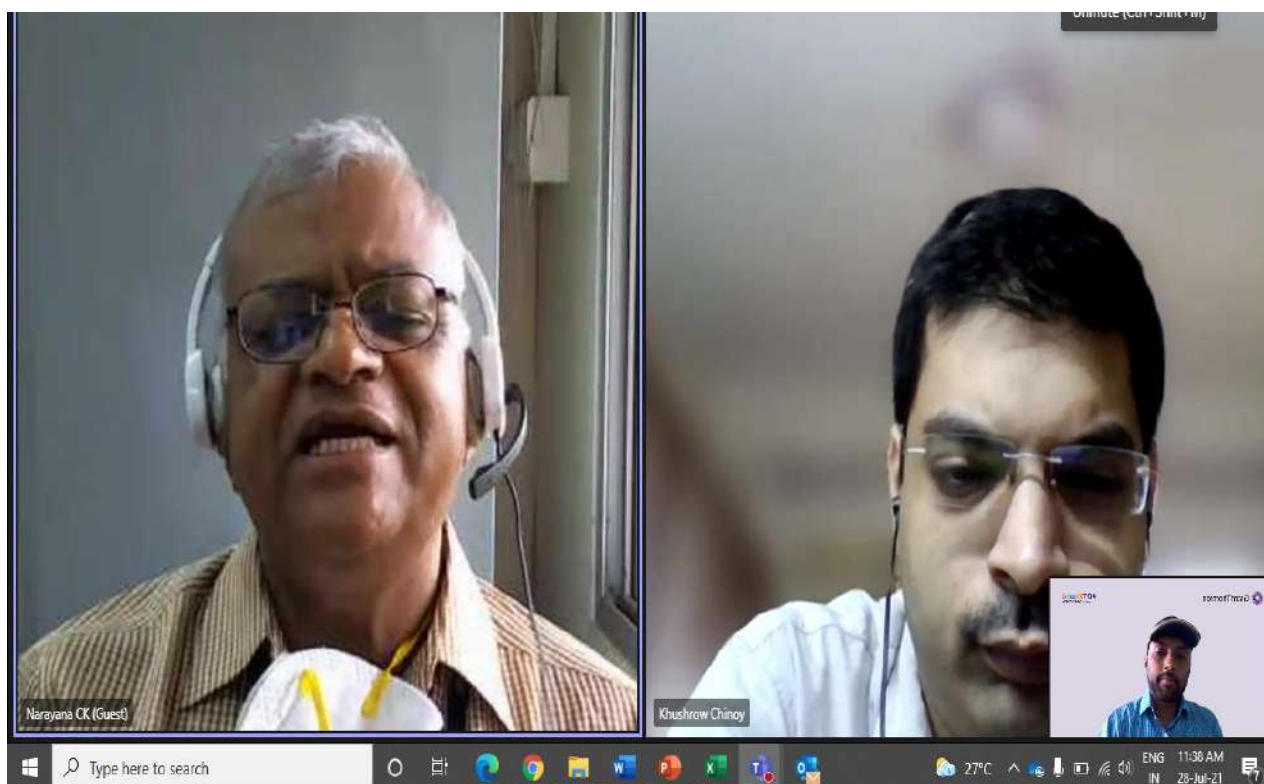
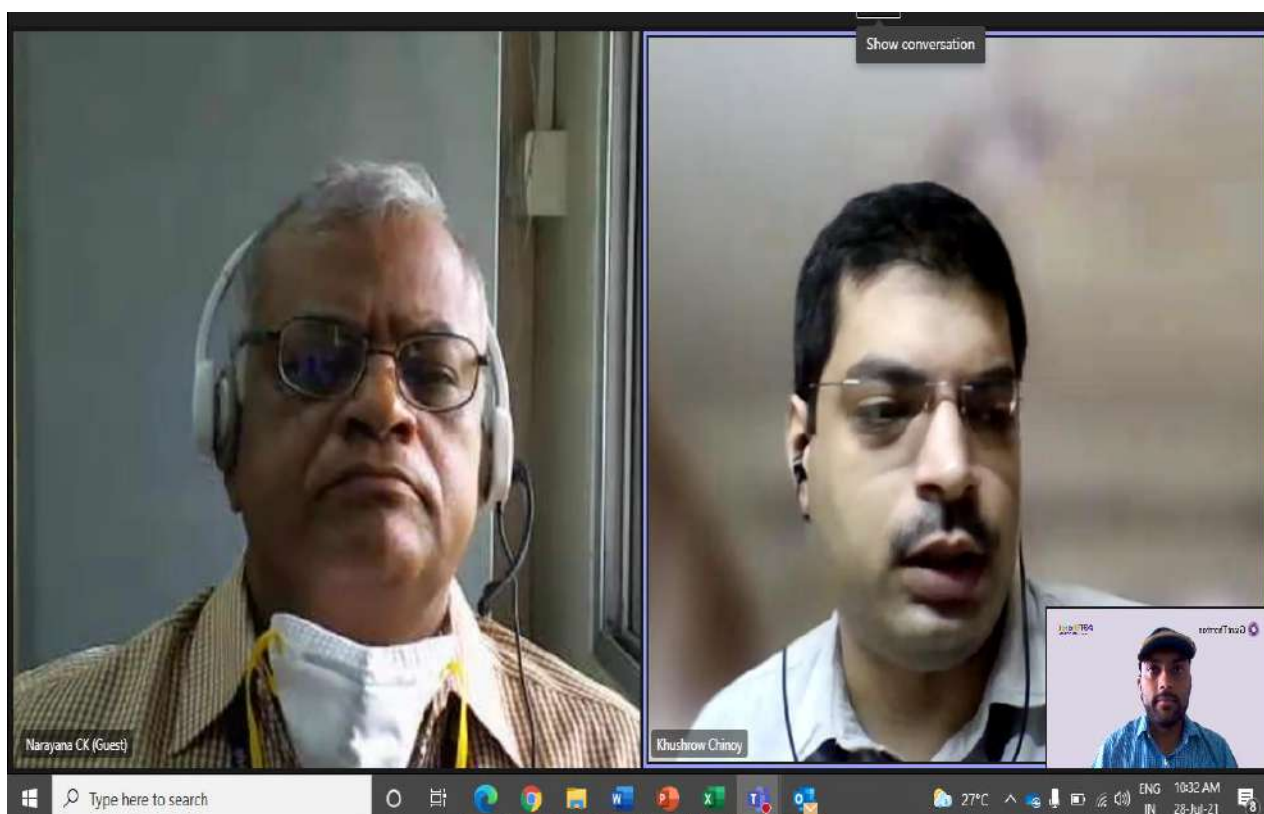
- Sold under the brand name "Keventer".
- Started operations from Kolkata and now they have presence in 4 states.
- Planning expansion to other states as well.



Brainstorming Workshop with Indian Institute of Horticulture Research (IIHR), Bengaluru

Key particulars	Details
Date and time of meeting	28 th July 2021
Key Participants	Dr. CK. Narayana, Head Incharge, Division of PHT & AE ICAR-IIHR, Bengaluru
Key discussion points	As per suggestion received from Honorable Director, IIFPT, Thanjavur. GT organized detailed consultative discussion with Indian Institute of Horticulture Research, Bengaluru. IIHR's Expertise • IIHR more than half a century old institute. Established in 1967 by Government of India • Largest institute working in Horticulture in Asia, in relation to manpower, quality and quantity of research • Has developed more than 200 varieties of fruits and vegetables hybrids • Developed more than 115 technologies related to production and post-harvesting • Research on increasing shelf life of crop produce • Have developed protocol for pre-harvest to post-harvest, packaging and transporting Quality Testing and Assurance • Quality standards more related more to safety standards • Standards of quality are set by FSSAI • Banana is it chemically and microbially safe for consumption • IIHR tests for 56 pesticides which are used by farmers during cultivation, sends the update on regular basis to ministry • For selling banana in domestic purpose nobody getting it tested • For exporting the quality requirements change from country to country Varietal Distribution and Particulars • Around 250 varieties of banana grown in India • 60% area is under Cavendish and rest 40% under non-cavendish varieties and plantains • Except TN majority of areas under G-9 cultivation. In TN apart from G-9, Poovan, Rasthali are majorly grown • Traditional varieties more nutritious, banana flour obtained from it more digestible compared to that obtained from G-9 • Banana flour used in making baby food and easily digestible • Nendran best for chips making Post Harvest Management and Value Addition • PH banana should be carried in crates which reduces transit loss to 1-2% • There is need of setting up banana flour making units at every integrated packhouse so that rejected banana can be converted to banana flour there itself • Banana flour can be used to make 'chapatis', awareness need to be created about its different usage • Its consumption can be scaled up as target is whole population • Apart from flour, its chips and puree are best VA products • Machinery cost very loess

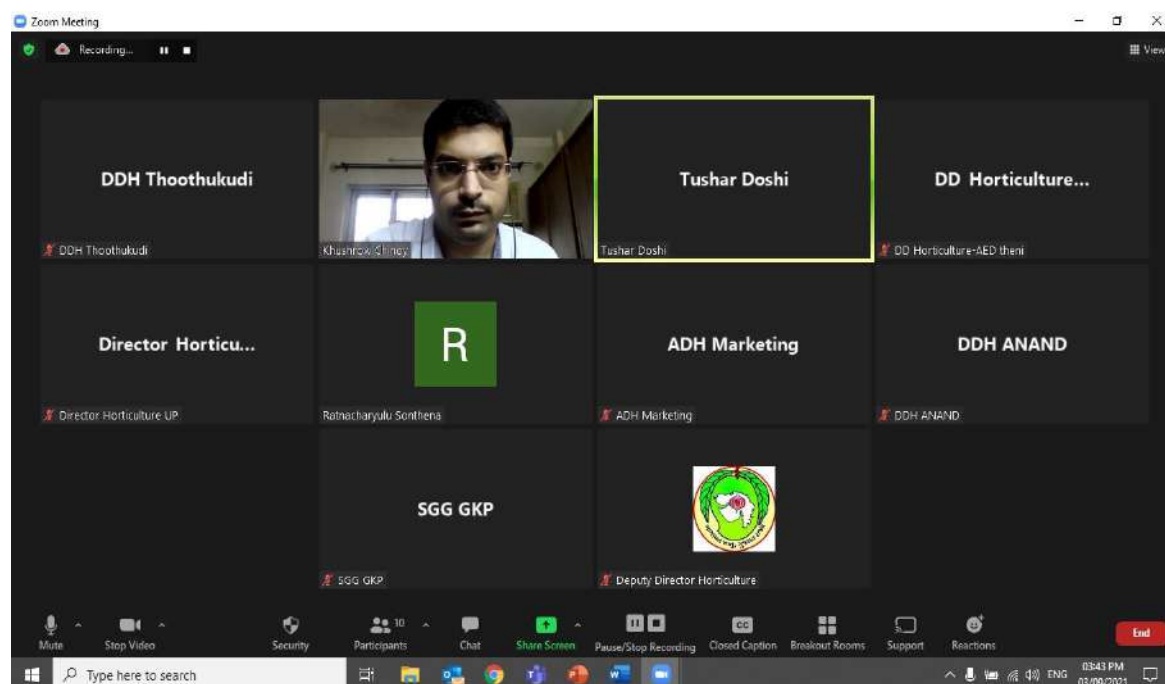
Snapshots



Validation Workshop with State Level Representatives

Key particulars	Details
Date and time of meeting	03/09/2021
Key Participants	Over 35 state and district level officials and representatives, processors from the states of Uttar Pradesh, Gujarat, Andhra Pradesh and Tamil Nadu
Key discussion points	Current banana ecosystem and recommendations 1. During the meeting, Grant Thornton Bharat shared cluster and state level findings with respective state level officials. 2. It was suggested that a packhouse facility should be present for every 3-4 villages in the cluster as they can be used for all horticulture crops and provides a facility for primary processing 3. Capacity building of farmers is required as they are lacking knowledge of fruit care practices and GAP and it was agreed that 50% of area should be under fruit care and GAP practices 4. Scope for value addition in banana puree and banana flour should be amplified. Current levels of processing is negligible and there is lack of estimates about respective data 5. There is a need for adoption of mobile packhouse as this will help in conducting primary processing at field itself. During rainy season it becomes difficult to perform primary processing at field. Mobile packhouses can solve the problem. 6. Frequent buyer seller meets should be conducted so that there is enhancement in exports and increase in sales from the region 7. Branding and GI tag initiatives should be undertaken in all states 8. There is immense potential in value addition in banana waste which is currently almost fully going to waste (banana pseudostem). Select avenues in this regard should be explored and expertise from SAUs should be taken on board (particularly Navsari Agriculture University)

Select Snapshots – Validation Workshop



Proposed interventions for Tamil Nadu Cluster-1

Value Chain Activity	Gaps Identified	Recommended Component	Numbers (A)	Units (MT, Ha, Sqm, set, participant etc.)	Indicative Per unit INR Lakh (B)	
Pre-Production and Production	Lack of Tissue Culture Planting Material	Tissue Culture Lab with Virus Indexing Facility & Hardening Center	9	1 with capacity of 2.3 Crore		5
	Limited number of Farmer Producer Organisations (FPOs)	FPOs formation and promotion	10	500 participants each	43	4.3
	Lack of GAP certification	Promotion and training of GAP	10000	Hectare	0.1	10
	Limited quality testing at farm and post farm levels	MRL, pesticide residue testing lab	1	Number	400	4
		Basic testing facility at district level	2	Number	25	0.5
Post-harvest Management and Value Addition	Dependency on traders for ripening	Ripening Chambers	6	10 MT each	10	0.6
	Lack of export promotion infrastructure	APEDA certified packhouse	0	At least 200 MT storage	400	0
	Limited packhouse	Farm level packing facility	50	50 MT PPC capacity	4	2
	Limited processing	Banana Chips/ Flour manufacturing unit	2	10 MT/day input capacity	75	1.5
		Banana Puree manufacturing unit	1	96 MT/day	1500	15
	Injury and loss due to improper packaging practices	Adoption of crates	100000	20 kg each	0.003	3
Logistics, Marketing and Branding	Limited availability of refrigerated transport	Refrigerated Transport Vehicles	10	15 MT reefer vans	30	3
	Limited branding, no GI tag	Branding, Digital and Marketing Initiatives	1	Approx. 10% investment towards holistic development of cluster brand	1.3295	1.32
Total						67.80

Introduction

Background of the study

- India's agriculture sector especially perishables like fruit and vegetables have grown significantly over the last two decades from a production of 43,001 million MT of Fruits and 88,662 million MT of Vegetables in 2001-02 to 99,069 million MT of Fruits and 191,760 million MT of Vegetables in 2019-20 (second advance estimate, NHB).
- However, the increase in production has not resulted in a proportionate increase in share in global trade which hovers around 1.7% and 0.5% for F&V respectively, significantly lower than many other countries.
- Amongst other reasons, a major reason is the insufficient and inefficient Infrastructure and Logistics Network across the supply chain. Farmers (especially S&M small and medium sized farmers) do not have access to proper post-harvest facilities like basic processing and storage infrastructure apart from lacking the global practices on harvest and post-harvest management.
- MoFPI is implementing Operation Greens scheme under which, as per Budget 2021, the scope has been expanded to 22 perishable products, which earlier was restricted only to onion, tomato and potato.
- In order to conduct an assessment in infrastructure and development of commodity value chain with regards to Banana, Grant Thornton Bharat has been appointed to carry out infrastructure gap assessment study encompassing 5 states and 8 clusters in India

Objectives of the project

Assess production and surplus quantity of produce in identified clusters	1	Availability of support facilities provided by State and Central Governments, availability of FPOs, cooperatives, SHGs and other groups	5
Current agri logistics and marketing channels of agriculture produce within district, cluster and outside state	2	Prominent markets within country and outside where agri products and processed products are being transported	6
Availability of primary and secondary processing infrastructure within the district/cluster	3	Availability of Rail Transport for movement of agri produce and level of utilization	7
Availability of appropriate testing facilities, their utilization and adequacy	4	Appropriateness of varieties and quality of agri produce being grown in cluster for food processing and suggestions for adoption of appropriate varieties	8

Proposed interventions for Tamil Nadu Cluster-1

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Pre-Production and Production	Lack of Tissue Culture Planting Material	Tissue Culture Lab with Virus Indexing Facility & Hardening Center	9	1 with capacity of 2.3 Crore		5
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	Lack of GAP certification	Promotion and training of GAP	10000	Hectare	0.1	10
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		Basic testing facility at district level	2	Number	25	0.5
Post-harvest Management and Value Addition	Dependency on traders for ripening	Ripening Chambers	6	10 MT each	10	0.6
	Lack of export promotion infrastructure	APEDA certified packhouse	0	At least 200 MT storage	400	0
	Limited packhouse	Farm level packing facility	50	50 MT PPC capacity	4	2
	Limited processing	Banana Chips/ Flour manufacturing unit	2	10 MT/day input capacity	75	1.5
		Banana Puree manufacturing unit	1	96 MT/day	1500	15
	Injury and loss due to improper packaging practices	Adoption of crates	100000	20 kg each	0.003	3
Logistics, Marketing and Branding	Limited availability of refrigerated transport	Refrigerated Transport Vehicles	10	15 MT reefer vans	30	3
	Limited branding, no GI tag	Branding, Digital and Marketing Initiatives	1	Approx. 10% investment towards holistic development of cluster brand	1.3295	1.32
Total						67.80

State level statistics gathered from primary and secondary sources

Gujarat

Annexure I: Production of horticultural crops in Gujarat (2019-20)

Table 1: Production of horticultural crops in Gujarat (2019-20) (Area in Ha., Production in MT)

S. No	District	FRUITS		VEGETABLES		SPICES		FLOWERS		TOTAL	
		Area	Prod	Area	Prod	Area	Prod	Area	Prod	Area	Prod
South Gujarat											
1	Surat	21114	714728	37783	600033	793	11529	417	4006	60107	1330297
2	Narmada	13603	709229	6207	82906	294	3685	172	1667	20276	797487
3	Bharuch	18450	995292	13288	205005	951	6146	1391	13160	34081	1219603
4	Dang	7058	40061	4191	67284	304	6178	220	1968	11773	115491
5	Navsari	45764	601435	23794	375249	1213	22764	2416	24452	73187	1023900
6	Valsad	44350	353330	12266	212898	251	2433	1402	14023	58269	582683
7	Tapi	10055	242665	22702	346134	1507	4793	459	4225	34723	597816
ZONE TOTAL		160394	3656739	120231	1889508	5313	57529	6477	63501	292415	5667277
Middle Gujarat											
1	Ahmedabad	4630	66071	26297	465101	30061	32652	2146	20583	63134	584407
2	Anand	22889	972603	41911	646497	318	5622	2106	20981	67224	1645703
3	Kheda	7527	168002	41408	839131	2547	9992	1937	19207	53419	1036332
4	Panchmahal	4797	55888	7300	109297	2410	19561	773	6930	15280	191676
5	Dahod	6511	58113	21922	328732	7556	86478	1564	14747	37553	488070
6	Vadodara	19441	672106	31274	577075	947	5972	1860	18038	53522	1273191
7	Mahisagar	4147	76570	7461	123237	2429	30517	330	3261	14367	233586
8	Chhota Udepur	12270	590684	14564	285428	790	11460	579	5589	28203	893160
ZONE TOTAL		82212	2660037	192137	3374498	47058	202253	11295	109337	332702	6346126
North Gujarat											
1	Banaskantha	18060	326843	91405	2385069	98606	114727	114	1068	208185	2827706
2	Patan	2665	34109	3401	74026	60669	42629	109	1002	66844	151765
3	Mehsana	21368	302620	28036	585791	16764	32444	202	1786	66370	922640
4	Sabarkantha	4371	89645	27032	803493	2031	6289	194	1849	33628	901277
5	Gandhinagar	5987	70625	21271	482731	1306	2652	401	3610	28965	559618
6	Aravalli	4888	122047	27130	830543	3587	13210	121	1181	35726	966982
ZONE TOTAL		57339	945889	198275	5161652	182963	211951	1141	10496	439717	6329988
Saurashtra-Kutch											
1	Kutch	56761	949115	13811	235166	72682	80407	176	1572	143430	1266260
2	Surendranagar	5899	64549	8430	142277	92671	117224	22	159	107022	324209
3	Rajkot	2374	26805	24701	436381	45094	83121	124	975	72293	547282
4	Jamnagar	2329	22401	6435	115195	30902	41843	350	3138	40016	182576
5	Porbandar	812	7896	1605	26242	42325	49549	44	352	44786	84040
6	Junagadh	15468	165884	13240	248332	35610	56947	265	2336	64583	473499
7	Amreli	9602	91693	14097	295533	3964	10542	35	249	27698	398016
8	Bhavnagar	24503	347673	34661	827795	1452	4383	164	1422	60780	1181273
9	Morbi	6279	82424	6036	96201	21286	32514	55	551	33656	211690
10	Botad	2246	29136	1591	27384	3275	4273	20	169	7132	60962
11	Gir Somnath	19104	178873	16276	306463	4681	8513	105	817	40167	494666
12	Devbhumi Dwarka	1119	31952	2796	47074	120092	135435	105	918	124112	215379
ZONE TOTAL		146495	1998400	143679	2804041	474034	624751	1465	12658	765674	5439850
GRAND TOTAL		446440	9261066	654322	13229700	709368	1096484	20378	195992	1830508	23783241

Source: Department of Horticulture, Gujarat

Table 2: District-wise estimated area & production of banana for the year 2019-20

S. No.	District	Banana	
		Area	Prod
South Gujarat			
1	Surat	8692	613829
2	Narmada	9240	662323
3	Bharuch	12286	896878
4	Dang	31	1208
5	Navsari	3183	176657
6	Valsad	1075	61006
7	Tapi	1293	77580
ZONE TOTAL		35800	2489481
Middle Gujarat			
1	Ahmedabad	149	7204
2	Anand	12710	826150
3	Kheda	990	56143
4	Panchmahal	417	15888
5	Dahod	7	182
6	Vadodara	6344	431963
7	Mahisagar	50	2075
8	Chhota Udepur	6950	483025
ZONE TOTAL		27617	1822630
North Gujarat			
1	Banaskantha	18	651
2	Patan	0	0
3	Mehsana	2	82
4	Sabarkantha	45	1925
5	Gandhinagar	0	0
6	Aravalli	75	3180
ZONE TOTAL		140	5838
Saurashtra-Kutch			
1	Kutch	2685	152777
2	Surendranagar	0	0
3	Rajkot	39	1400
4	Jamnagar	3	126
5	Porbandar	5	150
6	Junagadh	550	26538
7	Amreli	229	8006
8	Bhavnagar	1753	84109
9	Morbi	8	307
10	Botad	0	0
11	Gir Somnath	695	35590
12	Devbhumi Dwarka	13	572
ZONE TOTAL		5980	309574
GRAND TOTAL		69537	4627523

Source: Gujarat Horticulture Department

Table 3: Details of MIDH assisted Ripening Chambers in Gujarat

S. No.	District	No. of Ripening Chamber Units	Capacity (MT)
1	Ahmedabad	5	337
2	Amreli	0	0
3	Anand	7	603
4	Aravalli	0	0
5	Bharuch	2	165
6	Bhavnagar	2	37
7	Banaskantha	3	607
8	Botad	0	0
9	Chhota Udepur	0	0
10	Dahod	1	129
11	Dang	0	0
12	Devbhumi	0	0
13	Gandhinagar	2	70
14	Gir Somnath	1	302
15	Jamnagar	1	38
16	Junagadh	2	70
17	Kheda	0	0
18	Kutch	4	115
19	Mahisagar	0	0
20	Mehsana	1	51
21	Morbi	1	85
22	Narmada	3	277
23	Navsari	4	195
24	Panchmahals	0	0
25	Patan	1	56
26	Porbandar	0	0
27	Rajkot	4	199
28	Sabarkantha	9	998
29	Surat	0	0
30	Surendranagar	1	60
31	Tapi	0	0
32	Vadodara	9	419
33	Valsad	2	376
Total		65.00	5189

Source: Gujarat Horticulture Department

Table 4: Details of MIDH assisted Tissue Culture Labs in Gujarat

S. No.	District	No. of Tissue Culture Unit	Capacity (Lakh Tissue plants)
1	Ahmedabad	2	40
2	Amreli	0	0
3	Anand	6	125
4	Aravalli	0	0
5	Bharuch	2	40
6	Bhavnagar	0	0
7	Banaskantha	1	15
8	Botad	0	0
9	Chhotaudepur	0	0
10	Dahod	0	0
11	Dang	0	0
12	Devbhumi	0	0
13	Gandhinagar	2	40
14	GirSomnath	0	0
15	Jamnagar	0	0
16	Junagadh	0	0
17	Kheda	4	90
18	Kutch	2	30
19	Mahisagar	0	0
20	Mehsana	3	55
21	Morbi	0	0
22	Narmada	0	0
23	Navsari	2	40
24	Panchmahals	0	0
25	Patan	0	0
26	Porbandar	0	0
27	Rajkot	2	30
28	Sabarkantha	0	0
29	Surat	1	15
30	Surendranagar	0	0
31	Tapi	0	0
32	Vadodara	2	40
33	Valsad	0	0
Total		29.00	560.00

Source: Horticulture Department, Gujarat

Table 5: Details of Cold Storage units in Gujarat

S. No.	District	No. of Cold Storage	Capacity (MT)
1	Ahmedabad	13	32334.00
2	Amreli	1	1414.00
3	Anand	8	15316.00
4	Aravalli	21	100063.00
5	Bharuch	0	0.00
6	Bhavnagar	55	209988.00
7	Banaskantha	132	485061.00
8	Botad	1	3439.00
9	Chhotaudepur	0	0.00
10	Dahod	0	0.00
11	Dang	0	0.00
12	Devbhumi	0	0.00
13	Gandhinagar	49	197333.00
14	Gir Somnath	2	5483.00
15	Jamnagar	9	28580.00
16	Junagadh	6	16400.00
17	Kheda	6	16470.00
18	Kutch	2	2502.00
19	Mahisagar	0	0.00
20	Mehsana	15	64425.00
21	Morbi	6	14993.00
22	Narmada	0	0.00
23	Navsari	3	5780.00
24	Panchmahal	2	10000.00
25	Patan	0	0.00
26	Porbandar	1	1014.00
27	Rajkot	37	126324.00
28	Sabarkantha	49	195864.00
29	Surat	8	18741.00
30	Surendranagar	1	1981.00
31	Tapi	2	3750.00
32	Vadodara	6	14370.00
33	Valsad	6	15223.00
Total		441	1586848.00

Source: Horticulture Department, Gujarat

Table 6: Details of Integrated Packhouses in Gujarat

S. No.	District	No. of APEDA certified integrated packhouse	Fruits and Vegetables
1	Ahmedabad	1	Mango, Pomegranates, Other fruits, other vegetables
2	Amreli	0	NA
3	Anand	0	NA
4	Aravalli	0	NA
5	Bharuch	1	Mango, Pomegranate, Other Fruits, other vegetables
6	Bhavnagar	0	NA
7	Banaskantha	0	NA
8	Botad	0	NA
9	Chhotaudepur	0	NA
10	Dahod	0	NA
11	Dang	0	NA
12	Devbhumi	0	NA
13	Gandhinagar	0	NA
14	Gir Somnath	1	Mango, Pomegranate, Other Fruits, other vegetables
15	Jamnagar	0	NA
16	Junagadh	1	Mango, Pomegranate, Other Fruits, other vegetables
17	Kheda	0	NA
18	Kutch	0	NA
19	Mahisagar	0	NA
20	Mehsana	0	NA
21	Morbi	0	NA
22	Narmada	0	NA
23	Navsari	1	Mango, other vegetables
24	Panchmahal	0	NA
25	Patan	0	NA
26	Porbandar	0	NA
27	Rajkot	1	Grapes, Mango, Pomegranate, other fruits, other vegetables
28	Sabarkantha	1	Other Fruits, Okra, Chilly, Herbs, other vegetables
29	Surat	0	NA
30	Surendranagar	0	NA
31	Tapi	2	Mango, Pomegranate, Other Fruits, Okra, Chilly Other Vegetables
32	Vadodara	0	NA
33	Valsad	2	Mango, Pomegranate, Okra, Bitter Gourd, Chilly, other vegetables
Total		11	NA

Source: APEDA website (While making report and recommendations the data of primary survey has been taken into consideration)

Table 7: Details of FPOs in Gujarat (all commodities)

S. No.	District	No. of FPOs
1	Ahmedabad	6
2	Amreli	4
3	Anand	4
4	Aravalli	2
5	Bharuch	10
6	Bhavnagar	1
7	Banaskantha	3
8	Botad	2
9	Chhota Udepur	6
10	Dahod	32
11	Dang	0
12	Devbhumi	0
13	Gandhinagar	1
14	Gir Somnath	2
15	Jamnagar	0
16	Junagadh	3
17	Kheda	0
18	Kutch	3
19	Mahisagar	4
20	Mehsana	1
21	Morbi	0
22	Narmada	13
23	Navsari	6
24	Panchmahal	3
25	Patan	0
26	Porbandar	0
27	Rajkot	0
28	Sabarkantha	5
29	Surat	0
30	Surendranagar	5
31	Tapi	1
32	Vadodara	0
33	Valsad	0
Total		117

Source: NABARD website (For report purpose and analysis, data of primary survey has been taken into consideration)

Table 8: Drip Irrigation/MIS details of Gujarat

District	Area of banana cultivation under drip/MIS (Ha)							
	<=1 Ha		1 Ha & <= 2Ha		2 Ha & <= 5Ha		>5 Ha	
	No	Ha	No	Ha	No	Ha	No	Ha
Anand	10	308.97	171	599.89	420	491	457	59.52
Bharuch	40	1420.56	580	2098.11	1496	1641.43	2072	289.64
C. Udepur	2	112.03	80	268.07	160	222.15	193	12.29
Narmada	41	2072.98	636	3126.49	2228	1739.19	3114	260.05
Surat	18	841.45	436	1913.3	1143	1164.33	1356	104.71
Vadodara	48	629.65	463	1381.51	913	1328.75	966	348.47
Total	159	5385.64	2366	9387.37	6360	6586.85	8158	1074.68

Source: Gujarat Green Revolution Company (GGRC)

Uttar Pradesh

Table 9: District Wise Fruits and vegetables production in Uttar Pradesh (2019-20)

S. No.	District	Fruits		Vegetables		Total	
		Area	Production	Area	Production	Area	Production
1	AGRA	4334	90206	80636	1894299	84970	1984505
2	ALIGARH	12573	261263	32153	742422	44726	1003685
3	ALLAHABAD	5752	100062	24804	488665	30556	588727
4	AMBEDKAR NAGAR	5636	98164	12827	271803	18463	369967
5	AMETHI	4780	84049	13680	282641	18460	366690
6	AMROHA	10991	239282	11886	241666	22877	480948
7	AURAIYA	586	14405	8217	207776	8803	222181
8	AZAMGARH	3263	44629	14013	232526	17276	277155
9	BAGHPAT	2245	39531	4690	136050	6935	175581
10	BAHRAICH	7316	226365	8884	173519	16200	399884
11	BALLIA	5900	103770	15867	303415	21767	407186
12	BALRAMPUR	2471	51956	4041	77125	6512	129081
13	BANDA	621	8145	3674	81895	4295	90041
14	BARABANKI	5694	101815	23093	483959	28787	585774
15	BAREILLY	5682	106486	14002	310027	19684	416513
16	BASTI	5706	84771	8724	142319	14430	227090
17	BIJNOR	6844	138826	3746	102588	10591	241414
18	BUDAUN	7281	146525	29290	682101	36571	828626
19	BULANDSHAHR	19240	339331	19840	489123	39080	828455
20	CHANDAUJI	218	3372	3915	74925	4133	78297
21	CHITRAKOOT	605	5726	2693	66019	3298	71745
22	DEORIA	4265	46866	4916	95301	9181	142167
23	ETAH	8548	199520	22937	447998	31485	647518
24	ETAWAH	993	29613	20883	509989	21876	539602
25	FAIZABAD	9007	197559	14516	346511	23523	544070
26	FARRUKHABAD	9646	255206	47811	1192026	57457	1447232
27	FATEHPUR	10596	420945	24837	410507	35433	831452
28	FIROZABAD	3262	106326	59392	1321771	62654	1428097
29	GAUTAM BUDDHA NAGAR	231	5939	1176	31074	1407	37013
30	GHAZIABAD	532	9694	7030	211452	7562	221146
31	GHAZIPUR	3421	59918	17598	409377	21019	469295
32	GONDA	2803	41736	14201	282824	17005	324560
33	GORAKHPUR	25848	954242	10774	209991	36622	1164233
34	HAMIRPUR	346	4239	8254	138902	8601	143141
35	HAPUR	4196	69825	7405	172970	11601	242794
36	HARDOI	7485	139414	22875	520001	30360	659414
37	HATHRAS	6375	125994	55927	1253806	62301	1379800
38	JALAUN	285	5597	57227	869839	57512	875436
39	JAUNPUR	3053	52575	17775	382233	20828	434808
40	JHANSI	686	13799	37003	577542	37689	591341
41	KANNAUJ	2360	72240	61182	1479596	63542	1551836
42	KANPUR DEHAT	953	24660	10193	226681	11146	251341
43	KANPUR NAGAR	3636	83746	28071	581266	31707	665012
44	KASGANJ	11613	243370	15653	304952	27266	548322
45	KAUSHAMBI	11898	420904	8600	160604	20498	581508
46	KHERI	6765	175341	7381	171999	14146	347340
47	KUSHI NAGAR	21919	785180	6625	132333	28544	917513
48	LALITPUR	1186	18180	35120	506669	36305	524849
49	LUCKNOW	33152	700678	16840	382719	49992	1083397

S. No.	District	Fruits		Vegetables		Total	
		Area	Production	Area	Production	Area	Production
50	MAHARAJGANJ	10786	306278	7053	142924	17839	449202
51	MAHOBA	295	4054	15775	194749	16070	198803
52	MAINPURI	4012	141647	29477	701368	33489	843015
53	MATHURA	2810	43595	16667	391685	19477	435280
54	MAU	650	10855	3751	67769	4401	78624
55	MEERUT	10806	181194	16478	416423	27284	597617
56	MIRZAPUR	1668	27417	10466	176860	12134	204277
57	MORADABAD	5164	108988	10244	241615	15408	350603
58	MUZAFFARNAGAR	8440	142491	6372	155050	14812	297541
59	PILIBHIT	2513	75359	4778	120800	7291	196159
60	PRATAPGARH	19373	222506	12257	217824	31630	440330
61	RAE BARELI	1468	45882	12914	282000	14383	327882
62	RAMPUR	5273	121266	10595	262449	15868	383715
63	SAHARANPUR	32571	626911	7589	189834	40160	816745
64	SAMBHAL	2987	71674	13721	306377	16708	378051
65	SANT KABEER NAGAR	5860	117042	6759	98787	12619	215829
66	SANT RAVIDAS NAGAR	2395	41186	3150	62823	5545	104009
67	SHAHJAHANPUR	3116	50526	23406	552592	26522	603118
68	SHAMLI	5348	115573	1656	35600	7004	151173
69	SHRAVASTI	1718	34758	2290	51054	4008	85812
70	SIDDHARTH NAGAR	2843	42692	5647	84329	8490	127021
71	SITAPUR	18424	243601	11360	238407	29784	482007
72	SONBHADRA	75	1389	6471	114191	6546	115580
73	SULTANPUR	11169	182856	16811	336028	27980	518884
74	UNNAO	21531	484964	26930	647497	48461	1132461
75	VARANASI	2755	64152	10178	221272	12933	285424
	Total	496852	11060837	1275670	27126100	1772522	38186937

Source: Uttar Pradesh Horticulture Department, Area in ha and Production in MT

Table 10: District Wise Banana production in Uttar Pradesh (2019-20)

S. No.	Name of District	Banana	
		Area	Production
1	AGRA	8	374
2	ALIGARH	30	1517
3	ALLAHABAD	129	6168
4	AMBEDKAR NAGAR	114	4974
5	AMETHI	0	0
6	AMROHA	7	335
7	AURAIYA	12	418
8	AZAMGARH	11	513
9	BAGHPAT	28	1349
10	BAHRAICH	3354	151479
11	BALLIA	413	18869
12	BALRAMPUR	348	15703
13	BANDA	26	1249
14	BARABANKI	540	24160
15	BAREILLY	26	1062
16	BASTI	167	7938
17	BIJNOR	0	0
18	BUDAUN	82	3498
19	BULANDSHAHR	11	545
20	CHANDAULI	0	0
21	CHITRAKOOT	5	233
22	DEORIA	35	1463
23	ETAH	15	809
24	ETAWAH	11	361
25	FAIZABAD	666	29558
26	FARRUKHABAD	691	32549
27	FATEHPUR	8119	385099
28	FIROZABAD	10	505
29	GAUTAM BUDDHA NAGAR	0	0
30	GHAZIABAD	29	1419
31	GHAZIPUR	12	571
32	GONDA	75	3550
33	GORAKHPUR	19498	883624
34	HAMIRPUR	4	193
35	HAPUR	0	0
36	HARDOI	75	3462
37	HATHRAS	28	1412
38	JALAUN	0	0
39	JAUNPUR	41	1986
40	JHANSI	0	0
41	KANNAUJ	13	408
42	KANPUR DEHAT	79	3834
43	KANPUR NAGAR	0	0
44	KASGANJ	74	3494
45	KAUSHAMBI	6855	326727
46	KHERI	1943	89964
47	KUSHI NAGAR	15543	706390
48	LALITPUR	0	0
49	LUCKNOW	1307	60244

S. No.	Name of District	Banana	
		Area	Production
50	MAHARAJGANJ	5801	259371
51	MAHOBA	0	0
52	MAINPURI	1207	56569
53	MATHURA	11	560
54	MAU	13	601
55	MEERUT	77	3688
56	MIRZAPUR	28	1251
57	MORADABAD	65	3035
58	MUZAFFARNAGAR	20	965
59	PILIBHIT	1147	53235
60	PRATAPGARH	28	1399
61	RAE BARELI	773	34335
62	RAMPUR	8	372
63	SAHARANPUR	0	0
64	SAMBHAL	0	0
65	SANT KABEER NAGAR	1009	47778
66	SANT RAVIDAS NAGAR	0	0
67	SHAHJAHANPUR	12	786
68	SHAMLI	404	18585
69	SHRAVASTI	229	11004
70	SIDDHARTH NAGAR	77	3803
71	SITAPUR	604	28198
72	SONBHADRA	4	182
73	SULTANPUR	409	18200
74	UNNAO	209	9654
75	VARANASI	16	558
	Total	72585	3332135

Source: Uttar Pradesh, Horticulture Department, Area in ha and Production in MT

Table 11: Details of MIDH assisted Ripening Chamber units in Uttar Pradesh

District name	Number of ripening chambers	Capacity of ripening chamber (MT)
Fatehpur	-	-
Gorakhpur	02	69.13
Kaushambi	-	-
Kushinagar	-	-
Maharajganj	-	-
Total	02	69.13

Source: Department of Statistics, Uttar Pradesh

Table 12: Details of the MIDH assisted packhouses in the districts of the clusters of Uttar Pradesh

District name	Number of Packhouses pertaining to banana
Fatehpur	22
Gorakhpur	08
Kaushambi	24
Kushinagar	-
Maharajganj	-
Total	54

Source: Horticulture department, Uttar Pradesh

Table 13: Details of integrated packhouses in Uttar Pradesh

S. No.	District	No. of APEDA certified integrated packhouse	Fruits and Vegetables
1	Lucknow	1	Mango, other fruits, Okra, Chilly, other vegetables
2	Saharanpur	1	Mango, Other Fruits, Chilly, Other Vegetables
3	Gautam Buddha Nagar	1	Mango, Other Vegetables
4	Greater Noida	1	Grapes, Mango, Other Fruits
Total		4	NA

Source: APEDA (For recommendation purpose the data of primary survey has been used)

Table 14: Details of FPOs in Uttar Pradesh (all commodities)

S. No.	District	No. of FPOs
1	Agra	5
2	Allahabad	2
3	Bahraich	1
4	Balrampur	1
5	Barabanki	5
6	Bareilly	1
7	Bhadohi	1
8	Bijnor	5
9	Budaun	1
10	Deoria	2
11	Etah	1
12	Etawah	2
13	Farrukhabad	2
14	Fatehpur	4
15	Gautam Buddh Nagar	4
16	Ghaziabad	2
17	Ghazipur	1
18	Gorakhpur	2
19	Hamirpur	5
20	Jalaun	1
21	Jaunpur	2
22	Jhansi	6
23	Kanpur	6
24	Kanpur Dehat (Ramabai Nagar)	2
25	Kushinagar	2
26	Lakhimpur Kheri	1
27	Lalitpur	11
28	Lucknow	2
29	Maharajganj	1
30	Mahoba	4
31	Mathura	1
32	Meerut	4

S. No.	District	No. of FPOs
33	Mirzapur	1
34	Muzaffarnagar	2
35	Pilibhit	2
36	Pratapgarh	5
37	Raebareli	4
38	Shahjahanpur	1
39	Shamli	1
40	Siddharthnagar	1
41	Sitapur	4
42	Sonbhadra	4
43	Unnao	1
	Grand Total	116

Source: NABARD (For recommendation purpose the data of primary survey has been used)

Table 15: List of Cold storage facilities in cluster districts and their capacities in MT

Sr. No.	Name of District	Name of Facility	Capacity (MT)
1	Fatehpur	Uday Cold Storage and Ice Plant Private Limited	18620
2	Fatehpur	Uday Cold Storage and Ice Plant Private Limited	5875
3	Fatehpur	Sunder Cold Storage and Ice Plant Private Ltd	9909
4	Fatehpur	Rastogi and Ice and Cold Storage	12745
5	Fatehpur	Asia Cold Storage and Ice Plant Private Limited	10004
6	Fatehpur	Gopal Sheetgruh	10388
7	Fatehpur	Sadhu Cold Storage	6402
8	Fatehpur	Fatehpur Cold Storage	6036
9	Fatehpur	Rajrani Cold Storage and Ice Plant Private Ltd	5289
10	Fatehpur	Jawahar Cold Storage and Ice Factory	5179
11	Fatehpur	Jawahar Cold Storage and Ice Factory	5250
12	Fatehpur	Shankar Bardani Cold Storage Private Limited	13834
13	Fatehpur	Indra Sheetgruh and Ice plant	5972
14	Fatehpur	Mamta Agrotech Pvt Ltd	5914
15	Kaushambhi	Kaushambhi Cold Storage	11827
16	Kaushambhi	Himadri Cold Storage	17077
17	Kaushambhi	Sharda Cold Storage	11854
18	Kaushambhi	Shivsagar Cold Storage	6800
19	Kaushambhi	Sai Krupa Cold Storage	12610
20	Kaushambhi	Kamadgiri Cold Storage	5744
21	Kaushambhi	Hira Cold Storage and Ice Factory	13919
22	Kaushambhi	Raj Cold Storage	4945
23	Kaushambhi	Jed A Cold storage and Ice Factory	5424
24	Kaushambhi	Kamtanath Cold Storage and Ice Factory	5442

Sr. No.	Name of District	Name of Facility	Capacity (MT)
25	Gorakhpur	Vishwanath Sheetgruh and Ice Factory	9234
26	Gorakhpur	Janki Jeevan Sheetgruh	5931
27	Gorakhpur	Jaiswal Sheetgruh	4612
28	Gorakhpur	Purvanchal Sheetgruh	3445
29	Gorakhpur	V R Giri Cold Storage Kurauli	6548
30	Gorakhpur	Saadi Agro and Allied Industries	6526
31	Gorakhpur	Laxmi Cold Storage and Ice Industries	7980
32	Gorakhpur	Vijay Bahadur Sheetgruh	3765
33	Gorakhpur	Shivrupa Cold Storage and Ice Factory	8718
34	Maharajganj	Prem Cold Storage	3637
35	Maharajganj	Mathura Cold Storage	2494
36	Maharajganj	Gangotri Associates and Cold Storage	5446
37	Maharajganj	Jain Cold Storage and Ice Factory	5555
38	Maharajganj	Ramdoi Cold Storage	4271
39	Maharajganj	Nayak Cold Storage Private Ltd	8552
40	Maharajganj	PD Saraf Cold Storage Private Ltd	2750
41	Kushi Nagar	Laari Cold Storage, Kasia	12608
	TOTAL	-	319131

Source: Directorate of Horticulture and Food Processing, Uttar Pradesh

Maharashtra

Table 16: District Wise Production of fruits in Maharashtra (2018-19)

S. No.	District	Total	
		Area (Ha.)	Production (MT)
1	AHMEDNAGAR	46750	441109
2	AKOLA	12134	219476
3	AMRAVATI	77009	648986
4	AURANGABAD	33408	395796
5	BEED	12090	84641
6	BHANDARA	337	2983
7	BULDHANA	5853	73367
8	CHANDRAPUR	147	671
9	DHULE	12910	209940
10	GADCHIROLI	2604	20834
11	GONDIA	1085	9842
12	HINGOLI	1547	14422
13	JALGAON	56229	2800350
14	JALNA	24430	185404
15	KOLHAPUR	3340	12881
16	LATUR	4552	34624
17	NAGPUR	27331	162486
18	NANDED	11277	417854
19	NANDURBAR	7149	256703
20	NASHIK	102206	2019776
21	OSMANABAD	14726	163438
22	PALGHAR	13057	55663
23	PARBHANI	7452	78011
24	PUNE	38883	457173
25	RAIGAD	12692	40869
26	RATNAGIRI	61210	153601
27	SANGLI	31632	671101
28	SATARA	9172	96611
29	SINDHUDURG	33788	131607
30	SOLAPUR	75148	787716
31	THANE	2441	4776
32	WARDHA	7292	49069
33	WASHIM	3960	63517
34	YAVATMAL	3463	43478
	Total	757305	10808775

Source: Horticulture Department, Maharashtra

Table 17: District-wise banana production in Maharashtra (2018-19)

S. No.	District	Banana	
		Area (Ha.)	Production (MT)
1	AHMEDNAGAR	680	17000
2	AKOLA	2610	95265
3	AMRAVATI	627	5622
4	AURANGABAD	620	20460
5	BEED	148	5700
6	BHANDARA	30	926
7	BULDHANA	538	16152
8	CHANDRAPUR	0	0
9	DHULE	1384	73352
10	GADCHIROLI	6	210
11	GONDIA	78	2340
12	HINGOLI	550	1400
13	JALGAON	40112	2607280
14	JALNA	140	3243
15	KOLHAPUR	265	1750
16	LATUR	24	960
17	NAGPUR	29	966
18	NANDED	7520	315670
19	NANDURBAR	2766	96824
20	NASHIK	11	34
21	OSMANABAD	530	13100
22	PALGHAR	236	9205
23	PARBHANI	650	32500
24	PUNE	2415	89355
25	RAIGAD	70	2073
26	RATNAGIRI	121	150
27	SANGLI	1075	33852
28	SATARA	375	13126
29	SINDHUDURG	645	13314
30	SOLAPUR	3389	74725
31	THANE	16	17
32	WARDHA	230	6440
33	WASHIM	60	1800
34	YAVATMAL	186	5450
	Total	68136	3560261

Source: Horticulture Department, Maharashtra

Table 18: List of APEDA Certified Packhouses in Maharashtra

S. No.	District	No. of APEDA certified integrated packhouse	Fruits and Vegetables
1	AHMEDNAGAR	1	Mango, Grapes, Pomegranate, Other Fruits, Okra, Bitter Gourd, Chilly, Other Vegetables
2	AKOLA	0	NA
3	AMRAVATI	0	NA
4	AURANGABAD	0	NA
5	BEED	0	NA
6	BHANDARA	0	NA
7	BULDHANA	1	Grapes, Pomegranate
8	CHANDRAPUR	0	NA
9	DHULE	0	NA
10	GADCHIROLI	0	NA
11	GONDIA	0	NA
12	HINGOLI	0	NA
13	JALGAON	0	NA
14	JALNA	0	NA
15	KOLHAPUR	1	Herbs
16	LATUR	4	Grapes, Pomegranate, Other Fruits, Okra, Bitter Gourd, Chilly, Other Vegetables
17	MUMBAI CITY	2	Grapes, Pomegranate, Other Fruits, Okra, Other Vegetables
18	MUMBAI SUBURBAN	8	Mango, Grapes, Pomegranate, Other Fruits, Other Vegetables
19	NAGPUR	0	NA
20	NANDED	0	NA
21	NANDURBAR	0	NA
22	NASHIK	103	Pomegranate, Grapes
23	OSMANABAD	0	NA
24	PALGHAR	0	NA
25	PARBHANI	0	NA
26	PUNE	12	NA
27	RAIGAD	0	NA
28	RATNAGIRI	1	NA
29	SANGLI	20	NA
30	SATARA	3	NA
31	SINDHUDURG	0	NA
32	SOLAPUR	4	NA
33	THANE	1	NA
34	WARDHA	0	NA
35	WASHIM	0	NA
36	YAVATMAL	0	NA
Total		161	NA

Source: APEDA (For recommendation purpose, data and inputs gathered during stakeholder primary survey has been used)

Table 19: Details of FPOs in Maharashtra (all commodities)

S. No.	Name of District	No. of FPOs
1	Ahmednagar	4
2	Akola	1
3	Amravati	4
4	Aurangabad	19
5	Beed	11
6	Buldhana	5
7	Chandrapur	6
8	Dhule	3
9	Gadchiroli	2
10	Jalgaon	16
11	Jalna	2
12	Latur	3
13	Nagpur	1
14	Nanded	2
15	Nandurbar	2
16	Nashik	15
17	Osmanabad	3
18	Pune	1
19	Satara	3
20	Thane	2
21	Wardha	12
22	Washim	2
23	Grand Total	119

Source: NABARD (For recommendation purpose the data of primary survey has been used)

Andhra Pradesh

Table 20: District Wise Horticulture production in Andhra Pradesh (2019-20)

District	FRESH FRUITS		VEGETABLES		PLANTATIONS		SPICES		FLOWERS		TOTAL MEDICINAL & AROMATIC PLANTS		TOTAL	
	Area	Productn	Area	Prodn	Area	Prodn	Area	Prodn	Area	Prodn	Area	Prodn	Area	Prodn
Srikakulam	19073	448297	4040	105250	46766	96868	2098	14464	86	1516	0	0	72063	666395
Vizianagram	88492	2103546	30613	942744	19496	237789	7753	44582	2435	23858	244	488	149033	3353007
Visakhapatnam	21758	398830	7616	211063	114302	214353	31738	190975	453	5121	0	0	175867	1020342
East Godavari	38037	1322646	20523	432547	126716	633257	3932	30463	834	12436	24	78	190066	2431427
West Godavari	28651	1064886	6089	184768	157249	1730489	2254	14418	203	2587	23	154	194469	2997302
Krishna	81994	1179025	8739	281240	21170	343938	18889	133412	467	3823	0	0	131259	1941438
Guntur	11775	477424	10049	253222	234	115	85560	551016	205	1732	1115	5989	108938	1289498
Prakasam	25989	607770	6707	215499	997	957	35533	216210	544	4514	0	0	69770	1044950
Nellore	39226	778208	6630	189205	2204	16424	2982	24786	996	15663	182	1064	52220	1025350
Chittoor	133159	2532876	50568	1644012	2615	1399	8105	74657	9650	202777	1021	2164	205118	4457885
YSR Kadapa	81781	2870354	15292	459080	0	0	9334	96056	4870	164128	39	690	111316	3590308
Anantapur	152325	3917794	39499	1233661	2407	8744	10966	82681	3672	44707	248	496	209117	5288083
Kurnool	17880	504713	58478	1386184	0	0	34683	193858	2218	42432	1682	841	114941	2128028
Total	740140	18206369	264843	7538475	494156	3284333	253827	1667578	26633	525294	4578	11964	1784177	31234013

Source: Horticulture Department, Andhra Pradesh, Area in ha and Production in MT

Table 21: District Wise Production of banana in Andhra Pradesh (2019-20)

District	Banana	
	Area	Production
Srikakulam	3152	189120
Vizianagaram	5800	348000
Visakhapatnam	1383	82980
East Godavari	15769	946140
West Godavari	13125	787500
Krishna	4771	286260
Guntur	4996	299760
Prakasam	380	22800
Nellore	962	57720
Chittoor	2384	143040
YSR Kadapa	25887	1553220
Anantapur	15116	906960
Kurnool	3970	238200
Total	97695	5861700

Source: Horticulture Department, Andhra Pradesh, Area in ha and Production in MT

Table 22: District wise details of Cold Storage units in Andhra Pradesh (2019-20)

S. No.	Name of the District	Cold Storage units	
		(Nos.)	Capacity (MTs)
1	Srikakulam	4	15000
2	Vizianagaram	9	37500
3	Vizag	6	21500
4	East Godavari	10	42500
5	West Godavari	7	24350
6	Krishna	28	179000
7	Guntur	127	758500
8	Prakasham	104	561640
9	Nellore	7	37500
10	Chittoor	22	117500
11	Kadapa	2	13000
12	Ananthapur	14	84500
13	Kurnool	33	168000
Total		373	2060490

Source: Horticulture Department, Andhra Pradesh

Table 23: District wise details of Ripening Chamber in Andhra Pradesh

S. No	Name of the District	Ripening Chambers	
		(Nos.)	Capacity (MTs)
1	Srikakulam	2	20
2	Vizianagaram	6	110
3	Vizag	6	323
4	East Godavari	4	76
5	West Godavari	3	73
6	Krishna	16	1106
7	Guntur	8	304
8	Prakasham	15	717
9	Nellore	8	361
10	Chittoor	116	47969
11	Kadapa	29	985
12	Ananthapur	12	779
13	Kurnool	22	1494
Total		247	54316

Source: Andhra Pradesh Horticulture Department

Table 24: District wise details of Packhouses in Andhra Pradesh

S. No.	Name of the District	Packhouses	
		Nos.	Capacity (MTs)
1	Srikakulam	76	2280
2	Vizianagaram	206	6180
3	Vizag	129	3870
4	East Godavari	422	12660
5	West Godavari	279	8370
6	Krishna	12	360
7	Guntur	75	2250
8	Prakasham	171	5130
9	Nellore	438	13140
10	Chittoor	570	17100
11	Kadapa	1156	34680
12	Ananthapur	846	25380
13	Kurnool	382	11460
Total		4762	142860

Source: Andhra Pradesh Horticulture Department

Table 25: District wise details of Integrated Packhouses in Andhra Pradesh

S. No.	Name of the District	Integrated Packhouses	
		Nos.	Capacity
1	Srikakulam	-	-
2	Vizianagaram	1	16
3	Vizag	-	-
4	East Godavari	1	10
5	West Godavari	1	-
6	Krishna	2	32
7	Guntur	-	-
8	Prakasham	-	-
9	Nellore	-	-
10	Chittoor	4	64
11	Kadapa	1	16
12	Ananthapur	6	96
13	Kurnool	-	-

Total	16	234
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Source: Horticulture Department, Andhra Pradesh

Table 26: Details of collection centres/packhouses in Andhra Pradesh

S. No.	Name of the District	Collection Centres	
		Nos.	Capacity (MTs)
1	Srikakulam	9	3150
2	Vizianagaram	-	-
3	Vizag	-	-
4	East Godavari	1	350
5	West Godavari	-	-
6	Krishna	-	-
7	Guntur	-	-
8	Prakasham	-	-
9	Nellore	-	-
10	Chittoor	9	3150
11	Kadapa	3	1050
12	Ananthapur	1	350
13	Kurnool	-	-
Total		23	8050

Source: Horticulture Department, Andhra Pradesh

Table 27: Details of FPOs in Andhra Pradesh (all commodities)

S. No.	Name of District	No. of FPOs
1	Anantapur	5
2	Chittoor	4
3	East Godavari	5
4	Guntur	10
5	Kadapa	9
6	Krishna	9
7	Kurnool	10
8	Nellore	1
9	Prakasam	8
10	Sri Potti Sriramulu Nellore	9
11	Srikakulam	4
12	Visakhapatnam	9
13	Vizianagaram	6
14	West Godavari	6
	Total	95

Source: NABARD (For recommendation purpose the data of primary survey has been used)

Potential Raking Facilities

Post stakeholder meeting, GT had a detailed discussion with Mr. Pantulu, Andhra Pradesh Food Processing Society (APFPS) with regards to half and full raking facilities. Mr. Pantulu briefly summarised the concept and suggested raking facility at Railway stations - both half raking and full raking, at junctions for fruits and vegetables which would reduce time as well as cost of transportations

Half raking and full raking system of Indian Railways for carrying goods is the system wherein, the term half rake is used for a train consisting of 20 BCN wagons which can carry load of 1100 MT and full rake is used for a train consisting of 40 BCN wagons which can carry load of 2200 MT during one way transportation¹. Presently half rake goods sheds are 15 in number and full rake goods sheds are 54 in number². The cost of setting up a raking facility component wise is as follows:

Table 28: Costing heads for setting up full rake facility

Sr. No.	Cost Head	Approx. Cost (INR Cr.)
1	Abrasion resistance granolithic concrete floor 22500 sq mts. inside 720 mts x 30 mts shed	1.7
2	Shed 720 mts x 30 mts @ of Rs 14,500 per sq meter	31.32
3	7 EOT cranes of 30 T capacity @ 0.75 Cr.	5.25
4	Shed Illumination and Power Supply	2
5	Yard Alteration	2
6	Sick Line Extension and Upgradation	2.5
7	Other M&P	2
	Total	46.77

Source: Standardisation of Infrastructural Facilities for CC Rake Maintenance Depots on Indian Railways Depots³, Discussion with APFPS, Andhra Pradesh

This costing is for project which has 2 R&D lines and 720 mts * 30 mts shed complex which is sufficient for examination up to an average of 10 rakes per day (as of 2017). Considering inflation, the average cost as of 2021 is expected to rise to INR 60 Crore. It was suggested that a separate study may be initiated to assess the requirement of such a facility in Andhra Pradesh (and other states) as required.



Figure 1: Raking facility at Anantapur

¹ Freights train – general information, <https://www.irfca.org/fag/fag-freight.html>

² List of sheds, https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_tran/downloads/CR.pdf

³ Standardisation of infrastructural facilities for cc rake maintenance depots of IR depots <https://rdso.indianrailways.gov.in/works/uploads/File/Draft%20Report%20on%20standarization%20of%20facilities%20in%20Rake%20maintenance%20Depots%20of%20IR.pdf>

Tamil Nadu

Table 29: Area and production of horticultural crops in Tamil Nadu (2019-20)

Sl. No.	District	Fruits		Vegetables		Plantation Crops		SPICES		MEDICINAL CROPS AND AROMATIC CROPS		FLOWERS		Total	
		Area	Production	Area	Production	Area	Production	Area	Production	Area	Production	Area	Production	Area	Production
1	Ariyalur	1448	19294	2393	52434	30890	13570	446	769	17	5	244	3349	35,438	89,421
2	Coimbatore	10727	328965	7476	171045	104115	816564	3739	22914	12	71	550	6548	126,620	1,346,107
3	Cuddalore	7847	300758	8141	237998	31236	43027	947	2953	310	2231	772	8757	49,252	595,723
4	Dharmapuri	24145	112341	38102	994780	8789	105570	6542	19385	644	18153	9078	77517	87,300	1,327,745
5	Dindigul	28252	233272	15974	312677	40483	353795	7043	18081	2062	2336	1961	24679	95,775	944,840
6	Erode	16266	563392	15719	451412	15594	269836	5569	21849	368	5517	1876	19126	55,393	1,331,133
7	Kancheepuram	1430	8967	589	8730	650	8564	129	114	2	-	216	1919	3,016	28,293
8	Chengalpattu	8448	145862	1449	28357	1352	11192	317	203	-	-	145	1947	11,710	187,561
9	Kanniyakumari	6986	128092	1017	36908	55128	250768	2113	2543	23	321	80	1103	65,348	419,735
10	Karur	3855	114376	9598	132255	7051	58866	2275	9793	1075	1258	567	5547	24,421	322,095
11	Krishnagiri	38425	241579	37160	1070870	14926	163723	3174	22517	3	32	3228	45228	96,916	1,543,949
12	Madurai	11473	179443	3664	51916	11240	138233	1856	6334	255	3763	2022	23905	30,510	403,594
13	Nagapattinam	3219	42235	595	8354	5110	40186	269	521	18	271	281	2719	9,492	94,286
14	Namakkal	7055	117627	27363	750929	11939	88692	4407	12655	422	4303	1205	21707	52,391	995,912
15	Perambalur	720	11295	9096	132726	760	3233	795	2352	13	240	136	1782	11,520	151,628
16	Pudukkottai	5173	158628	2364	56367	15569	80743	654	555	1	1	371	3378	24,132	299,671
17	Ramanathapuram	801	10320	249	3781	10414	76239	17691	11823	54	810	103	823	29,312	103,795

Sl. No.	District	Fruits		Vegetables		Plantation Crops		SPICES		MEDICINAL CROPS AND AROMATIC CROPS		FLOWERS		Total	
		Area	Production	Area	Production	Area	Production	Area	Production	Area	Production	Area	Production	Area	Production
18	Salem	10118	104374	26028	728191	24699	554769	2578	7508	836	11213	3495	46589	67,754	1,452,644
19	Sivagangai	4457	71865	2934	45040	10452	34496	8342	29520	37	450	107	1410	26,329	182,781
20	Thanjavur	5586	200856	1107	39053	41347	410991	700	2920	37	296	178	1707	48,955	655,824
21	Theni	20293	502871	8590	194729	29505	141337	2305	10943	36	524	477	4829	61,206	855,233
22	The Nilgiris	1091	29108	9695	229399	63220	671092	3484	2810	20	301	69	8340	77,579	941,049
23	Thiruvallur	14166	96385	2940	39514	800	5467	955	4622	29	410	1714	21211	20,604	167,609
24	Thiruvannamalai	5714	131626	9575	252614	643	4398	1589	1035	5322	121796	4686	67737	27,529	579,206
25	Thiruvarur	1119	19386	751	13485	6225	68364	571	11730	7	105	41	525	8,714	113,595
26	Thoothukudi	10832	423613	8575	157693	9337	323593	13940	17151	428	365	280	2800	43,392	925,214
27	Thirupur	4767	104319	9617	201725	62112	363689	1056	6447	1792	1440	54	602	79,398	678,223
28	Thiruchirapalli	9805	330390	10553	302990	6066	35911	2319	9616	77	664	1063	11460	29,883	691,031
29	Thirunelveli	8433	161228	1744	45007	7780	50913	1298	2232	15	225	576	5331	19,846	264,936
30	Thenkasi	9721	107412	3592	51002	13372	45289	718	2048	69	3773	652	5528	28,124	215,052
31	Vellore	10797	74796	3123	51144	8143	73460	963	1000	-	-	588	5913	23,614	206,313
32	Ranipettai	1395	43543	2061	15618	954	10448	1118	2455	-	-	445	5254	5,973	77,317
33	Thirupathur	5428	31556	3045	58141	11375	113790	971	1714	-	-	270	2457	21,089	207,658
34	Villupuram	5694	109281	4663	64875	4844	16391	688	452	23	303	843	10722	16,755	202,024

Sl. No.	District	Fruits		Vegetables		Plantation Crops		SPICES		MEDICIANAL CROPS AND AROMATIC CROPS		FLOWERS		Total	
		Area	Production	Area	Production	Area	Production	Area	Production	Area	Production	Area	Production	Area	Production
35	Kallakurichi	857	23134	14399	440591	898	2138	1790	4619	1366	33314	201	2598	19,511	506,396
36	Virudhunagar	7082	57412	3547	43323	10425	87101	3156	10117	655	3914	597	6664	25,462	208,531
	Total	313625	5339103	307490	7475670	677442	5536436	106507	284300	16027	218405	39172	461712	1,460,264	19,315,626

Source: Horticulture Department, Tamil Nadu, Area in Ha and production in MT

Table 30: District wise banana production in Tamil Nadu

S. No.	District	Area (Ha.)	Production (MT)
1	Ariyalur	165	7128
2	Coimbatore	7071	312131
3	Cuddalore	4945	258854
4	Dharmapuri	318	13702
5	Dindigul	3475	109894
6	Erode	14116	533602
7	Kancheepuram	114	4932
8	Chengalpattu	240	10354
9	Kanniyakumari	4644	110658
10	Karur	2420	104331
11	Krishnagiri	376	16234
12	Madurai	3003	153332
13	Nagapattinam	634	27345
14	Namakkal	2592	100704
15	Perambalur	231	9948
16	Pudukkottai	2550	137473
17	Ramanathapuram	148	6377
18	Salem	1711	53182
19	Sivagangai	1166	50311
20	Thanjavur	4282	192416
21	Theni	5966	409571
22	The Nilgiris	621	26782
23	Thiruvallur	594	25639
24	Thiruvannamalai	2685	93183
25	Thiruvarur	372	16043
26	Thoothukudi	8237	408553
27	Thirupur	1980	78286
28	Thiruchirapalli	6335	315093
29	Thirunelveli	4587	124258
30	Thenkasi	1338	57708
31	Vellore	1252	54015
32	Ranipettai	726	41127
33	Thirupathur	1067	17811
34	Villupuram	1050	45306
35	Kallakurichi	453	19524
36	Virudhunagar	949	40922
	Total	92413	3986729

Source: Horticulture Department, Tamil Nadu

S. No.	District	No. of APEDA certified integrated packhouse	Fruits and Vegetables
1	Ariyalur	0	NA
2	Coimbatore	0	NA
3	Cuddalore	0	NA
4	Dharmapuri	0	NA
5	Dindigul	0	NA
6	Erode	0	NA
7	Kancheepuram	1	Other Fruits, Okra, Chilly, Herbs, Other Vegetables
8	Chengalpattu	0	NA
9	Kanniyakumari	2	Other Fruits, Other Vegetables,
10	Karur	0	NA
11	Krishnagiri	0	NA
12	Madurai	0	NA
13	Nagapattinam	0	NA
14	Namakkal	0	NA
15	Perambalur	0	NA
16	Pudukkottai	0	NA
17	Ramanathapuram	0	NA
18	Salem	0	NA
19	Sivagangai	0	NA
20	Thanjavur	0	NA
21	Theni	0	NA
22	The Nilgiris	0	NA
23	Thiruvallur	0	NA
24	Thiruvannamalai	0	NA
25	Thiruvarur	0	NA
26	Thoothukudi	0	NA
27	Thirupur	0	NA
28	Thiruchirapalli	1	Other Fruits, Okra, Chilly, Herbs, other vegetables
29	Thirunelveli	2	Pomegranate, Other Fruits, Other Vegetables
30	Thenkasi	0	NA
31	Vellore	0	NA
32	Ranipettai	0	NA
33	Thirupathur	0	NA
34	Villupuram	0	NA
35	Kallakurichi	0	NA
36	Virudhunagar	0	NA
Total		6	NA

Source: APEDA (For recommendation purpose the data of primary survey has been used)

Table 31: Details of FPOs in Tamil Nadu (all commodities)

S. No.	Name of District	Count of District
1	Ariyalur	2
2	Chengalpet	5
3	Coimbatore	5
4	Cuddalore	4
5	Dharmapuri	4
6	Dindigul	13
7	Erode	9
8	Kanchipuram	2
9	Kanyakumari	2
10	Karur	3
11	Krishnagiri	6
12	Madurai	11
13	Nagapattinam	3
14	Namakkal	3
15	Nilgiris	3
16	Perambalur	2
17	Pondicherry	1
18	Pudukkottai	3
19	Ramanathapuram	7
20	Salem	7
21	Sivaganga	4
22	Thanjavur	4
23	Theni	5
24	Thoothukudi	7
25	Tiruchirappalli	6
26	Tirunelveli	7
27	Tirupur	3
28	Tiruvallur	5
29	Tiruvannamalai	9
30	Vellore	6
31	Viluppuram	7
32	Virudhunagar	12
	Grand Total	170

Source: NABARD (For recommendation purpose the data of primary survey has been used)

Note on Mobile food Processing Unit - IIFPT

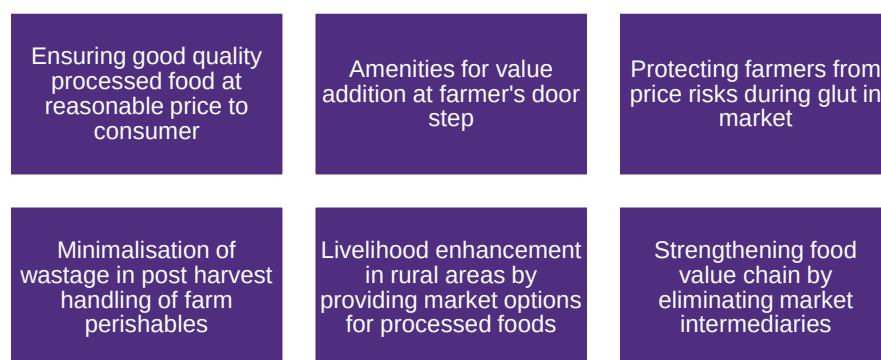
India is one of the top producers of agricultural and horticultural crops which is plagued by production and post-harvest losses. due to lack of post-harvest management and best practices. This is mainly due to lack of knowhow/availability of technology/ food processing units nearer to the production areas. To address this issue Indian Institute of Food Processing Technology (IIFPT), Thanjavur has developed the technology of Mobile Food Processing Unit (MFPU). Such units aid in value addition at places where setting up a permanent cottage level processing unit is nearly impossible due to various reasons such as non-availability of continuous power supply, lack of repair and service facilities near the farms. It is a unit which is housed in a mobile van so that the unit has access to fields in remote location covering vast areas of agricultural farms. In this unit one can process around half a tonne of raw fruit or vegetable in an 8-hour shift. The unit requires 2 personnel for operations, one skilled and one semi/non-skilled. It can perform on-farm processing primarily for the following fruits/vegetables.

- Garlic, Grapes, Apple, Sapota, Tomato, Onion, Ginger, Mango, Guava, Banana

Currently, this is being used primarily in tomato processing. The unit houses the following equipment facilities:

- Washer: to wash harvested fresh produce
- Blancher: to soften outer skin and tissues of the produce for further pulping
- Pulper: to extract fresh pulp from blanched produce separated from skin and seeds
- Steam jacket kettle: to boil and concentrate fresh pulp to required consistency
- Filling machine: to fill concentrate in cans/bottles
- Sterilising unit: for sterilisation of filled produce
- Cooling tank: for cooling sterilized produce
- Storage rack: for storage of processed products

Benefits of MPU



Case Study: Comparative economics of marketing raw and processed tomato (per hectare basis)

I. For Raw Tomato

- Farmer Net Profit = Sale price – (Cost of Crop Production + Marketing Cost)
- Farmer Net Profit (INR) = 500000 – (75000 + 10000) = 425000 /-

II. For Processed Tomato in MFPU

- Net Profit = Sale price of processed puree – (Cost of Crop Production + Processing Cost + Marketing Cost)
- Farmer Net Profit (INR) = 1312500 – (75000 + 262500 + 10000) = 965000 /-

Additional profit through MFPU (INR) = 965000 /- (more than doubled the farmer income). For the above calculations below figures has been taken as basis:

Assumptions: Yield of tomato @50 MT/Ha; Price for raw tomato - INR 5 /kg; Price of processed tomato (Puree)@ INR 75 /kg; Processing cost at MFPU @ INR 15 /kg of Puree and Recovery of Puree - 35%

Source: IIFPT, Thanjavur

Methodology Considered for Calculations

1. Requirement of Farmer Producer Organisations (FPOs)

- In every district of the cluster, top 3 banana growing talukas/tehsils/Revenue Division to have at least 2 FPOs for banana having membership of at least 500 members each.
- In each state the basis of division of district is different. For Gujarat, taluka; for Maharashtra and Uttar Pradesh, tehsil, for Andhra Pradesh, Revenue Division and for Tamil Nadu taluka has been considered.

2. Coverage under Good Agricultural Practices (GAP)

- GAP to be practiced in at least 50% of area under banana cultivation.

3. Requirement of Tissue Culture labs

- The requirement of planting material and requisite tissue culture laboratories have been calculated on the basis of saplings requirement of a cluster.
- Requirement of saplings has been calculated on basis of 3000 saplings in one hectare of farmland.
- Once tissue culture sapling is utilized, in the next cropping cycle, a farmer conventionally uses ratoons. Post this, in the next subsequent cropping cycle, he/she opts again to go for tissue culture-based planting material. Assuming 80% of farmers utilize tissue culture-based planting material, the number of required sapling and laboratory facilities has been calculated at cluster level.

4. Quality Testing Infrastructure

- In every district, 1 basic testing laboratory is proposed where farmers can get crop advisory, sample testing etc conducted. Atleast 1 high end laboratory facility with metal residue, pesticide residue, microbiological testing facilities should be present at cluster level.

5. Requirement of Collection Centres

- For every taluka/tehsil/taluk/Revenue Division, it is suggested to have at least 2 packhouses each with a processing capacity 50 MT/day.

6. Requirement of Crates for handling of banana

- Annual production of banana has been considered along with major seasons of trade.
- In Gujarat, major season of trade is April-July, for Maharashtra it is April-September, for Andhra Pradesh it is November to April and for Uttar Pradesh it is September to December. For Tamil Nadu there is no major peak season of harvesting and trade happens throughout the year.
- Majority of the harvesting and trade happens during the months in peak season. Based on the volumes handled during peak season, capacity in kg per crate (20 kg) as well as frequency of use per crate, an estimate with regard to the number of crates required to be deployed in the cluster has been calculated.

7. Requirement of Ripening Chambers

- From the primary survey as well as data collected from respective state horticulture departments Gujarat and Andhra Pradesh, it was found that there are sufficient ripening chambers to cater to the local demand of ripened banana. However, from survey of FPOs in both these states, it was found that there is a requirement of ripening facilities to cater to demand of ripened banana themselves instead of relying on third parties or market

intermediaries for the same. Hence, a 10 MT handling capacity/day of ripening facilities is proposed for each FPO dealing in banana.

- For the states of Uttar Pradesh, Maharashtra and Tamil Nadu, presently, household level ripening is prevalent in the clusters. This is carried out by use of ice, carbide and liquid ethylene. Ripening Chambers are required in these areas to cater to the local requirement of scientifically ripened banana. For this ripening chamber with 50 MT handling capacity/day are being proposed. This has been estimated based on primary survey, fresh banana consumption in concerned districts and per capita consumption levels.

8. Requirement of Integrated Packhouse Facilities (APEDA Certified)

- 1 integrated packhouse facility per cluster to cater to the demand is proposed. (Except for Jalgaon district, where there are 3 facilities already operational)

9. Suggestion for banana processing facilities at cluster level

- Currently, the level of secondary processing in banana is minimal across the study region. Cottage level processing is taking place which seldom follows best practices. It is proposed to set up 10MT/day capacity chips manufacturing unit at district level; along with an integrated banana pulp/puree facility at cluster level. With interventions at pre-production, production and post-harvest level, as well as availability of good quality raw material, price competitiveness and best trade practices, the scenario with regards to banana processing in the future can grow positively.

10. Reefer vans

- Utilization of reefer van transport is imperative for exports. It is also recommended to increase its use for domestic transfer of produce as well. Based on number of round trips undertaken by a van, peak season volumes of banana in a cluster/district as well as exports seasonality, the number of reefer van requirement has been calculated. Each reefer van is expected to have a capacity of 15 MT.

Meetings with Key Stakeholders - State Governments

Gujarat - Directorate of Horticulture and Food Processing

Key particulars	Details
Date and time of meeting	06/07/2021
Key Participants	Ms. Mittalben Dudhat – Deputy Director Horticulture
Key discussion points	Current banana ecosystem in Gujarat • Most of the farmers practicing precision farming • High use of tissue culture plants due to government initiatives and PSUs • Micro irrigation and drip irrigation availability high, particularly in banana sector (Gujarat Green Revolution Company) • High use water soluble fertilizer • Farmers use intercrop, crop rotation and use ratoons also • Major produce transferred through roadways. Truck is preferred mode of transport as number of loading and unloading instances are only 2 whereas in case of train, such instances are 3 to 4 times. Last mile connectivity in train transport is a challenge • Exports from Gujarat look low, but it is on par with other major exporting states • Exporters in Gujarat follow different method of exporting, they order a reefer container in Maharashtra, on arrival of which the container is loaded and then sent back to JNPT • Exports from the state of Gujarat are registered as exports of Maharashtra. Moreover, JNPT port is user friendly, it is preferred by exporters • There is a need to have cost and power efficient cold storage and ripening chamber facilities, for which research is ongoing.

Gujarat Agro Industries Corporation (GAIC)

Key particulars	Details
Date and time of meeting	06/07/2021
Key Participants	Meher Kumar – Assistant Manager, Marketing
Key discussion points	Current Value addition ecosystem in Gujarat • Chips is the major value-added product made in Gujarat • GAIC provides subsidy for projects which are less capital intensive and cost less than INR 25 lakhs. For projects greater than 25 lakhs subsidy is provided by MoFPI • GAIDC has assisted in building an APEDA certified packhouse which has been given on lease to ABNN exports and right now being used for mango exports in Bharuch. • Value addition in banana needs to be increased. This will help in enhancing shelf life and utilisation. During the time of pandemic, the banana was trading at the rate of INR 1 per kg. If value addition of banana is promoted, then in future such scenario can be avoided. • Few banana processing units (cottage level) are present in Bharuch and Narmada districts. Challenges

	The market of chips and flour is in nascent stage in India. Need to develop market further. Banana flour has huge potential for consumption due to health benefits. Needs better marketing and visibility amongst customers.
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Gujarat Green Revolution Corporation (GGRC)

Key particulars	Details
Date and time of meeting	17/08/2021 at 2 PM
Key Participants	Ashutosh Vadawale – Manager (Tech and MIS)
Key discussion points	Current Micro Irrigation scenario in Gujarat, especially in banana cultivation - Farmers receive subsidy under PMKSY PDMC (per drop more crop) scheme. Centre provides 45% of subsidy and total 70% provided. So, rest 25% is provided by state government. - Cost of setting up drip irrigation comes out to be INR 1-1.2 lakh + 12% GST per Ha, depending on Drip Irrigation design which changes from field-to-field conditions. So, a farmer receives subsidy to the tune of 70,000/unit. - When drip is installed then there is a tripartite agreement between farmers, service providers and GGRC. There is a trial run during which training is given to farmers regarding its usage. This is individual farmer-based scheme. So, if an FPO wants to apply for this scheme then it's not possible. Moreover, the complete process of filling the application and then actually getting drip in the field it takes 30-60 days. - Companies arrange training for farmers before, during and after installation of MIS in field. Also, with drip farmers are suggested that to take maximum advantage of drip Water Soluble Fertilizer should be used. That's why with increase in drip the consumption of WSF also increases. Challenges: - Currently, saturation in terms of land availability is a challenge. There is saturation achieved in the company as 24 lakh Ha is the potential are of Micro irrigation system (MIS) and 20.35 lakhs has already been covered under MIS in the state. Broadening of scope to canal management of dams is being looked into.

Gujarat State Fertilizers and Chemicals Limited (GSFC)

Key particulars	Details
Date and time of meeting	08/07/2021 at 11 AM
Key Participants	GSFC, Government of Gujarat - M R Dhukate – Assistant Manager – Tissue Culture Horticulture Department, Vadodara - Yogesh Khant, Horticulture Officer, Vadodara - Rajendra Parmar, Horticulture Officer, Vadodara
Key discussion points	Tissue Culture scenario in cluster GSFC (Gujarat State Fertilizers and Chemicals Ltd.) is a joint initiative with Gujarat government. The company provides Tissue Culture saplings in all the major banana growing

	districts of the state. The tissue culture which the company supplies is of G-9 variety. Their brand name is Sardar. 2. Total volume of tissue cultured saplings of G-9 developed and sold by the company is 15 lakhs nos. annually. To prepare a sapling, the company incurs a total cost of INR 12 per sapling. 3. A plant which comes fresh out of lab is not fit for use in field because it's grown in sanitized environment. Before the sapling is sold it is hardened in hardening centre for around 1-2 months. 4. Challenges faced in banana cultivation are that sometimes yield is not as per expectation. It is so because agriculture practices followed are not good and sometimes environmental conditions also affect the yield. Some farmers complain that they should get compensation for low yield but that is not possible. It needs to be understood by them that reason for low yield is not following of good agricultural practices. They pour Urea, NPK according to their own ideas without knowing the actual requirement of it in the field. So, awareness and education is extremely necessary.
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Gujarat Narmada Valley Fertilizers and Chemicals (GNFC)

Key particulars	Details
Date and time of meeting	09/07/2021 at 4 PM
Key Participants	Gujarat Narmada Valley Fertilisers & Chemicals (GNFC) 1. Senior Manager, Tissue Culture, Horticulture Department, Bharuch 1. Jenish Parekh – DDH, Bharuch 2. Atik Mullah – Horticulture Officer, Bharuch
Key discussion points	Tissue Culture scenario in cluster • GNFC (Gujarat State Fertilizers and Chemicals Ltd.) is a joint initiative with Gujarat government. The company provides Tissue Culture saplings in all the major banana growing districts of the state. The tissue culture which the company supplies is of G-9 variety. Under the brand name Narmada TC saplings are sold. • Total volume of tissue cultured saplings of G-9 developed and sold by the company is 10-12 lakhs nos. annually. To prepare a sapling the company incurs a total cost of INR 18 per sapling due to higher labour cost incurred. • Organic fertilizers are seldom used much by farmers and farmers are not applying fertilizer and/or DAP according to the requirement of the field. They just apply it on their own whims and fancies. Some farmers just put more fertilizer and DAP because their neighboring farmers has put more. Moreover, in their mind there is a notion that more the fertilizer more the crop which is ultimately damaging their own fields. • Recommendations from the person received were that farmers should not put more than required fertilizers. Instead, organic fertilizer should be used. Farmers need to be educated about balanced application of nutrients in the field. • Another recommendation received was E-mandi is the best concept for giving access of market to farmers and it need to be created for horticulture crops also.

	Cooperative/ FPOs need to be supported in terms of technology so they have access to terminal market.
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Directorate of Horticulture and Food Processing Department, Uttar Pradesh

Key particulars	Details
Date and time of meeting	15/07/2021 at 2 PM
Key Participants	Dr. Ravindra Tomar – Director Horticulture
Key discussion points	Current banana ecosystem of banana in Uttar Pradesh - Drip irrigation adoption is low in cluster - Landholding is small and surface water is available in plenty - After the introduction of new farm laws, trade of banana through APMC has completely stopped. This has led to an issue for a farmer as currently it has become difficult to discover price of the produce. Earlier it used to be rate being offered in APMC. A different mechanism of price discovery is required for farmers. - Recommendations from the person received were that there is a need to grow different varieties of banana in UP - An e-platform is the need of the hour as it will not only help farmer in selling the produce but also in discovering price - Farmers who have land more than 2 Ha need to develop collection centre on their land so that primary processing can happen near to field itself - There is requirement of Tissue Culture labs in the state as farmers will be getting saplings at lower price if lab will be local - Lack of knowledge regarding fruit care practices but interest in farmers to adopt such practices, training need to be imparted

Tamil Nadu Horticulture Development Agency (TANHODA)

Key particulars	Details
Date and time of meeting	15/07/2021 at 2 PM
Key Participants	- Smt. Brinda Devi, Director, TANHODA - Mr. Tamil Vendhan, Assistant Director, TANHODA - Mr. Arumugam, Assistant Horticulture Director, Theni - Smt. Thangam, APEDA, State Incharge - Smt. Shobana, APEDA, Field Officer - Mr. Chirag Jain, Director, Grant Thornton - Mr. Khushrow Chinoy, Manager, Grant Thornton - Mr. Ravindiran, Consultant, Grant Thornton
Key discussion points	Challenges in Banana Sector in the cluster - Lack of Tissue Culture Lab and residue testing laboratory facilities in the cluster - Improvement in penetration of GAP in banana cultivation needs to improve - Negligible banana processing or secondary value addition taking place in the cluster. - Negligible number of Reefer Vans - Need for more Integrated Packhouse in the cluster - FPOs has to be created with banana as Focus Crop - Below 5% of Farmers follow Fruit Care Practices

	• Less Branding, Marketing & Export Promotion Interventions Recommended ✓ Tissue Culture unit should be established as farmers using TC more and they have to procure it from Coimbatore & Hosur (The Cost Increases) ✓ MRL Should be Established ✓ Mobile Soil Testing Lab should also be established at Cluster Level ✓ Upcoming Mega Food Park in Theni Which would provide facilities for processing banana ✓ More FPOs should be created and they should be encouraged to create farm machinery Bank ✓ Measures should be taken to Increase GAP in the cluster ✓ Fruit Care Practices should expand to more area (As most of the losses happening during Harvesting) ✓ Should Create Awareness about IP/INM and increase the implementing area ✓ Effort needed to be taken to develop the Cluster Brand. ✓ More analysis needed to be done understand the status –Quo of the infrastructure and Demand
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Snapshots – Field Visits and Stakeholder Consultations



District Horticulture
Dept - Gorakhpur



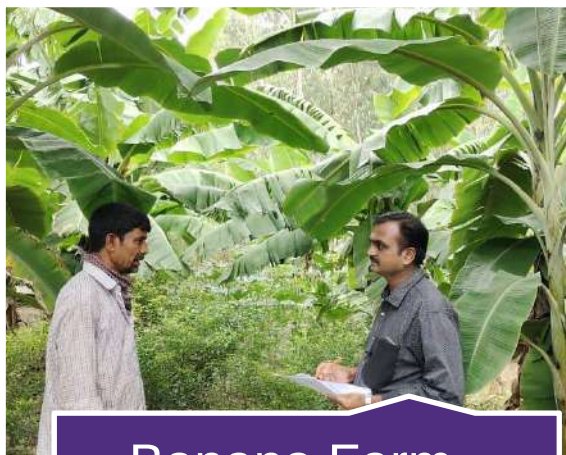
District Horticulture
Dept - Tuticorin



District Horticulture
Dept - Anantapur



District Horticulture
Dept - Theni



Banana Farm -
Vizag



Open Market -
Madurai



Banana farm -
Fatehpur



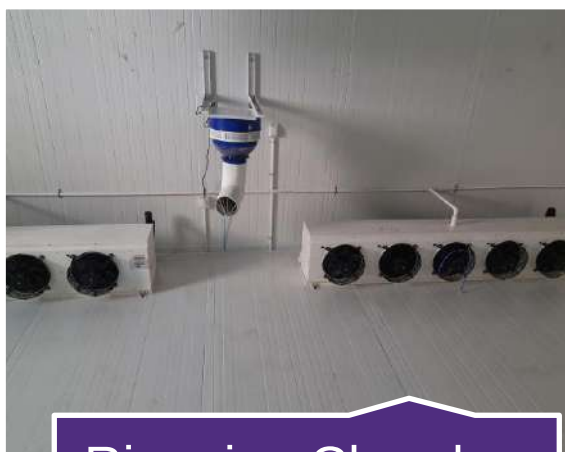
Micro Irrigation -
Vadodara



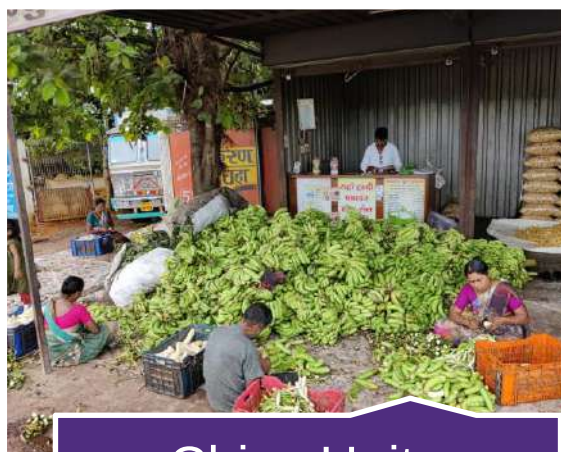
FPC - Jalgaon



Banana Fibre Unit - Anand



Ripening Chamber - Bharuch



Chips Unit - Jalgaon